An explicit uniform spectral gap for the Laughlin state parent Hamiltonian on rectangular tori

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Preprint — 23 October 2026

Abstract

We consider spinless fermions in the lowest Landau level of a rectangular torus with periods L_x, L_y, interacting through the untruncated V_1 pseudopotential, normalized so that every pair of relative angular momentum one pays energy one. We prove a transfer principle: every planar certificate of the form used by OpenAI on the sphere, with constant γ, gives $H_{\mathbb{T}}^2 \geq (\gamma - \epsilon_{\mathrm{tr}}(\mathbb{T}))H_{\mathbb{T}}$ for the Hamiltonian $H_{\mathbb{T}}$ on the whole Fock space of every rectangular torus, where $\epsilon_{\mathrm{tr}}(\mathbb{T})$ is bounded by an explicit function of the shorter period L that tends to zero at the Gaussian rate $O(L^4\, 2^{-L^2/20})$. For OpenAI's planar certificate the bounds are explicit: $H_{\mathbb{T}}^2 \geq \gamma_{\mathbb{T}} H_{\mathbb{T}}$, with a constant $\gamma_{\mathbb{T}}$ that depends on the torus only through its lattice of periods, and for every rectangular torus whose shorter period is at least L, with $L \geq 16$ (in units of the magnetic length), one has $\gamma_{\mathbb{T}} \geq \underline{\gamma}(L)$, where $\underline{\gamma}$ is nondecreasing and tends at the same rate to $\gamma_* = 4616733319001/10^{14}$, the constant of OpenAI's certificate; numerically $\underline{\gamma}(L) > 0$ for $L \geq 16.457$, $\underline{\gamma}(L) > 1/25$ for $L \geq 16.857$ and $\underline{\gamma}(L) \geq \gamma_* - 10^{-5}$ for $L \geq 18.685$. The bound is uniform in the particle number and in the aspect ratio. Hence the spectrum of $H_{\mathbb{T}}$ in every particle sector lies in $\{0\} \cup [\gamma_{\mathbb{T}}, \infty)$. At $N_\phi = 3N$ flux quanta the kernel on N particles is spanned by the three centre-of-mass copies of the Laughlin state, and no sector with more particles has a zero mode; the same three-dimensional count holds, for every modular parameter, for the explicit theta-function pair operators $B_n^{(\tau)}$, whose identification with the V_1 Hamiltonian of an oblique torus is not proved here. The proof transcribes OpenAI's spherical argument to the torus and consumes the planar certificate as data. On the torus the averaging over rotations is replaced by an average over magnetic translations, and torus Moyal identities make the pair-level and four-body parts of the comparison operator exact. The three-body coefficient of the normal-ordered square of $H_{\mathbb{T}}$ is exactly the periodized planar three-body target, so the three-body target transfers without error; the comparison operator's three-body part carries one non-orthonormality term, bounded by $C_3\mathcal{E}_8(1+\mathcal{E}_8)H_{\mathbb{T}}$. The remaining errors are Gaussian in the shorter period.

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1 Introduction

We prove that spectral-gap certificates for the Laughlin V_1 model pass from the plane to rectangular tori with explicit errors. The main result is a transfer principle (Theorem 1.1): every planar certificate of the form introduced by OpenAI [1], with constant γ, gives $H_{\mathbb{T}}^2 \geq (\gamma - \epsilon_{\mathrm{tr}}(\mathbb{T}))H_{\mathbb{T}}$ on the whole Fock space for the untruncated V_1 Hamiltonian $H_{\mathbb{T}}$ of every rectangular torus, where $\epsilon_{\mathrm{tr}}(\mathbb{T})$ is bounded by an explicit function of the shorter period that tends to zero at a Gaussian rate. For the certificate of [1] we evaluate these bounds. On every rectangular torus whose shorter period is at least L, with $L \geq 16$, the inequality $H_{\mathbb{T}}^2 \geq \gamma_{\mathbb{T}} H_{\mathbb{T}}$ holds with a constant $\gamma_{\mathbb{T}} \geq \underline{\gamma}(L)$, uniformly in the particle number and in the aspect ratio, where $\underline{\gamma}$ is an explicit nondecreasing function that tends to the planar constant γ_* of [1] at a Gaussian rate in L; both $\gamma_* - \underline{\gamma}(L)$ and the bound on $\epsilon_{\mathrm{tr}}(\mathbb{T})$ are $O(L^4\, 2^{-L^2/20})$ (Remark 9.3). The thresholds are explicit: a numerical evaluation of $\underline{\gamma}$ (Remark 9.4) gives $\underline{\gamma}(L) > 0$ for $L \geq 16.457$, $\underline{\gamma}(L) > 1/25$ for $L \geq 16.857$ and $\underline{\gamma}(L) \geq \gamma_* - 10^{-5}$ for $L \geq 18.685$, in units of the magnetic length (Theorem 1.2).

Laughlin's wave function [2] is the exact zero-energy ground state of Haldane's V_1 pseudopotential Hamiltonian [3], in which a pair of spinless fermions in the lowest Landau level pays energy one if its relative angular momentum is one and nothing otherwise. Whether this Hamiltonian has a spectral gap above its zero modes that stays bounded below as the system grows is a long-standing question [11, Appendix A]. For the untruncated interaction it was answered on the sphere and in the plane by OpenAI [1]: with $\gamma_* = 4616733319001/10^{14} > 1/25$, the spherical Hamiltonian H_Q satisfies $H_Q^2 \geq \gamma H_Q$ on the whole Fock space for every $\gamma < \gamma_*$ and all sufficiently large flux Q, and the planar Hamiltonian satisfies $(H_N^{\mathrm{pl}})^2 \geq \gamma_* H_N^{\mathrm{pl}}$ for every particle number. The proof compares H_Q^2 with a positive operator built from seven explicit rows, averaged over rotations; the rows satisfy finitely many planar inequalities, verified by exact rational arithmetic. Uniform gaps for truncated versions of the pseudopotential were obtained earlier, on thin cylinders and thin tori [6, 7]. Two further works treat the untruncated interaction on the sphere: Gao [8] raised the constant to about 0.219, on the sphere and in the plane, by a semidefinite search, and Scheer and Khalaf [9] constructed a semidefinite hierarchy of lower bounds on the sphere, with exact rational certificates and a rigorous lower bound on the limit inferior of the gap as the flux grows, and leave the torus to future work. In all three spherical results the flux threshold is existential: it is shown to exist, not computed.

A uniform bound of this kind, for every flat torus whose shortest lattice vector exceeds an existential threshold, has also been claimed in a recent candidate proof prepared by Hou with advice from Wu [10]; the present argument, obtained independently, is restricted to rectangular tori but makes the threshold, the constant $\underline{\gamma}(L)$ and its Gaussian rate of convergence to γ_* explicit. That candidate proof, which is unrefereed and not on arXiv, states the bound $H^2 \geq \gamma H$ for every $\gamma < \gamma_*$, uniformly in the particle number, on tori of every lattice shape and with every boundary twist, and also claims that the kernel at filling one third is three-dimensional for every shape. Where the two overlap — the uniform bound on rectangular tori and the three-dimensional

kernel — the statements agree. It transfers fixed certificates and states no principle for a class of certificates; Theorem 1.1 below is such a principle, with its error explicit.

1.1 The transfer principle

Let $\mathbb{T}$ be a rectangular torus with periods L_x, L_y (in units of the magnetic length) and $N_\phi = L_xL_y/2\pi$ flux quanta, let $\mathfrak{h}$ be its lowest Landau level and $\mathcal{F}$ the fermionic Fock space over $\mathfrak{h}$, and let $H_{\mathbb{T}}$ be the untruncated V_1 Hamiltonian: the second quantization of the periodization, over the lattice of periods, of the planar projector onto pairs of relative angular momentum one (Definition 2.10). Let $L := \min\{L_x, L_y\}$ be the shorter period.

The certificate of [1] is a finite family of rows together with planar inequalities that they satisfy. On the torus, at a base point R, row ν is the operator

$$F_\nu(R) = \lambda_\nu\, B(\tilde{u}_{t_\nu}(R)) + \sum_{(p,j)\in\mathcal{S}_\nu} \alpha^\nu_{pj}\, c(\chi^R_{p+j-t_\nu})^*\, c(\chi^R_j)\, B(\tilde{u}_p(R))$$

of Definition 6.7, where $B(\tilde{u}_p(R))$ removes a pair in the periodized pair state of label p about R and $c(\chi^R_m)$ a particle in the periodized orbital of degree m about R (Section 5). The proof uses the certificate only through the data of the rows and the inequalities, which define the following class.

Definition (planar certificates of the same form)**.** A *planar certificate of the same form* as that of [1] consists of finitely many rows ν, each with an integer $t_\nu \geq 0$, a real λ_ν and real coefficients α^ν_{pj} supported in a finite set $\mathcal{S}_\nu \subset \{(p,j) : 0 \leq p \leq P_c,\ 0 \leq j \leq M_c,\ 0 \leq p+j-t_\nu \leq M_c\}$, where $P_c, M_c \geq 0$ are integers, and an allowance $\varepsilon \geq 0$, such that, with the cutoffs $Z_c := \max\{p+j : (p,j) \in \bigcup_\nu \mathcal{S}_\nu\}$ and $D_c := \max\{p+j+(q+l-t_\nu)\}$ over $(p,j),(q,l) \in \mathcal{S}_\nu$, the inequalities

$$e_z \leq \tfrac{3}{2} q_z \quad (z \leq Z_c,\ z \notin \{0,1,3\}), \qquad G_D(2|e_1\rangle\langle e_1| - E_D + \varepsilon \mathbf{1})G_D \geq 0 \quad (1 \leq D \leq D_c) \tag{1.1}$$

hold. Here $q_z = (3z-1)(-\frac{1}{2})^z$ are the planar three-body coefficients (Lemma 7.3); e_z and E_D are the three-body scalars and the four-body copy matrices of the rows, defined by (6.3) and (6.4) from the planar coupling coefficients (the sum over T in (6.3) running to Z_c, and (6.4) taken for $1 \leq D \leq D_c$); G_D is the Gram matrix of the planar four-body relative states $W_4\omega_{D,r}$, $r \in \mathcal{R}_D$, defined with them in Section 6; $\mathcal{R}_D$ is the set of odd integers in $[1, D]$; and $|e_1\rangle\langle e_1|$ is the matrix with $(1,1)$ entry 1 and all other entries zero. The *constant* of the certificate is

$$\gamma := 1 - \eta_* - \delta_\infty(Z_c) - \varepsilon\, N_4, \qquad \eta_* = \sum_\nu \lambda_\nu^2,$$

$$\delta_\infty(Z_c) := \frac{3}{2} \sum_{\substack{z > Z_c \\ z \text{ odd}}} \frac{(3z-1)(z+1)}{2^z}, \qquad N_4 := \sum_{D=1}^{D_c} \Big\lfloor \frac{(D+1)^2}{4} \Big\rfloor.$$

We require that some $\mathcal{S}_\nu$ be nonempty, that $M_c \geq 1$ and that $D_c \geq 2$, so that Z_c is defined and the bounds of Lemma 5.3(b) are used only for $m \geq 1$; the certificate of [1] has $M_c = 8$ and $D_c = 23$.

Every row of this form ends in a pair annihilator and has a definite weight $t_\nu + 1$ (each of its summands removes two particles and lowers the total orbital degree by $t_\nu + 1$), and these two features are all that the proof uses of a row; that is why the principle is a property of the form, not of the numbers in the certificate of [1].

Theorem 1.1 (the transfer principle)**.** *Given a planar certificate of the same form with constant γ, for every rectangular torus, on $\mathcal{F}$,*

$$H_{\mathbb{T}}^2 \geq (\gamma - \epsilon_{\mathrm{tr}}(\mathbb{T}))\, H_{\mathbb{T}},$$

where, with $\mathcal{E}_m = \mathcal{E}_m(\mathbb{T})$ and $\sigma_z = \sigma_z(\mathbb{T})$,

$$\epsilon_{\mathrm{tr}}(\mathbb{T}) := (1-\lambda_{\min}) + \frac{3}{2} \sum_{\substack{z>Z_c \\ z \text{ odd}}} \frac{(3z-1)(z+1)}{2^z}\sigma_z + (M_c+1)(P_c+1)\sum_{\nu}\|\alpha^\nu\|^2\, \mathcal{E}_{M_c}(1+\mathcal{E}_{M_c}) \\ + \varepsilon N_4((1+\mathcal{E}_{D_c})^2 - 1) + 2N_4\, \mathcal{E}_{D_c-1}(1+\mathcal{E}_{D_c})^2.$$

Here $\lambda_{\min}$ is the smallest eigenvalue of the Gram matrix of the torus pair functions (Lemma 4.2), $\sigma_z(\mathbb{T}) = \sup_R \max_{m\leq z}(\|\chi_m^R\|^2 - 1)$ (Lemma 5.3(c)), $\mathcal{E}_m(\mathbb{T})$ is the lattice sum, over the nonzero periods, of the norms of the truncated displacement matrices $\mathcal{D}^{(m)}$ (Section 5), and $\|\alpha^\nu\|^2 = \sum_{p,j}(\alpha_{pj}^\nu)^2$. For every rectangular torus whose shorter period is at least L, $\epsilon_{\mathrm{tr}}(\mathbb{T}) \leq \epsilon_{\mathrm{tr}}(L)$ with $\epsilon_{\mathrm{tr}}(L) \to 0$ as $L \to \infty$. In particular, if $\gamma > 0$, then for every $\gamma' < \gamma$ there is L_0 such that $H_{\mathbb{T}}^2 \geq \gamma' H_{\mathbb{T}}$ on the whole Fock space of every rectangular torus whose shorter period is at least L_0, uniformly in the particle number and the aspect ratio.

The function $\epsilon_{\mathrm{tr}}(L)$, obtained in the proof from explicit bounds on the terms of $\epsilon_{\mathrm{tr}}(\mathbb{T})$, is $O(L^4\, 2^{-L^2/20})$ as $L \to \infty$, with a constant that depends on the certificate (Remark 9.3). The theorem does not require the planar certificate to come from a spherical proof: any finite family of rows of this form satisfying (1.1) with $\gamma > 0$ yields a gap on rectangular tori, uniform above a threshold period, with that rate.

For the certificate of [1] (Section 6 and Appendix A), $P_c = 7$, $M_c = 8$, $Z_c = 15$, $D_c = 23$, $\delta_\infty(15) = 61/4096$ and $N_4 = 1222$, with $\eta_* = 93527408868499/10^{14}$ and $\varepsilon = 3/10^6$; its constant is $\gamma = \gamma_* = 4616733319001/10^{14} > 1/25$, and $\gamma_* - \epsilon_{\mathrm{tr}}(\mathbb{T})$ is, term by term, the constant $\gamma_{\mathbb{T}}$ of (9.1). Theorem 1.1 for this certificate, together with an evaluation of ϵ_{tr} for it, gives the following explicit bounds.

Theorem 1.2 (OpenAI's certificate on rectangular tori)**.** *For every rectangular torus $\mathbb{T}$, on the whole Fock space,*

$$H_{\mathbb{T}}^2 \geq \gamma_{\mathbb{T}}\, H_{\mathbb{T}},$$

with the explicit constant $\gamma_{\mathbb{T}}$ of (9.1), which depends on the torus only through its lattice of periods. For $L \geq 16$ let $\underline{\gamma}(L)$ be the explicit function of Corollary 9.2. Then $\gamma_{\mathbb{T}} \geq \underline{\gamma}(L)$ for every rectangular torus whose shorter period is at least L, so that $H_{\mathbb{T}}^2 \geq \underline{\gamma}(L) H_{\mathbb{T}}$ uniformly in the particle number and in the aspect ratio; $\underline{\gamma}$ is nondecreasing on $[16,\infty)$, and $\underline{\gamma}(L) \to \gamma_*$ as $L \to \infty$, with $\gamma_* - \underline{\gamma}(L) = O(L^4\, 2^{-L^2/20})$. Numerically, $\underline{\gamma}(L) > 0$ for $L \geq 16.457$, $\underline{\gamma}(L) > 1/25$ for $L \geq 16.857$ and $\underline{\gamma}(L) \geq \gamma_* - 10^{-5}$ for $L \geq 18.685$, in units of the magnetic length; these values are floating-point evaluations of rigorous closed-form bounds, not interval-arithmetic certificates (Remark 9.4).

Theorem 1.2 is thus the case of Theorem 1.1 for the certificate of [1], with ϵ_{tr} evaluated. Section 9 proves the two in the opposite order: Theorem 9.1 and Corollary 9.2, with Remark 9.3 and the thresholds evaluated in Section 9.2, prove Theorem 1.2 directly, and Section 9.4 then proves Theorem 1.1, restated as Theorem 9.6, by the same argument with the data of the certificate left general.

Remark 1.3 (scope). Theorem 1.1 is proved for the form of the definition: rows each of whose summands is a pair annihilator, alone or dressed by one creation and one annihilation operator; the two-body V_1 interaction of a single species of spinless fermions; and rectangular tori without boundary twists. Three natural extensions are not covered. Rows of higher degree in the fermion operators have squares with parts on five or more particles; they would need the torus Moyal identity on that many particles, which Theorem 6.2 states and proves for every number of particles, and a comparison term for those parts, which is not constructed here. Two species, as in the spinful hollow-core model of Section 10, and general lattices of periods with boundary twists are not treated. Whether these extensions hold is not claimed here.

By Theorem 1.2, the spectrum of $H_{\mathbb{T}}$ in every particle sector lies in $\{0\} \cup [\gamma_{\mathbb{T}}, \infty)$. At $N_\phi = 3N$ the kernel on N particles is three-dimensional, spanned by the three centre-of-mass copies of the Laughlin state, and no sector with more than N particles has a zero mode (Theorem 3.1, Proposition 3.2); the same count holds, for every modular parameter, for the explicit theta-function pair operators $B_n^{(\tau)}$ of (3.2), whose identification with the V_1 Hamiltonian of an oblique torus is not proved here. So $H_{\mathbb{T}} \geq \gamma_{\mathbb{T}}(\mathbf{1} - P_N)$ on N particles, with P_N of rank three, and $H_{\mathbb{T}} \geq \gamma_{\mathbb{T}}$ on more particles (Corollary 9.5).

At the square aspect ratio $L = \sqrt{2\pi N_\phi}$, so the bound is positive for $N_\phi \geq 44$ and exceeds $1/25$ for $N_\phi \geq 46$. At filling one third ($N_\phi = 3N$) this means a gap for $N \geq 15$ ($\underline{\gamma}(L) \geq 0.0385$ at $N = 15$) and a gap larger than $1/25$ for $N \geq 16$ ($\underline{\gamma}(L) \geq 0.0456$ at $N = 16$). For elongated tori the condition is on the shorter period, so larger systems are needed. All error terms decay like e^{-cL^2} in the shorter period; the binding one at the thresholds is a lattice sum of norms of truncated displacement matrices, bounded by a multiple of $L^{44}e^{-L^2/4}$ (Lemma 5.3), and the thresholds are where it becomes small. The slowest-decaying bound is that on the three-body tail beyond the blocks covered by the certificate, $O(L^4\, 2^{-L^2/20})$ (Proposition 7.5); it sets the rate of Remark 9.3.

1.2 From the sphere to the torus

The torus proof follows the architecture of [1]. One expands $H_{\mathbb{T}}^2$ in normal order into two-, three- and four-body terms, and compares them with the corresponding terms of a positive comparison operator $K_{\mathbb{T}} = \frac{1}{\pi}\int_{\mathbb{T}} \sum_\nu F_\nu(R)^* F_\nu(R)\, \mathrm{d}R$, where the rows $F_\nu(R)$ of Section 1.1 are the planar rows of the certificate, those of [1] for Theorem 1.2, built at the base point R from periodized orbitals and pair states. The planar certificate is used as data, without change: for that of [1], the row coefficients, the thirteen three-body inequalities and the twenty-three four-body matrix inequalities. Section, lemma and equation numbers of [1] refer to its version of 24 September 2026, and its lemmas are also cited by name. On the sphere the comparison operator is averaged over rotations and the finite-flux corrections are controlled by a relative-error transfer lemma; on the torus the average over the base point R replaces the rotations, and the corrections take a different and simpler form. Four facts, proved here, make the transcription work:

(1) *The projector replacement.* On two particles $H_{\mathbb{T}}$ is not a projection. The Gram matrix of the torus pair functions is computed in closed form as a lattice sum of image overlaps (Lemma 4.1); its eigenvalues are $1 \pm \Delta(L_y) \pm \Delta(L_x)$ for even N_ϕ and $1 \pm (\Delta(L_x)^2 + \Delta(L_y)^2)^{1/2}$ for odd N_ϕ, up to a smaller remainder, with $\Delta(\ell) = 2(\ell^2/4 - 1)e^{-\ell^2/8}$ (Lemma 4.2). Only the smallest eigenvalue enters the budget.

(2) *Periodized local modes.* The periodized orbitals about a point have an explicit Gram matrix, a lattice sum of displacement matrices with phases (Lemma 5.2), whose deviation from the

identity is bounded uniformly by lattice sums of norms of truncated displacement matrices (Lemma 5.3).

(3) *Torus Moyal identities.* Averaging a projector onto periodized k-particle states with a fixed relative part and a centre-of-mass orbital about R over $R \in \mathbb{T}$ gives exactly the periodized image of the relative projector, and different centre-of-mass degrees average to zero (Theorem 6.2). This replaces Schur's lemma for $SU(2)$; the proof uses Schur's lemma for the magnetic translations of the centre-of-mass torus. With it, the pair-level part of $K_{\mathbb{T}}$ is exactly $\eta_* H_{\mathbb{T}}$, and its three- and four-body parts are exactly the periodized images of the planar partial traces that define the certificate, up to one small non-orthonormality term (Proposition 6.8).

(4) *The square of $H_{\mathbb{T}}$.* The three-body coefficient of $H_{\mathbb{T}}^2$ is a sum of products of two pair operators sharing exactly one particle, and periodization is multiplicative on such products (Lemma 7.2). Hence it is *exactly* the periodized planar three-body target, with the planar coefficients $q_z = (3z-1)(-\frac{1}{2})^z$ on every block (Theorem 7.4). The three-body target therefore transfers without error, and the tail beyond the blocks covered by the certificate is bounded with an explicit rate (Proposition 7.5). The four-body coefficient involves products on disjoint pairs, and a Bessel-type bound for the localized channels of the relative coordinate of the two pair centres closes the budget (Lemma 8.1).

The resulting error terms are all relative to $H_{\mathbb{T}}$ (they vanish on its kernel and carry no factor of the particle number), and all are Gaussian in the shorter period. They are collected in the error $\epsilon_{\mathrm{tr}}(\mathbb{T})$ of Theorem 1.1, and for the certificate of [1] in the constant $\gamma_{\mathbb{T}}$ of (9.1).

Section 3 determines the kernel at $N_\phi = 3N$, for the pair operators $B_n^{(\tau)}$ of (3.2) at every modular parameter and hence for $H_{\mathbb{T}}$ on every rectangular torus. The threefold degeneracy of the torus Laughlin states is classical [4]; that these states are the only zero modes of the torus V_1 Hamiltonian is stated for rectangular tori in [5, Section II.2], and a proof for every lattice shape and boundary twist is claimed in [10]. The argument of Section 3 uses third-order vanishing at coincidences, division by the cube of the theta-function Jastrow factor, and Liouville's theorem.

1.3 Structure of the paper

Section 2 sets up the torus, the planar lowest Landau level, the periodization and the periodized image of planar operators, and defines $H_{\mathbb{T}}$; Proposition 2.11 gives its orbital form. Section 3 determines the kernel. Section 4 computes the pair Gram matrix and records the normal-ordered square of a lifted pair operator. Section 5 introduces the periodized local modes and bounds their Gram matrix. Section 6 proves the torus Moyal identities, states the certificate, and decomposes the comparison operator. Section 7 proves the exact three-body identity and the three-body comparison, and Section 8 the four-body Bessel bound. Section 9 proves Theorem 1.1 and evaluates it for the certificate of [1]: it assembles the budget and evaluates the constants for that certificate, which proves Theorem 1.2, derives the spectral consequences, and then proves Theorem 1.1 by the same argument with the data of the certificate left general. Section 10 discusses the transfer principle as a basis for further work and lists what is not proved. Appendix A records the certificate data, and Appendix B lists the notation.

2 Setting

We use units in which the magnetic field and the magnetic length are equal to one ($\hbar = e = 1$). For $a = (a_x, a_y)$ and $b = (b_x, b_y)$ in $\mathbb{R}^2$ we write $a \wedge b := a_x b_y - a_y b_x$, and for $r = (x, y) \in \mathbb{R}^2$ we write $w := x + \mathrm{i}y$ for the corresponding complex coordinate. Inner products are linear in the second argument.

2.1 The torus and its lowest Landau level

Definition 2.1 (the torus and its lowest Landau level)**.** Let $N_\phi \in \mathbb{N}$ and $L_x, L_y > 0$ with $L_x L_y = 2\pi N_\phi$; put $\kappa := 2\pi/L_y$, so that $L_x = \kappa N_\phi$. Let $\Lambda := L_x\mathbb{Z} \times L_y\mathbb{Z}$, $\mathbb{T} := \mathbb{R}^2/\Lambda$, $\mathsf{C} := [0, L_x) \times [0, L_y)$, and let $L := \min\{L_x, L_y\}$ be the *shorter period*. Let

$$D_x := \partial_x, \qquad D_y := \partial_y - \mathrm{i}x$$

be the covariant derivatives of the vector potential $(0, x)$. A smooth function f on $\mathbb{R}^2$ is an *LLL function* if $(D_x + \mathrm{i}D_y)f = 0$. The *lowest Landau level of the torus*, $\mathfrak{h}$, is the space of LLL functions with

$$f(x, y + L_y) = f(x, y), \qquad f(x + L_x, y) = e^{\mathrm{i}L_x y} f(x, y), \tag{2.1}$$

with the inner product $\langle f, g\rangle := \int_{\mathsf{C}} \overline{f}\, g \,\mathrm{d}r$.

An LLL function is exactly a function of the form $f = e^{-x^2/2} g(w)$ with g entire, since $(D_x + \mathrm{i}D_y)(e^{-x^2/2}g) = e^{-x^2/2}(\partial_x + \mathrm{i}\partial_y)g$. The functions $|f|^2$ and $\overline{f}g$ are Λ-periodic for $f, g \in \mathfrak{h}$, so the inner product is an integral over $\mathbb{T}$. All results concern these boundary conditions, without Aharonov–Bohm phases through the cycles.

Lemma 2.2 (Landau-gauge orbitals)**.** *For $n \in \mathbb{Z}$ let $\phi_n(x, y) := (L_y\sqrt{\pi})^{-1/2} e^{\mathrm{i}\kappa n y} e^{-(x-\kappa n)^2/2}$ and for $j \in \mathbb{Z}_{N_\phi}$ let $\psi_j := \sum_{m\in\mathbb{Z}} \phi_{\tilde{\jmath}+mN_\phi}$, where $\tilde{\jmath} \in \mathbb{Z}$ is any representative of j. Write $[n]$ for the class of $n \in \mathbb{Z}$ in $\mathbb{Z}_{N_\phi}$.*

(i) The series converges absolutely and locally uniformly, ψ_j does not depend on $\tilde{\jmath}$, and $\psi_j \in \mathfrak{h}$.

(ii) $(\psi_j)_{j\in\mathbb{Z}_{N_\phi}}$ is an orthonormal basis of $\mathfrak{h}$; in particular $\dim \mathfrak{h} = N_\phi$, and every $f \in \mathfrak{h}$ is bounded.

Proof. (i) $|\phi_n(x, y)| = (L_y\sqrt{\pi})^{-1/2} e^{-(x-\kappa n)^2/2}$, and these Gaussians are summable over $n \in \tilde{\jmath} + N_\phi\mathbb{Z}$ locally uniformly in x. Each ϕ_n is $e^{-x^2/2}$ times an entire function of w (namely $(L_y\sqrt{\pi})^{-1/2} e^{\kappa n w - \kappa^2 n^2/2}$), so $e^{x^2/2}\psi_j$ is a locally uniform limit of entire functions, hence entire, and ψ_j is an LLL function. Replacing $\tilde{\jmath}$ by $\tilde{\jmath} + N_\phi$ relabels the sum. Since $\kappa L_y = 2\pi$ each ϕ_n is L_y-periodic in y, and since $\kappa N_\phi = L_x$, $\phi_n(x + L_x, y) = e^{\mathrm{i}L_x y}\phi_{n-N_\phi}(x, y)$; summing over the class gives (2.1).

(ii) For $n, n' \in \mathbb{Z}$ the y-integral of $\overline{\phi_n}\phi_{n'}$ over $[0, L_y)$ is $\delta_{nn'}\pi^{-1/2}e^{-(x-\kappa n)^2}$. Hence $\langle \psi_i, \psi_j\rangle = 0$ for $i \neq j$, and $\langle \psi_j, \psi_j\rangle = \sum_m \int_0^{L_x} \pi^{-1/2} e^{-(x-\kappa\tilde{\jmath}-mL_x)^2}\mathrm{d}x = 1$. Let $f \in \mathfrak{h}$. By the first condition in (2.1), $f(x, y) = \sum_n f_n(x) e^{\mathrm{i}\kappa n y}$ with smooth f_n; the LLL condition reads $f_n' + (x - \kappa n)f_n = 0$, so $f_n = a_n e^{-(x-\kappa n)^2/2}$, and the second condition in (2.1) gives $a_n = a_{n-N_\phi}$. Hence $f = (L_y\sqrt{\pi})^{1/2} \sum_{j=0}^{N_\phi - 1} a_j \psi_j$. Each ψ_j is bounded by (i). □

2.2 The planar lowest Landau level and magnetic translations

Let $\mathfrak{h}_{\mathrm{pl}}$ be the Hilbert space of square-integrable LLL functions on $\mathbb{R}^2$. For $a \in \mathbb{R}^2$ the *magnetic translation* t_a acts on functions on $\mathbb{R}^2$ by

$$(t_a f)(r) := e^{-\mathrm{i}a_x a_y/2}\, e^{\mathrm{i}a_x y}\, f(r-a). \tag{2.2}$$

For $\gamma = (jL_x, kL_y) \in \Lambda$ put $\tilde{t}_\gamma := (-1)^{jkN_\phi} t_\gamma$, and on the torus put $\theta_x := t_{(\kappa,0)}$ and $\theta_y := t_{(0,-L_y/N_\phi)}$, that is $(\theta_x f)(x,y) = e^{\mathrm{i}\kappa y} f(x-\kappa, y)$ and $(\theta_y f)(x,y) = f(x, y + L_y/N_\phi)$.

Lemma 2.3 (magnetic translations)**.** *(i) t_a maps LLL functions to LLL functions, $|(t_a f)(r)| = |f(r-a)|$, and t_a is unitary on $\mathfrak{h}_{\mathrm{pl}}$ with $t_a^* = t_{-a}$.*

(ii) $t_a t_b = e^{\mathrm{i}(a\wedge b)/2}\, t_{a+b}$, hence $t_a t_b = e^{\mathrm{i}\, a\wedge b}\, t_b t_a$.

(iii) $\gamma \mapsto \tilde{t}_\gamma$ is a representation of Λ, and $\mathfrak{h}$ is the space of LLL functions f with $\tilde{t}_\gamma f = f$ for all $\gamma \in \Lambda$.

(iv) For $\delta \in \Lambda/N_\phi := (L_x/N_\phi)\mathbb{Z} \times (L_y/N_\phi)\mathbb{Z}$, t_δ maps $\mathfrak{h}$ onto $\mathfrak{h}$ unitarily. In particular $\theta_x \psi_j = \psi_{j+1}$, $\theta_y \psi_j = e^{2\pi \mathrm{i} j/N_\phi}\psi_j$, $\theta_y\theta_x = e^{2\pi\mathrm{i}/N_\phi}\theta_x\theta_y$ and $\theta_x^{N_\phi} = \theta_y^{N_\phi} = \mathbf{1}$ on $\mathfrak{h}$.

Proof. (i) t_a is multiplication by a unimodular function composed with a translation, so it preserves absolute values and L^2 norms; a direct computation gives $(D_x + \mathrm{i}D_y)t_a f = t_a(D_x + \mathrm{i}D_y)f$, and $t_a t_{-a} = \mathbf{1}$ by (ii). (ii) Expanding (2.2) twice, $t_a t_b f(r)$ has the phase $-\frac{1}{2}a_x a_y - \frac{1}{2}b_x b_y + (a_x + b_x)y - b_x a_y$ and $t_{a+b}f(r)$ has the phase $-\frac{1}{2}(a_x+b_x)(a_y+b_y) + (a_x+b_x)y$; the difference is $\frac{1}{2}(a_x b_y - a_y b_x)$. (iii) For $\gamma = (jL_x, kL_y)$ and $\gamma' = (j'L_x, k'L_y)$, $\gamma \wedge \gamma' = 2\pi N_\phi(jk' - kj')$, so by (ii) $\tilde{t}_\gamma \tilde{t}_{\gamma'} = (-1)^{(jk + j'k' + jk' - kj')N_\phi} t_{\gamma+\gamma'} = \tilde{t}_{\gamma+\gamma'}$, since $jk + j'k' + jk' - kj' \equiv (j+j')(k+k') \pmod 2$. The two conditions (2.1) are $\tilde{t}_{(0,L_y)}f = f$ and $\tilde{t}_{(L_x,0)}f = f$. (iv) By (ii), $t_\delta \tilde{t}_\gamma = e^{\mathrm{i}\,\delta\wedge\gamma}\tilde{t}_\gamma t_\delta$ with $\delta \wedge \gamma \in 2\pi\mathbb{Z}$ for $\delta \in \Lambda/N_\phi$, so t_δ preserves $\mathfrak{h}$; it is isometric for the inner product of $\mathfrak{h}$ by (i), because $|t_\delta f|^2$ is a translate of the Λ-periodic function $|f|^2$. The formulas for θ_x, θ_y follow from $\theta_x \phi_n = \phi_{n+1}$ and $\theta_y \phi_n = e^{2\pi \mathrm{i} n/N_\phi}\phi_n$. □

Bargmann form. Let $\mathcal{B}$ be the Bargmann space of entire functions g with

$$\|g\|_{\mathcal{B}}^2 := \frac{1}{\pi}\int_{\mathbb{C}} |g(z)|^2 e^{-|z|^2}\, \mathrm{d}^2 z < \infty;$$

the monomials $z^m/\sqrt{m!}$ form an orthonormal basis, written $|m\rangle$, and multiplication by z and ∂_z are mutually adjoint. For $\beta \in \mathbb{C}$ the *displacement operator* is

$$(\mathcal{D}(\beta)g)(z) := e^{-|\beta|^2/2} e^{\beta z} g(z - \bar{\beta}), \tag{2.3}$$

that is $\mathcal{D}(\beta) = e^{-|\beta|^2/2} e^{\beta z} e^{-\bar{\beta}\partial_z}$, and for $m \geq 0$ we write $\mathcal{D}^{(m)}(\beta) := [\langle a|\mathcal{D}(\beta)|b\rangle]_{0\leq a,b\leq m}$ for its truncation to the first $m+1$ levels. $\mathrm{L}_n^{(\alpha)}$ denotes the generalized Laguerre polynomial.

Lemma 2.4 (symmetric-gauge orbitals and displacements)**.** *(i) The map $\mathcal{U} : \mathcal{B} \to \mathfrak{h}_{\mathrm{pl}}$ defined by*

$$(\mathcal{U}g)(r) := (2\pi)^{-1/2} e^{\mathrm{i}xy/2} g(w/\sqrt{2}) e^{-|w|^2/4}$$

is unitary. Put $\eta_m := \mathcal{U}(z^m/\sqrt{m!})$, so that

$$\eta_m(r) = \frac{e^{\mathrm{i}xy/2}\, w^m\, e^{-|w|^2/4}}{\sqrt{2\pi\, 2^m\, m!}}, \qquad m \geq 0.$$

(ii) $\mathcal{U}^{-1}t_a\mathcal{U} = \mathcal{D}(\beta(a))$ *with* $\beta(a) := (a_x - \mathrm{i}a_y)/\sqrt{2}$, *so that* $|\beta(a)|^2 = |a|^2/2$ *and* $\langle \eta_m, t_a\eta_n\rangle = \langle m|\mathcal{D}(\beta(a))|n\rangle$.

(iii) For $m \geq n$ *and* $x := |\beta|^2$,

$$\langle m|\mathcal{D}(\beta)|n\rangle = \sqrt{\tfrac{n!}{m!}}\,\beta^{m-n}e^{-x/2}\,\mathrm{L}_n^{(m-n)}(x),$$
$$\langle n|\mathcal{D}(\beta)|m\rangle = \sqrt{\tfrac{n!}{m!}}\,(-\bar\beta)^{m-n}e^{-x/2}\,\mathrm{L}_n^{(m-n)}(x).$$

In particular $|\langle m|\mathcal{D}(\beta)|n\rangle| \leq 1$, *and* $\|\mathcal{D}^{(m)}(\beta)\|$ *depends only on* $|\beta|$.

(iv) The orthogonal projection of $L^2(\mathbb{R}^2)$ *onto* $\mathfrak{h}_{\mathrm{pl}}$ *has the kernel* $P_{\mathrm{pl}}(r,r') = \sum_m \eta_m(r)\overline{\eta_m(r')}$, *with* $|P_{\mathrm{pl}}(r,r')| = (2\pi)^{-1}e^{-|r-r'|^2/4}$, *and* $\int P_{\mathrm{pl}}(r,r')f(r')\mathrm{d}r' = f(r)$ *for every bounded LLL function* f. *Every* $f \in \mathfrak{h}_{\mathrm{pl}}$ *satisfies* $|f(r)| \leq (2\pi)^{-1/2}\|f\|$.

Proof. (i) Write $z = w/\sqrt{2}$. Then $|\mathcal{U}g|^2 = (2\pi)^{-1}|g(z)|^2e^{-|z|^2}$ and $\mathrm{d}r = 2\,\mathrm{d}^2z$, so $\mathcal{U}$ is isometric. Since $e^{\mathrm{i}xy/2-|w|^2/4} = e^{-x^2/2}e^{w^2/4}$, $\mathcal{U}g$ is an LLL function; conversely an LLL function $e^{-x^2/2}\tilde g(w)$ equals $\mathcal{U}g$ with $g(z) = (2\pi)^{1/2}e^{-z^2/2}\tilde g(\sqrt{2}z)$, and square integrability is $g \in \mathcal{B}$. (ii) Put $b := (a_x + \mathrm{i}a_y)/\sqrt{2}$. Then $e^{-|z-b|^2/2} = e^{-|z|^2/2}e^{(\bar z b + z\bar b)/2 - |b|^2/2}$, and the phases in $t_a\mathcal{U}g$ combine to $e^{\mathrm{i}xy/2}e^{\mathrm{i}(a_xy - a_yx)/2}$ with $\mathrm{i}(a_xy - a_yx)/2 = (z\bar b - \bar z b)/2$. Hence $\mathcal{U}^{-1}t_a\mathcal{U}g(z) = e^{z\bar b - |b|^2/2}g(z-b)$, which is (2.3) with $\beta = \bar b$. (iii) By the Baker–Campbell–Hausdorff formula ($[\beta z, -\bar\beta\partial_z] = |\beta|^2$), $\mathcal{D}(\beta) = e^{-x/2}e^{\beta z}e^{-\bar\beta\partial_z}$, and

$$\langle m|\mathcal{D}(\beta)|n\rangle = e^{-x/2}\sum_{k\leq n}\frac{\sqrt{m!\,n!}\;\beta^{m-k}(-\bar\beta)^{n-k}}{(m-k)!\,(n-k)!\,k!} = \sqrt{\tfrac{n!}{m!}}\,\beta^{m-n}e^{-x/2}\sum_{i=0}^{n}\frac{m!\,(-x)^i}{(m-n+i)!\,i!\,(n-i)!},$$

with $i = n-k$; the last sum is $\mathrm{L}_n^{(m-n)}(x)$. The second formula follows from $\langle n|\mathcal{D}(\beta)|m\rangle = \overline{\langle m|\mathcal{D}(-\beta)|n\rangle}$, since $\mathcal{D}(\beta)^* = \mathcal{D}(-\beta)$ by (i)–(ii) and Lemma 2.3(i). The bound $|\langle m|\mathcal{D}|n\rangle| \leq 1$ holds because $\mathcal{D}(\beta)$ is unitary. By the two formulas, $\langle a|\mathcal{D}(e^{\mathrm{i}\theta}\beta)|b\rangle = e^{\mathrm{i}\theta(a-b)}\langle a|\mathcal{D}(\beta)|b\rangle$ for real θ, so the truncations at β and $e^{\mathrm{i}\theta}\beta$ are conjugate by a diagonal unitary.

(iv) $\sum_m \eta_m(r)\overline{\eta_m(r')} = (2\pi)^{-1}e^{\mathrm{i}(xy-x'y')/2}e^{z\bar z' - (|z|^2+|z'|^2)/2}$, of modulus $(2\pi)^{-1}e^{-|z-z'|^2/2}$, and $|z-z'|^2 = |r-r'|^2/2$. For $f = \mathcal{U}g$ bounded, $|g(z)| \leq Ce^{|z|^2/2}$, and $\int P_{\mathrm{pl}}(r,r')f(r')\mathrm{d}r' = (\mathcal{U}h)(r)$ with $h(z) = \pi^{-1}\int g(z')e^{z\bar z'}e^{-|z'|^2}\mathrm{d}^2z'$. The substitution $z' = z+u$ gives $h(z) = \pi^{-1}\int g(z+u)e^{-\bar z u}e^{-|u|^2}\mathrm{d}^2u$, an absolutely convergent integral; in polar coordinates the mean value of the entire function $u \mapsto g(z+u)e^{-\bar z u}$ over each circle is $g(z)$, so $h(z) = g(z)$. For $f = \mathcal{U}g \in \mathfrak{h}_{\mathrm{pl}}$, that is $g \in \mathcal{B}$, both integrals converge absolutely by Cauchy–Schwarz, since $z' \mapsto g(z')e^{-|z'|^2/2}$ and $z' \mapsto e^{z\bar z'}e^{-|z'|^2/2}$ are square-integrable, and the same computation proves the reproducing identity on $\mathfrak{h}_{\mathrm{pl}}$. Finally, for $f \in \mathfrak{h}_{\mathrm{pl}}$, $|f(r)| = |\langle P_{\mathrm{pl}}(\cdot,r), f\rangle| \leq P_{\mathrm{pl}}(r,r)^{1/2}\|f\|$. □

For $R = (R_x, R_y) \in \mathbb{R}^2$ and $m \geq 0$ we put

$$\eta_m^R := e^{-\mathrm{i}R_xR_y/2}\,t_R\,\eta_m, \qquad \eta_m^R(r) = \frac{e^{\mathrm{i}(y-R_y)(x+R_x)/2}\,w_R^m\,e^{-|w_R|^2/4}}{\sqrt{2\pi\,2^m\,m!}}, \quad w_R := w - (R_x + \mathrm{i}R_y), \tag{2.4}$$

the symmetric-gauge orbitals about the point R, written in the gauge $(0,x)$. They form an orthonormal basis of $\mathfrak{h}_{\mathrm{pl}}$ for every R.

The Landau representation. For $s \in \mathbb{R}$ let $\phi_s^\delta(x,y) := (2\pi)^{-1/2}\pi^{-1/4}e^{\mathrm{i}sy}e^{-(x-s)^2/2}$ (δ-normalized Landau orbitals) and let $\mathcal{H}_m(s) := (2^m m!\sqrt{\pi})^{-1/2}H_m(s)e^{-s^2/2}$ be the normalized Hermite functions. A direct Gaussian integration with the generating functions of η_m and $\mathcal{H}_m$ gives

$$\langle \phi_s^\delta, \eta_m^R\rangle_{\mathbb{R}^2} = e^{-\mathrm{i}sR_y}\,\mathcal{H}_m(s-R_x), \tag{2.5}$$

the integral being absolutely convergent. For $R=0$ use the generating function

$$\sum_m \frac{t^m}{m!}\sqrt{2\pi 2^m m!}\,\eta_m = e^{\mathrm{i}xy/2}e^{tw-|w|^2/4};$$

the y-integral and then the x-integral are Gaussian:

$$\int_{\mathbb{R}^2} e^{-\mathrm{i}sy-(x-s)^2/2}\,e^{\mathrm{i}xy/2}e^{tw-|w|^2/4}\,\mathrm{d}r = 2\pi e^{-s^2/2+2ts-t^2} = 2\pi e^{-s^2/2}\sum_m H_m(s)\frac{t^m}{m!},$$

and comparing coefficients gives (2.5); the case of general R follows from $t_R\phi_s^\delta = e^{-\mathrm{i}R_y(s+R_x/2)}\phi_{s+R_x}^\delta$. Since $\phi_n = \sqrt{\kappa}\,\phi_{\kappa n}^\delta$,

$$\langle \phi_n, \eta_m^R\rangle_{\mathbb{R}^2} = \sqrt{\kappa}\,e^{-\mathrm{i}\kappa nR_y}\mathcal{H}_m(\kappa n - R_x).$$

2.3 Periodization and unfolding

For $k \geq 1$ we realize $\mathfrak{h}^{\otimes k}$ as the space of functions on $(\mathbb{R}^2)^k$ that are LLL functions of each variable and invariant under $\tilde{t}_\gamma^{(i)}$ (the operator $\tilde{t}_\gamma$ acting on the i-th variable) for every i and every $\gamma \in \Lambda$, with the inner product $\int_{\mathsf{C}^k}$. This space is spanned by the products $\psi_{j_1}(r_1)\cdots\psi_{j_k}(r_k)$: expanding such a function Ψ in its first variable in the basis (ψ_j) of Lemma 2.2, the coefficients $\langle\psi_j, \Psi(\cdot, r_2, \dots, r_k)\rangle$ are functions of the same kind in $k-1$ variables (Morera's theorem), and one inducts. In particular its elements are bounded. Permutations P_σ of the variables act on it unitarily.

Lemma 2.5 (periodization and unfolding)**.** *Let f be a function on $(\mathbb{R}^2)^k$ which is an LLL function of each variable and satisfies $|f(r)| \leq Ce^{-c\sum_i |r_i - r_0|^2}$ for some $C, c > 0$ and $r_0 \in \mathbb{R}^2$. Then*

$$\Pi^{(k)} f := \sum_{\gamma\in\Lambda^k} \tilde{t}_{\gamma_1}^{(1)}\cdots\tilde{t}_{\gamma_k}^{(k)} f$$

converges absolutely and locally uniformly, $\Pi^{(k)}f \in \mathfrak{h}^{\otimes k}$, and for every $\Psi \in \mathfrak{h}^{\otimes k}$

$$\langle \Pi^{(k)} f, \Psi\rangle = \int_{(\mathbb{R}^2)^k} \overline{f}\,\Psi. \tag{2.6}$$

$\Pi^{(k)}$ commutes with permutations of the variables. We write $\Pi := \Pi^{(1)}$.

Proof. $|\tilde{t}_{\gamma_i}^{(i)} f(r)|$ is $|f|$ translated by γ_i in the i-th variable, so the series converges absolutely and locally uniformly by the Gaussian bound. Each term is $\prod_i e^{-x_i^2/2}$ times an entire function of $(w_1, \dots, w_k)$, so the limit is of the same form and is an LLL function of each variable. Reindexing and Lemma 2.3(iii) give $\tilde{t}_\gamma^{(i)}\Pi^{(k)}f = \Pi^{(k)}f$. For (2.6), use $\Psi = \tilde{t}_{\gamma_1}^{(1)}\cdots\tilde{t}_{\gamma_k}^{(k)}\Psi$ and the fact that for any two functions g_1, g_2, $\overline{\tilde{t}_\gamma g_1}\,\tilde{t}_\gamma g_2$ is the translate of $\overline{g_1}g_2$ by γ; thus the term γ of $\int_{\mathsf{C}^k}\overline{\Pi^{(k)}f}\Psi$ is $\int_{\mathsf{C}^k-\gamma}\overline{f}\Psi$, and the translates of C^k tile $(\mathbb{R}^2)^k$. All rearrangements are justified by absolute convergence, since Ψ is bounded. Permutations commute with the sum because all variables carry the same boundary conditions. □

2.4 Fock space and lifts

Let $\mathcal{F} := \bigoplus_{n=0}^{N_\phi} \bigwedge^n \mathfrak{h}$, with the convention that wedges of orthonormal vectors $e_{i_1} \wedge \cdots \wedge e_{i_n}$, $i_1 < \cdots < i_n$, have norm one. For $v \in \bigwedge^k \mathfrak{h}$ let $B(v)$ be the adjoint of left wedge multiplication by v; $B(v)$ is conjugate-linear in v and

$$B(v \wedge w) = B(w)B(v). \tag{2.7}$$

For $f \in \mathfrak{h}$, $c(f) := B(f)$ is the annihilation operator; it is conjugate-linear in f, satisfies the canonical anticommutation relations $\{c(f), c(g)^*\} = \langle f, g\rangle$, $\{c(f), c(g)\} = 0$, and $\|c(f)\| = \|f\|$. We put $c_j := c(\psi_j)$. For a unitary u on $\mathfrak{h}$, $\mathcal{F}(u)$ is its second quantization. The map $\iota_n(f_1 \wedge \cdots \wedge f_n) := \sqrt{n!}\, A_n(f_1 \otimes \cdots \otimes f_n)$, $A_n := (n!)^{-1} \sum_\sigma \operatorname{sgn}(\sigma) P_\sigma$, identifies $\bigwedge^n \mathfrak{h}$ isometrically with the antisymmetric elements of $\mathfrak{h}^{\otimes n}$; we use it without further mention, so that $\bigwedge^n \mathfrak{h}$ is the space of antisymmetric functions in $\mathfrak{h}^{\otimes n}$.

For an operator X on $\bigwedge^k \mathfrak{h}$ its *lift* is defined by linear extension of

$$\mathcal{L}_k(|v\rangle\langle w|) := B(v)^* B(w), \qquad v, w \in \textstyle\bigwedge^k \mathfrak{h}. \tag{2.8}$$

We extend X to $\mathfrak{h}^{\otimes k}$ by XA_k; this extension commutes with permutations, and for a k-subset S of $\{1, \dots, n\}$ we write X_S for it acting on the tensor positions in S.

Lemma 2.6 (lifts)**.** *(i) For $v \in \bigwedge^k \mathfrak{h}$, $\Phi \in \bigwedge^{n-k} \mathfrak{h}$ and $\Psi \in \bigwedge^n \mathfrak{h}$: $v \wedge \Phi = \binom{n}{k}^{1/2} A_n(v \otimes \Phi)$ and $B(v)\Psi = \binom{n}{k}^{1/2} (\langle v| \otimes \mathbf{1})\Psi$.*

(ii) On $\bigwedge^n \mathfrak{h}$, $\mathcal{L}_k(X) = \sum_{|S|=k} X_S$ for $n \geq k$, and $\mathcal{L}_k(X) = 0$ for $n < k$; in particular $\mathcal{L}_k(X)$ restricted to $\bigwedge^k \mathfrak{h}$ is X.

(iii) $\mathcal{L}_k$ is linear, $\mathcal{L}_k(X)^* = \mathcal{L}_k(X^*)$, and $X \geq 0$ implies $\mathcal{L}_k(X) \geq 0$. Operators Z_k on $\bigwedge^k \mathfrak{h}$ with $\sum_{k=0}^m \mathcal{L}_k(Z_k) = 0$ vanish.

(iv) The wedge maps $W_3 : \bigwedge^2 \mathfrak{h} \otimes \mathfrak{h} \to \bigwedge^3 \mathfrak{h}$, $W_3(v \otimes u) := v \wedge u$, and $W_4 : \bigwedge^2 \mathfrak{h} \otimes \bigwedge^2 \mathfrak{h} \to \bigwedge^4 \mathfrak{h}$, $W_4(v \otimes v') := v \wedge v'$, are $\sqrt{3}A_3$ and $\sqrt{6}A_4$ restricted to these subspaces of $\mathfrak{h}^{\otimes 3}$, $\mathfrak{h}^{\otimes 4}$.

Proof. (i) For orthonormal basis vectors, $v = e_1 \wedge \cdots \wedge e_k$ and $\Phi = e_{k+1} \wedge \cdots \wedge e_n$, the first identity reads $\sqrt{n!} A_n(e_1 \otimes \cdots \otimes e_n) = \binom{n}{k}^{1/2} \sqrt{k!(n-k)!}\, A_n(A_k \otimes A_{n-k})(e_1 \otimes \cdots \otimes e_n)$, which holds since $A_n(A_k \otimes A_{n-k}) = A_n$; it extends by multilinearity. Then $\langle \Phi, B(v)\Psi\rangle = \langle v \wedge \Phi, \Psi\rangle = \binom{n}{k}^{1/2} \langle v \otimes \Phi, \Psi\rangle$, because A_n is an orthogonal projection fixing Ψ. (ii) By (i), $B(v)^* B(w)\Psi = \binom{n}{k} A_n(Y \otimes \mathbf{1})\Psi$ with $Y = |v\rangle\langle w|$, which commutes with permutations of its k positions. For antisymmetric Ψ, $A_n(Y \otimes \mathbf{1})\Psi = (n!)^{-1} \sum_\sigma (P_\sigma (Y \otimes \mathbf{1}) P_\sigma^{-1})\Psi$, and each k-subset occurs for $k!(n-k)!$ permutations; this gives $\binom{n}{k}^{-1} \sum_S Y_S \Psi$. For $n < k$, $B(w)\Psi = 0$. (iii) Linearity and the adjoint are clear; a positive X is a sum $\sum |v\rangle\langle v|$, and $B(v)^* B(v) \geq 0$. If $\sum_k \mathcal{L}_k(Z_k) = 0$, restricting to $k = 0, 1, \dots$ particles successively gives $Z_0 = 0$, $Z_1 = 0$, and so on, by (ii). (iv) is (i) with $n = 3, 4$. □

2.5 Jacobi coordinates and the periodized image

For $k \geq 1$ fix a real orthogonal $k \times k$ matrix O whose first row is $k^{-1/2}(1, \dots, 1)$, and let it act on $(r_1, \dots, r_k) \in (\mathbb{R}^2)^k$ componentwise. The new variables are the *centre-of-mass coordinate* $\zeta := k^{-1/2} \sum_i r_i \in \mathbb{R}^2$ and *relative coordinates* $\rho \in (\mathbb{R}^2)^{k-1}$. The specific choices used below are, for $k = 2$, $\rho = (r_1 - r_2)/\sqrt{2}$; for $k = 3$, $\rho = (\rho_1, \rho_2)$ with $\rho_1 = (r_1 - r_2)/\sqrt{2}$, $\rho_2 = (r_1 + r_2 - 2r_3)/\sqrt{6}$; for $k = 4$, $\rho = (\rho_1, \rho_2, \rho_3)$ with $\rho_1 = (r_1 - r_2)/\sqrt{2}$, $\rho_2 = (r_3 - r_4)/\sqrt{2}$, $\rho_3 = (r_1 + r_2 - r_3 - r_4)/2$.

Lemma 2.7 (Jacobi coordinates)**.** *(i) A function on $(\mathbb{R}^2)^k$ is an LLL function of each r_i if and only if, written in (ζ,ρ), it is an LLL function of each new variable. The change of variables is unitary on L^2; it identifies $\mathfrak{h}_{\mathrm{pl}}^{\otimes k}$ with $\mathfrak{h}_{\mathrm{pl}}\otimes\mathfrak{h}_{\mathrm{pl}}^{\otimes(k-1)}$ (centre of mass $\otimes$ relative).*

(ii) The simultaneous translation $\prod_i t_a^{(i)}$ acts as $t_{\sqrt{k}\,a}$ on ζ and trivially on ρ.

(iii) A permutation of the particles acts on (ζ,ρ) by an orthogonal map that fixes ζ.

Proof. (i) With $(\xi,\upsilon)=(Ox,Oy)$, $\partial_{x_i}+\mathrm{i}\partial_{y_i}+x_i=\sum_j O_{ji}(\partial_{\xi_j}+\mathrm{i}\partial_{\upsilon_j}+\xi_j)$, and O is invertible; the Jacobian is one; and $\sum_i x_iy_i$, $\sum_i|w_i|^2$ are invariant, so the Bargmann factorization of Lemma 2.4(i) is preserved. (ii) $\prod_i t_a^{(i)}f(r)=e^{-\mathrm{i}ka_xa_y/2}e^{\mathrm{i}a_x\sum_i y_i}f(r-a\mathbf{1})$; here $\sum_i y_i=\sqrt{k}\,\zeta_y$, and $r\mapsto r-a\mathbf{1}$ is $\zeta\mapsto\zeta-\sqrt{k}a$ with ρ fixed, because the rows of O other than the first are orthogonal to $(1,\dots,1)$; this is $t_{\sqrt{k}a}$ in ζ. (iii) A permutation matrix fixes $(1,\dots,1)$. □

Let $\mathbb{T}_k:=\mathbb{R}^2/\sqrt{k}\Lambda$, a rectangular torus with periods $\sqrt{k}L_x$, $\sqrt{k}L_y$ and kN_ϕ flux quanta, and let $\mathfrak{u}_k$ be its lowest Landau level (Definition 2.1 with these periods), the *centre-of-mass torus space*; by Lemma 2.2, $\dim\mathfrak{u}_k=kN_\phi$. Its lattice translations are $\tilde{t}'_{\gamma'}$, $\gamma'\in\sqrt{k}\Lambda$. For $\gamma=(jL_x,lL_y)\in\Lambda$, Lemma 2.7(ii) and $(-1)^{jlN_\phi\cdot k}=(-1)^{jl(kN_\phi)}$ show that the simultaneous lattice translation $\tilde{t}_\gamma^{\otimes k}:=\prod_i\tilde{t}_\gamma^{(i)}$ acts as $\tilde{t}'_{\sqrt{k}\gamma}$ on ζ and trivially on ρ.

Definition 2.8 (admissible operators, the periodized image)**.** A *k-body admissible operator* is an integral operator Y on functions on $(\mathbb{R}^2)^k$ with a continuous kernel $Y(r;r')$ such that

(Y1) for each r', $Y(\cdot\,;r')$ is an LLL function of each variable, and for each r, $\overline{Y(r;\cdot)}$ is;

(Y2) Y commutes with the simultaneous magnetic translations $\prod_i t_a^{(i)}$, $a\in\mathbb{R}^2$;

(Y3) $|Y(r;r')|\le C\exp(-c\sum_i(|r_i-r_1|^2+|r'_i-r_1|^2))$ for some $C,c>0$.

For admissible Y the *periodized image* $\mathfrak{P}_k(Y)$ is the operator on $\mathfrak{h}^{\otimes k}$ defined by

$$\langle\Phi,\mathfrak{P}_k(Y)\Psi\rangle:=\int_{\mathsf{D}}\overline{\Phi}\,(Y\Psi),\qquad\Phi,\Psi\in\mathfrak{h}^{\otimes k},\tag{2.9}$$

where $\mathsf{D}\subset(\mathbb{R}^2)^k$ is a fundamental domain for the diagonal action $r\mapsto r+\gamma\mathbf{1}$ of Λ, for instance $\{r_1\in\mathsf{C}\}$ or, in Jacobi coordinates, $\{\zeta\in\sqrt{k}\mathsf{C},\ \rho\text{ arbitrary}\}$.

Lemma 2.9 (the periodized image)**.** *Let Y be admissible and $\Phi,\Psi\in\mathfrak{h}^{\otimes k}$.*

(a) $Y\Psi(r)=\int Y(r;r')\Psi(r')\mathrm{d}r'$ converges absolutely, is an LLL function of each variable, satisfies $\tilde{t}_\gamma^{\otimes k}(Y\Psi)=Y\Psi$ for $\gamma\in\Lambda$, and $|Y\Psi(r)|\le C'\exp(-c'\sum_i|r_i-r_1|^2)$. The integral (2.9) converges absolutely and does not depend on D.

(b) $\mathfrak{P}_k$ is linear, $\mathfrak{P}_k(Y)^*=\mathfrak{P}_k(Y^*)$, and $P_\sigma\mathfrak{P}_k(Y)P_\sigma^{-1}=\mathfrak{P}_k(P_\sigma YP_\sigma^{-1})$, $\mathfrak{P}_k(Y)P_\sigma=\mathfrak{P}_k(YP_\sigma)$. In particular $\mathfrak{P}_k(Y)$ restricted to $\bigwedge^k\mathfrak{h}$ equals $\mathfrak{P}_k(A_kYA_k)$ restricted to $\bigwedge^k\mathfrak{h}$.

(c) (Fibering.) *For a Gaussian-decaying relative LLL function a (that is, $|a(\rho)|\le Ce^{-c|\rho|^2}$) and a bounded function Ψ on $(\mathbb{R}^2)^k$ put $(\mathcal{I}_a\Psi)(\zeta):=\int\overline{a(\rho)}\,\Psi(\zeta,\rho)\,\mathrm{d}\rho$. Then $\mathcal{I}_a$ maps $\mathfrak{h}^{\otimes k}$ into $\mathfrak{u}_k$; the operator $|a\rangle\langle b|\otimes\mathbf{1}$, with kernel $a(\rho)\overline{b(\rho')}P_{\mathrm{pl}}(\zeta,\zeta')$, is admissible; and $\mathfrak{P}_k(|a\rangle\langle b|\otimes\mathbf{1})=\mathcal{I}_a^*\mathcal{I}_b$. Consequently $\mathfrak{P}_k(Y_{\mathrm{rel}}\otimes\mathbf{1})\ge0$ for every positive finite-rank operator Y_{rel} on the relative*

variables whose range consists of Gaussian-decaying relative LLL functions. Moreover, for $g \in \mathfrak{u}_k$,

$$\mathcal{I}_a^* g = \sum_{\gamma \in \Lambda^{k-1}} \Big(\prod_{i=2}^{k} \tilde{t}_{\gamma_i}^{(i)} \Big) (a \otimes g), \qquad (a \otimes g)(\zeta, \rho) := a(\rho) g(\zeta), \tag{2.10}$$

the series converging absolutely and locally uniformly.

Proof. (a) The kernel decays in $r_i' - r_1$ for every i and Ψ is bounded, so the integral converges, locally uniformly in r, with the stated Gaussian bound. By (Y1), $e^{x_i^2/2} Y(r; r')$ is an entire function of w_i for fixed values of the other variables, so by Morera's theorem the locally uniformly convergent integral $e^{x_i^2/2} Y\Psi(r)$ is entire in w_i, and $Y\Psi$ is an LLL function of each variable. By (Y2) and $\tilde{t}_\gamma^{\otimes k} = (\pm 1) \prod_i t_\gamma^{(i)}$, $\tilde{t}_\gamma^{\otimes k} Y \Psi = Y \tilde{t}_\gamma^{\otimes k} \Psi = Y\Psi$. The integrand of (2.9) is therefore invariant under $r \mapsto r + \gamma \mathbf{1}$ (the phases of $\tilde{t}_\gamma^{\otimes k}$ cancel in $\overline{\Phi}\, Y \Psi$) and decays like a Gaussian in the relative coordinates, so it is integrable over a fundamental domain, and any two fundamental domains give the same integral. (b) Linearity is clear. For the adjoint, let $G(r, r') := \overline{\Phi(r)}\, Y(r; r')\, \Psi(r')$. By (Y2), conjugation by $\tilde{t}_\gamma^{\otimes k}$ multiplies the kernel by the phase of $\tilde{t}_\gamma^{\otimes k}$ at r and its inverse at r', and Φ, Ψ are $\tilde{t}_\gamma^{\otimes k}$-invariant; hence $G(r + \gamma \mathbf{1}, r' + \gamma \mathbf{1}) = G(r, r')$. By (Y3) and the boundedness of Φ, Ψ, G is absolutely integrable over $\mathsf{D} \times (\mathbb{R}^2)^k$ and over $(\mathbb{R}^2)^k \times \mathsf{D}$, and both integrals equal the integral of G over a fundamental domain of the joint diagonal action on $(\mathbb{R}^2)^k \times (\mathbb{R}^2)^k$. The first is $\int_{\mathsf{D}} \overline{\Phi}\, Y \Psi$ and the second is $\int_{\mathsf{D}} \overline{Y^* \Phi}\, \Psi$, where Y^* has the kernel $\overline{Y(r'; r)}$, which is again admissible. For permutations: P_σ preserves $\mathfrak{h}^{\otimes k}$ and maps fundamental domains to fundamental domains. The last statement follows from $A_k \Psi = \Psi$ for $\Psi \in \bigwedge^k \mathfrak{h}$. (c) $\mathcal{I}_a \Psi$ is an LLL function of ζ by Lemma 2.7(i) and Morera's theorem as in (a), it is bounded, and it is $\tilde{t}'$-invariant because $\tilde{t}_\gamma^{\otimes k}$ acts on ζ alone; so $\mathcal{I}_a \Psi \in \mathfrak{u}_k$. The kernel $a(\rho)\overline{b(\rho')} P_{\mathrm{pl}}(\zeta, \zeta')$ satisfies (Y1) and (Y2) (Lemma 2.7(ii), and P_{pl} commutes with t_a), and (Y3) holds because $|\rho|^2 + |\rho'|^2 + |\zeta - \zeta'|^2$ dominates a multiple of $\sum_i (|r_i - r_1|^2 + |r_i' - r_1|^2)$, all these differences being linear combinations of ρ, ρ' and $\zeta - \zeta'$. By Lemma 2.4(iv) applied in ζ to the bounded LLL function $\mathcal{I}_b \Psi$, $((|a\rangle\langle b| \otimes \mathbf{1})\Psi)(\zeta, \rho) = a(\rho)(\mathcal{I}_b \Psi)(\zeta)$, and with $\mathsf{D} = \{\zeta \in \sqrt{k}\mathsf{C}\}$ the integral (2.9) becomes $\int_{\sqrt{k}\mathsf{C}} \overline{\mathcal{I}_a \Phi}\, \mathcal{I}_b \Psi = \langle \mathcal{I}_a \Phi, \mathcal{I}_b \Psi \rangle_{\mathfrak{u}_k}$. A positive finite-rank Y_{rel} is $\sum_\mu |b_\mu\rangle\langle b_\mu|$ with b_μ in its range, and $\mathfrak{P}_k$ depends only on the kernel of its argument, so $\mathfrak{P}_k(Y_{\mathrm{rel}} \otimes \mathbf{1}) = \sum_\mu \mathcal{I}_{b_\mu}^* \mathcal{I}_{b_\mu} \geq 0$. For (2.10): $a \otimes g$ is an LLL function of each particle, invariant under the diagonal action (because $g \in \mathfrak{u}_k$), and decays like a Gaussian in $r_i - r_1$. For $\Psi \in \mathfrak{h}^{\otimes k}$, $\langle \Psi, \mathcal{I}_a^* g \rangle = \langle \mathcal{I}_a \Psi, g \rangle = \int_{\mathsf{D}} \overline{\Psi}\, (a \otimes g)$ with $\mathsf{D} = \{\zeta \in \sqrt{k}\mathsf{C}\}$; the integrand is invariant under the diagonal action, so we may take $\mathsf{D} = \{r_1 \in \mathsf{C}\}$ instead, and unfolding the variables $r_2, \dots, r_k$ into C as in Lemma 2.5 shows that the right side of (2.10), which lies in $\mathfrak{h}^{\otimes k}$ by the argument of Lemma 2.5, has the same inner product with every Ψ. □

2.6 The Hamiltonian

Two-particle LLL functions are, by Lemma 2.7, functions of the relative coordinate $\rho = (r_1 - r_2)/\sqrt{2}$ and of $\zeta = (r_1 + r_2)/\sqrt{2}$, and the relative angular momentum of the pair is the degree in ρ. Let

$$P_1 := |\eta_1\rangle\langle\eta_1| \otimes \mathbf{1},$$

with η_1 in the relative variable ρ: the orthogonal projection of $\mathfrak{h}_{\mathrm{pl}}^{\otimes 2}$ onto the pairs of relative angular momentum one, with every centre-of-mass state. The planar N-particle operator $\sum_{i<j} (P_1)_{ij}$ is the V_1 pseudopotential Hamiltonian in which every pair of relative angular momentum one pays energy one and every other pair pays nothing, which is the normalization of [1] and of the planar spectral-gap conjecture [11, Appendix A].

Definition 2.10 (the torus V_1 Hamiltonian)**.** $h := \mathfrak{P}_2(P_1)$, an operator on $\mathfrak{h}^{\otimes 2}$ which preserves $\bigwedge^2 \mathfrak{h}$ (Lemma 2.9(b)); we use the same letter for its restriction to $\bigwedge^2 \mathfrak{h}$. The *torus V_1 Hamiltonian* is $H_{\mathbb{T}} := \mathcal{L}_2(h)$ on $\mathcal{F}$. We put $V_{\mathbb{T}} := \operatorname{ran} h \subset \bigwedge^2 \mathfrak{h}$.

By Lemma 2.9(c), $h \geq 0$, so $H_{\mathbb{T}} \geq 0$; $H_{\mathbb{T}}$ preserves the particle number, and on n particles $H_{\mathbb{T}} = \sum_{i<j} h_{ij}$ by Lemma 2.6(ii). The next proposition gives its orbital form; the pair functions v_n are the image sums of the planar pair states.

Proposition 2.11 (orbital form)**.** *Let $c_0 := \kappa^3/\sqrt{2\pi}$ and $f(d) := d\, e^{-\kappa^2 d^2/4}$. For $n \in \mathbb{Z}$ put*

$$v_n := \sqrt{c_0} \sum_{A+B=n} f(A-B)\, \psi_{[A]} \wedge \psi_{[B]} \in \bigwedge^2 \mathfrak{h}, \tag{2.11}$$

the sum running over $(A,B) \in \mathbb{Z}^2$; it converges absolutely and $v_{n+2N_\phi} = v_n$. Then

$$h = \sum_{n \in \mathbb{Z}_{2N_\phi}} |v_n\rangle\langle v_n|, \qquad H_{\mathbb{T}} = c_0 \sum_{n \in \mathbb{Z}_{2N_\phi}} b_n^* b_n, \qquad b_n := \sum_{\substack{d \geq 1 \\ d \equiv n \ (2)}} 2d\, e^{-\kappa^2 d^2/4}\, c_{[(n-d)/2]}\, c_{[(n+d)/2]}. \tag{2.12}$$

Moreover, with $\mathcal{I} := \mathcal{I}_{\eta_1} : \mathfrak{h}^{\otimes 2} \to \mathfrak{u}_2$ and $(\psi'_n)_{n \in \mathbb{Z}_{2N_\phi}}$ the orbitals of $\mathbb{T}_2 = \mathbb{R}^2/\sqrt{2}\Lambda$ (Lemma 2.2 with periods $\sqrt{2}L_x$, $\sqrt{2}L_y$), $h = \mathcal{I}^*\mathcal{I}$ and

$$v_n = \mathcal{I}^* \psi'_n = \sum_{\gamma \in \Lambda} \tilde{t}_\gamma^{(2)} (\eta_1(\rho)\, \psi'_n(\zeta)), \tag{2.13}$$

the image sum over the second particle of the planar pair state with relative orbital η_1 and centre-of-mass state ψ'_n (in the tensor realization of $\bigwedge^2 \mathfrak{h}$).

Proof. By Lemma 2.9(c), $h = \mathcal{I}^*\mathcal{I}$. For $A, B \in \mathbb{Z}$ write $d = A - B$ and $m = A + B$. In the coordinates $\rho = (r_1 - r_2)/\sqrt{2}$, $\zeta = (r_1 + r_2)/\sqrt{2}$, orthogonality of the coordinate change gives

$$\phi_A(r_1)\phi_B(r_2) = (L_y\sqrt{\pi})^{-1}\, e^{\mathrm{i}\kappa d \rho_y/\sqrt{2}} e^{-(\rho_x - \kappa d/\sqrt{2})^2/2}\; e^{\mathrm{i}\kappa m \zeta_y/\sqrt{2}} e^{-(\zeta_x - \kappa m/\sqrt{2})^2/2}.$$

By (2.5) at $R = 0$, $\int \overline{\eta_1(\rho)}\, e^{\mathrm{i}s\rho_y} e^{-(\rho_x - s)^2/2} \mathrm{d}\rho = (2\pi)^{1/2} \pi^{1/4} \mathcal{H}_1(s)$, with $\mathcal{H}_1(s) = \pi^{-1/4}\sqrt{2}\, s\, e^{-s^2/2}$. The orbitals ϕ'_m, ψ'_n of $\mathbb{T}_2$ are built with $\kappa' := \kappa/\sqrt{2} = 2\pi/(\sqrt{2}L_y)$ and $2N_\phi$ flux quanta, and $e^{\mathrm{i}\kappa m \zeta_y/\sqrt{2}} e^{-(\zeta_x - \kappa m/\sqrt{2})^2/2} = (\sqrt{2}L_y\sqrt{\pi})^{1/2} \phi'_m(\zeta)$. Hence

$$\mathcal{I}(\phi_A \otimes \phi_B) = 2^{1/4}\sqrt{\kappa}\; \mathcal{H}_1(\kappa d/\sqrt{2})\, \phi'_m = 2^{1/4}\pi^{-1/4}\kappa^{3/2} f(A-B)\, \phi'_{A+B}.$$

Since $\sum_{A,B} |\phi_A(r_1)\phi_B(r_2)|$ is bounded and η_1 is integrable, dominated convergence gives $\mathcal{I}(\psi_a \otimes \psi_b) = \sum_{A \equiv a,\, B \equiv b} \mathcal{I}(\phi_A \otimes \phi_B)$ for $a, b \in \mathbb{Z}_{N_\phi}$. The coefficient of ϕ'_m in this sum depends on m only through its class modulo $2N_\phi$ (shift A and B by N_ϕ), so, grouping the m by classes,

$$\mathcal{I}(\psi_a \otimes \psi_b) = 2^{1/4}\pi^{-1/4}\kappa^{3/2} \sum_{n \in \mathbb{Z}_{2N_\phi}} C_n(a,b)\, \psi'_n, \qquad C_n(a,b) := \sum_{\substack{A+B=n \\ A \equiv a,\ B \equiv b}} f(A-B),$$

for any representative n of the class. As (ψ'_n) is orthonormal and the C_n are real, $\langle \Phi, h\Psi\rangle = \langle \mathcal{I}\Phi, \mathcal{I}\Psi\rangle = \sum_n \overline{\langle w_n, \Phi\rangle}\, \langle w_n, \Psi\rangle$ with $w_n := 2^{1/4}\pi^{-1/4}\kappa^{3/2} \sum_{A+B=n} f(A-B)\, \psi_{[A]} \otimes \psi_{[B]}$, and $w_n = \mathcal{I}^* \psi'_n$. The tensor $\sum_{A+B=n} f(A-B) \psi_{[A]} \otimes \psi_{[B]}$ is antisymmetric (f is odd), and it equals $2^{-1/2}$ times the antisymmetric tensor of $\sum_{A+B=n} f(A-B) \psi_{[A]} \wedge \psi_{[B]}$; since $2^{1/4}\pi^{-1/4}\kappa^{3/2} 2^{-1/2} = c_0^{1/2}$, $w_n = v_n$. This proves $h = \sum_n |v_n\rangle\langle v_n|$. Then $H_{\mathbb{T}} = \sum_n B(v_n)^* B(v_n)$, and $B(\psi_{[A]} \wedge \psi_{[B]}) = c_{[B]} c_{[A]}$ by (2.7); pairing (A,B) with (B,A) gives $B(v_n) = \sqrt{c_0}\, b_n$. Finally, (2.13) is (2.10) with $k = 2$, $a = \eta_1$ and $g = \psi'_n$. □

Remark 2.12. The operator $\sum_n b_n^* b_n$ differs from $H_{\mathbb{T}}$ by the factor $c_0^{-1} = \sqrt{2\pi}/\kappa^3$, which grows like $N_\phi^{3/2}$ at fixed aspect ratio; every uniform statement below refers to $H_{\mathbb{T}}$, the normalization in which a pair of relative angular momentum one has energy one. The kernel of $H_{\mathbb{T}}$ is the joint kernel of the operators b_n, which does not depend on any normalization.

3 The kernel

The kernel of $H_{\mathbb{T}}$ is the joint kernel of the pair operators b_n (Remark 2.12). We determine it for a family of operators that contains the b_n and is defined for every modular parameter. The argument is elementary: a zero mode vanishes to third order at every coincidence of two particles, so it is divisible by the cube of the theta-function Jastrow factor; at $N_\phi = 3N$ the quotient depends on the centre of mass only, and the space of possible quotients is three-dimensional.

3.1 Pair operators for every modular parameter

Let $\tau \in \mathbb{C}$ with $\operatorname{Im}\tau > 0$ and $N_\phi \in \mathbb{N}$. For $a \in \mathbb{R}$, $b \in \mathbb{R}$ and t with $\operatorname{Im} t > 0$ let

$$\vartheta\begin{bmatrix} a \\ b \end{bmatrix}(z \,|\, t) := \sum_{n\in\mathbb{Z}} \exp\bigl(\mathrm{i}\pi t(n+a)^2 + 2\pi\mathrm{i}(n+a)(z+b)\bigr), \qquad \vartheta_1(z) := \vartheta\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}(z \,|\, \tau).$$

The series converge absolutely and locally uniformly, and directly from them

$$\vartheta\begin{bmatrix} a \\ b \end{bmatrix}(z+1|t) = e^{2\pi\mathrm{i}a}\vartheta\begin{bmatrix} a \\ b \end{bmatrix}(z|t), \qquad \vartheta\begin{bmatrix} a \\ b \end{bmatrix}(z+t|t) = e^{-\mathrm{i}\pi t - 2\pi\mathrm{i}(z+b)}\vartheta\begin{bmatrix} a \\ b \end{bmatrix}(z|t). \tag{3.1}$$

For $j \in \mathbb{Z}_{N_\phi}$ let $\varpi_j(u) := \vartheta\bigl[\begin{smallmatrix} j/N_\phi \\ 0 \end{smallmatrix}\bigr](N_\phi u \,|\, N_\phi\tau) = \sum_{n\equiv j} \exp(2\pi\mathrm{i}nu + \mathrm{i}\pi\tau n^2/N_\phi)$. Let $\Lambda_u := \mathbb{Z} + \tau\mathbb{Z}$.

Let $(\mathsf{e}_j)_{j\in\mathbb{Z}_{N_\phi}}$ be the standard basis of $\mathbb{C}^{N_\phi}$, c_j the corresponding annihilation operators on $\bigwedge \mathbb{C}^{N_\phi}$, and for $n \in \mathbb{Z}$

$$B_n^{(\tau)} := \sum_{A+B=n} (A-B)\,\mathsf{q}^{(A-B)^2}\, c_{[B]}c_{[A]}, \qquad \mathsf{q}^{d^2} := e^{\mathrm{i}\pi\tau d^2/(2N_\phi)}, \tag{3.2}$$

the sum running over $(A,B) \in \mathbb{Z}^2$; $B_{n+2N_\phi}^{(\tau)} = B_n^{(\tau)}$. Let T_x, T_y be the unitaries $\mathsf{e}_j \mapsto \mathsf{e}_{j+1}$ and $\mathsf{e}_j \mapsto e^{2\pi\mathrm{i}j/N_\phi}\mathsf{e}_j$.

Relation to the torus. For the rectangular torus of Definition 2.1 put $\tau := \mathrm{i}L_x/L_y$ and $u := (y - \mathrm{i}x)/L_y$. Then $e^{\mathrm{i}\kappa n y}e^{-(x-\kappa n)^2/2} = e^{-x^2/2}e^{2\pi\mathrm{i}nu + \mathrm{i}\pi\tau n^2/N_\phi}$ (use $\kappa(y-\mathrm{i}x) = 2\pi u$ and $\kappa^2/2 = \pi L_x/(L_y N_\phi)$), so

$$\psi_j(x,y) = (L_y\sqrt{\pi})^{-1/2}\,\varpi_j(u)\,e^{-x^2/2}, \tag{3.3}$$

$\mathsf{q}^{d^2} = e^{-\kappa^2 d^2/4}$, and under $\mathsf{e}_j \mapsto \psi_j$ the operators $B_n^{(\tau)}$ are the b_n of Proposition 2.11 (pair (A,B) with (B,A)), while T_x, T_y are θ_x, θ_y. In particular the kernel of $H_{\mathbb{T}}$ on $\bigwedge^N \mathfrak{h}$ is the joint kernel of the $B_n^{(\tau)}$.

Holomorphic parts. For $S = \{s_1 < \cdots < s_N\}$ let $\mathsf{e}_{s_1} \widehat{\wedge\cdots\wedge} \mathsf{e}_{s_N}(u_1,\dots,u_N) := \det[\varpi_{s_\alpha}(u_\beta)]_{\alpha,\beta}$, and extend linearly to a map $\Psi \mapsto \widehat{\Psi}$ from $\bigwedge^N \mathbb{C}^{N_\phi}$ to functions on $\mathbb{C}^N$ (for $N=0$, $\widehat{1} := 1$). By (3.3), for a rectangular torus the wave function of Ψ is $(N!)^{-1/2}(L_y\sqrt{\pi})^{-N/2}\prod_k e^{-x_k^2/2}\,\widehat{\Psi}(u)$. We write $U := \sum_k u_k$, e_k for the k-th unit vector of $\mathbb{C}^N$, and $\mathbf{1} := (1,\dots,1)$.

Theorem 3.1 (the kernel at $N_\phi = 3N$)**.** *Let* $\operatorname{Im}\tau > 0$*,* $N \geq 1$ *and* $N_\phi = 3N$*. The joint kernel* $\mathcal{K}_N$ *of the operators* $B_n^{(\tau)}$*,* $n \in \mathbb{Z}_{2N_\phi}$*, on* $\bigwedge^N \mathbb{C}^{N_\phi}$ *is three-dimensional. It is spanned by the states* Ψ_0, Ψ_1, Ψ_2 *with*

$$\widehat{\Psi_j}(u) = A_j(U)\, J(u)^3, \quad J(u) := \prod_{1\leq i<k\leq N} \vartheta_1(u_i - u_k), \quad A_j(s) := \vartheta\begin{bmatrix} j/3 + (N-1)/6 \\ (N-1)/2 \end{bmatrix}(3s \,|\, 3\tau). \tag{3.4}$$

Moreover $\mathcal{F}(T_y)\Psi_j = e^{2\pi \mathrm{i} m_j/N_\phi}\Psi_j$ *with* $m_j = N(N-1)/2 + jN$*, and* $\mathcal{F}(T_x)\Psi_j = e^{-\mathrm{i}\pi(N-1)/3}\Psi_{j+1}$ *(indices modulo 3): the three states are the three centre-of-mass copies of one state. For a rectangular torus,* $\mathcal{K}_N = \ker(H_{\mathbb{T}}|_{\bigwedge^N \mathfrak{h}})$*.*

Proposition 3.2 (no zero modes above filling one third)**.** *If* $N' \geq 2$ *and* $N_\phi < 3N'$*, the joint kernel of the* $B_n^{(\tau)}$ *on* $\bigwedge^{N'} \mathbb{C}^{N_\phi}$ *is zero. In particular, at* $N_\phi = 3N$ *no sector with more than* N *particles contains a zero mode.*

For $N \geq 2$ the functions (3.4) are the torus Laughlin states of [4]. That they are the only zero modes of the torus V_1 Hamiltonian is stated for rectangular tori in [5, Section II.2], and a proof for every lattice shape and boundary twist is claimed in [10]. Theorem 3.1 proves the count for the operators $B_n^{(\tau)}$ at every modular parameter, and hence for $H_{\mathbb{T}}$ on every rectangular torus.

3.2 Theta functions

For $\ell \in \mathbb{Z}$, $a \in \mathbb{R}$ and $\beta \in \mathbb{C} \setminus \{0\}$ let $\Theta_\ell[a;\beta]$ be the space of entire functions g on $\mathbb{C}$ with $g(s+1) = e^{2\pi \mathrm{i} a} g(s)$ and $g(s+\tau) = \beta e^{-2\pi \mathrm{i}\ell s} g(s)$.

Lemma 3.3. *(a)* $\dim \Theta_\ell[a;\beta] = \ell$ *for* $\ell \geq 1$*, and* $\Theta_\ell[a;\beta] = 0$ *for* $\ell \leq -1$*.*

(b) A nonzero $g \in \Theta_\ell[a;\beta]$*,* $\ell \geq 1$*, has exactly* ℓ *zeros, counted with multiplicity, in every period parallelogram whose boundary avoids its zeros.*

(c) ϑ_1 *is odd,* $\vartheta_1(s+1) = -\vartheta_1(s)$*,* $\vartheta_1(s+\tau) = -e^{-\mathrm{i}\pi\tau - 2\pi \mathrm{i} s}\vartheta_1(s)$*, and its zeros are exactly the points of* Λ_u*, all simple.*

(d) $J(u_\pi) = \operatorname{sgn}(\pi) J(u)$ *for permutations* π*;* $J(u + c\mathbf{1}) = J(u)$ *for* $c \in \mathbb{C}$*;* $J(u + e_k) = (-1)^{N-1} J(u)$*;* $J(u + \tau e_k) = (-1)^{N-1} e^{-\mathrm{i}\pi(N-1)\tau - 2\pi \mathrm{i}(N u_k - U)} J(u)$*; and the factor* $\vartheta_1(u_i - u_k)$ *vanishes, simply, exactly on the hyperplanes* $D_{ik}^{\lambda} := \{u_i - u_k = \lambda\}$*,* $\lambda \in \Lambda_u$*.*

(e) If f *is a function on* $\mathbb{C}^N$ *with* $f(u + e_k) = f(u)$ *and* $f(u + \tau e_k) = e^{-\mathrm{i}\pi N_\phi(2u_k + \tau)} f(u)$ *for all* k*, then* $\mathcal{A} := f/J^3$ *satisfies, where defined,* $\mathcal{A}(u + e_k) = (-1)^{N-1}\mathcal{A}(u)$ *and*

$$\mathcal{A}(u + \tau e_k) = (-1)^{N-1} \exp\bigl(-2\pi \mathrm{i}(N_\phi - 3N)u_k - 6\pi \mathrm{i} U - \mathrm{i}\pi(N_\phi - 3N + 3)\tau\bigr)\, \mathcal{A}(u).$$

At $N_\phi = 3N$ *the factor does not depend on* u_k*:* $\mathcal{A}(u + \tau e_k) = (-1)^{N-1} e^{-3\pi \mathrm{i}(2U + \tau)} \mathcal{A}(u)$*.*

(f) The space $\Theta_3^{(N)} := \{A \text{ entire} : A(s+1) = (-1)^{N-1} A(s),\ A(s+\tau) = (-1)^{N-1} e^{-3\pi \mathrm{i}(2s+\tau)} A(s)\}$ *is three-dimensional, with basis* A_0, A_1, A_2 *of* (3.4)*, and* $A_j(s + 1/3) = e^{2\pi \mathrm{i} a_j} A_j(s)$ *with* $a_j := j/3 + (N-1)/6$*.*

Proof. (a) $g(s)e^{-2\pi i a s}$ is entire and 1-periodic, hence a Laurent series in $e^{2\pi i s}$, absolutely and locally uniformly convergent on $\mathbb{C}^\times$: $g(s) = \sum_n c_n e^{2\pi i(n+a)s}$. Comparing coefficients in the τ-relation gives $c_n e^{2\pi i(n+a)\tau} = \beta c_{n+\ell}$, and by iteration, for $k \geq 0$, $|c_{m+k\ell}| = |\beta|^{-k}\exp(-2\pi \operatorname{Im}\tau[k(m+a) + \ell k(k-1)/2])|c_m|$. If $\ell \geq 1$ this is Gaussian decay in both directions, so any $c_0, \dots, c_{\ell-1}$ determine a convergent series in the space, and conversely; the dimension is ℓ. If $\ell \leq -1$ and some $c_m \neq 0$, then $|c_n|$ grows like $\exp(\pi \operatorname{Im}\tau|\ell|k^2)$ along $n = m + k\ell \to -\infty$, so the terms of the series at any point do not tend to zero, a contradiction; hence $g = 0$. (b) g'/g is 1-periodic and $(g'/g)(s+\tau) = (g'/g)(s) - 2\pi i\ell$, so the integral of g'/g around the parallelogram with vertices $s_0, s_0+1, s_0+1+\tau, s_0+\tau$ is $2\pi i\ell$; apply the argument principle. (c) The relations are (3.1) with $a = b = 1/2$; oddness follows by $n \mapsto -n-1$ in the series. The Fourier coefficients do not vanish, so $\vartheta_1 \not\equiv 0$, and $\vartheta_1 \in \Theta_1[\frac{1}{2}; -e^{-i\pi\tau}]$ has exactly one zero in every period parallelogram, by (b). Since $\vartheta_1(0) = 0$ by oddness, the zeros are the points of Λ_u, each simple. Consequently $\vartheta_1(s-1) = -\vartheta_1(s)$ and $\vartheta_1(s-\tau) = -e^{-i\pi\tau+2\pi i s}\vartheta_1(s)$. (d) A transposition of neighbouring variables changes the sign of one factor (oddness) and permutes the others. The shift $u \mapsto u + e_k$ changes the sign of the $N-1$ factors containing u_k. Under $u \mapsto u+\tau e_k$ the factor with partner j is multiplied by $-e^{-i\pi\tau-2\pi i(u_k-u_j)}$ in both orders of the pair (by (c)), and $\sum_{j\neq k}(u_k - u_j) = Nu_k - U$. The last statement is (c). (e) Cube the relations in (d), divide, and use $-i\pi N_\phi\tau + 3i\pi(N-1)\tau = -i\pi(N_\phi - 3N + 3)\tau$. (f) By (a) with $\ell = 3$, $e^{2\pi i a} = (-1)^{N-1}$ and $\beta = (-1)^{N-1}e^{-3\pi i\tau}$, the dimension is 3. By (3.1) with $t = 3\tau$, $A_j(s+1) = e^{6\pi i a_j}A_j(s) = (-1)^{N-1}A_j(s)$ and $A_j(s+\tau) = e^{-2\pi i(N-1)/2}e^{-3\pi i\tau - 6\pi i s}A_j(s)$; the A_j have nonzero Fourier coefficients on the disjoint frequency sets $j + (N-1)/2 + 3\mathbb{Z}$, so they are independent. The last formula is the first relation of (3.1) at $z = 3s$ with $t = 3\tau$, shifted by 1. □

3.3 Holomorphic parts and pair coefficients

Lemma 3.4. *(a) $\Psi \mapsto \widehat{\Psi}$ is injective, and its image on $\bigwedge^N \mathbb{C}^{N_\phi}$ is the space of antisymmetric entire functions f on $\mathbb{C}^N$ with $f(u+e_k) = f(u)$ and $f(u+\tau e_k) = e^{-i\pi N_\phi(2u_k+\tau)}f(u)$ for every k.*

(b) For $N \geq 2$ and $\Psi \in \bigwedge^N \mathbb{C}^{N_\phi}$ write $u = (u_1, u_2, u')$. Then

$$\widehat{\Psi}(u_1, u_2, u') = \sum_{a,b\in\mathbb{Z}_{N_\phi}} \varpi_a(u_1)\varpi_b(u_2)\,\widehat{c_b c_a \Psi}(u').$$

(c) $\widehat{\mathcal{F}(T_x)\Psi}(u) = e^{2\pi i U + i\pi N\tau/N_\phi}\,\widehat{\Psi}(u + \frac{\tau}{N_\phi}\mathbf{1})$ and $\widehat{\mathcal{F}(T_y)\Psi}(u) = \widehat{\Psi}(u + \frac{1}{N_\phi}\mathbf{1})$.

Proof. (a) Each ϖ_j lies in $\Theta_{N_\phi}[0; e^{-i\pi N_\phi\tau}]$ (from the series, as in Lemma 3.3(f)), and the ϖ_j have disjoint Fourier supports $j + N_\phi\mathbb{Z}$, so they form a basis of this N_ϕ-dimensional space. Slater determinants therefore have the stated properties. Products $\prod_\beta \varpi_{j_\beta}(u_\beta)$ for distinct index tuples are linearly independent (fix all variables but one and induct), and distinct sets S give disjoint sets of tuples, so the map is injective. Conversely let f have the properties. For $N = 1$ use the basis. For $N \geq 2$, choose points $s_1, \dots, s_{N_\phi}$ with $M := [\varpi_j(s_i)]$ invertible (the evaluation functionals separate the span of the ϖ_j); for fixed $u' = (u_2, \dots, u_N)$, $u_1 \mapsto f(u_1, u')$ lies in the one-particle space, so $f(u_1, u') = \sum_j c_j(u')\varpi_j(u_1)$ with $c(u') = M^{-1}(f(s_i, u'))_i$, entire and with the one-particle relations in each remaining variable. By induction on the number of variables (without antisymmetry) f is a finite combination of products of orbitals; the coefficients are unique, and antisymmetry of f makes them antisymmetric, so f is a combination of Slater determinants. (b) By linearity take $\Psi = \mathsf{e}_{s_1} \wedge \cdots \wedge \mathsf{e}_{s_N}$ and $a = s_p$, $b = s_q$, $p \neq q$. The Laplace expansion of $\det[\varpi_{s_\alpha}(u_\beta)]$ along the columns $1, 2$ gives the coefficient of $\varpi_{s_p}(u_1)\varpi_{s_q}(u_2)$ as $(-1)^{p+q+1}\widehat{\Psi_{pq}}$ if

$p<q$ and $(-1)^{p+q}\widehat{\Psi_{pq}}$ if $p>q$, where Ψ_{pq} is the wedge with $\mathsf{e}_{s_p},\mathsf{e}_{s_q}$ removed. On the Fock side, c_{s_p} acts first with the sign $(-1)^{p-1}$, then c_{s_q} with the sign $(-1)^{q-2}$ if $q>p$ and $(-1)^{q-1}$ if $q<p$; the totals agree. Terms with $a=b$ or $\{a,b\}\not\subset S$ vanish on both sides. (c) T_x and T_y act on one-particle holomorphic parts by $f(u)\mapsto e^{\mathrm{i}\pi(2u+\tau/N_\phi)}f(u+\tau/N_\phi)$ and $f(u)\mapsto f(u+1/N_\phi)$, as the series of ϖ_j show; apply this to each row of the determinant. □

3.4 Zero modes vanish to third order

Lemma 3.5. *Let $N\geq 2$, any $N_\phi\geq 1$, $\Psi\in\bigwedge^N\mathbb{C}^{N_\phi}$ and $f:=\widehat{\Psi}$. The following are equivalent: (i) $B_n^{(\tau)}\Psi=0$ for every $n\in\mathbb{Z}_{2N_\phi}$; (ii) $[(\partial_{u_1}-\partial_{u_2})f](w,w,u')=0$ for all $w\in\mathbb{C}$, $u'\in\mathbb{C}^{N-2}$; (iii) $f=(u_1-u_2)^3g$ with g entire; (iv) for all $i<k$ and $\lambda\in\Lambda_u$, $f=(u_i-u_k-\lambda)^3g_{ik}^{\lambda}$ with g_{ik}^{λ} entire.*

Proof. (i)⇔(ii). With $\varpi_a'(w)=\sum_{A\equiv a}2\pi\mathrm{i}A\,e^{2\pi\mathrm{i}Aw+\mathrm{i}\pi\tau A^2/N_\phi}$ and Lemma 3.4(b),

$$[(\partial_1-\partial_2)f](w,w,u')=\sum_{a,b}\big[\varpi_a'\varpi_b-\varpi_a\varpi_b'\big](w)\,\widehat{c_bc_a\Psi}(u')=2\pi\mathrm{i}\sum_{n\in\mathbb{Z}}e^{2\pi\mathrm{i}nw+\mathrm{i}\pi\tau n^2/(2N_\phi)}\,\widehat{B_n^{(\tau)}\Psi}(u'),$$

using $A^2+B^2=\frac12(A+B)^2+\frac12(A-B)^2$ and grouping by $n=A+B$; all rearrangements are justified by absolute convergence. At fixed $\operatorname{Im}w$ the right side is an absolutely convergent Fourier series in $\operatorname{Re}w$ in which each frequency $n\in\mathbb{Z}$ occurs once, with coefficient $e^{-2\pi n\operatorname{Im}w}e^{\mathrm{i}\pi\tau n^2/(2N_\phi)}\widehat{B_n^{(\tau)}\Psi}(u')$. Hence (ii) holds iff $\widehat{B_n^{(\tau)}\Psi}\equiv 0$ for all n, iff (i), by injectivity (Lemma 3.4(a)). (ii)⇔(iii). Put $\tilde f(R,v,u'):=f(R+v/2,R-v/2,u')$, entire and odd in v. Then $\tilde f$ and $\partial_v^2\tilde f$ vanish at $v=0$, and (ii) says $\partial_v\tilde f|_{v=0}=0$. So the first three Taylor coefficients in v vanish and $\tilde f=v^3\tilde g$ with $\tilde g(R,v,u')=\frac12\int_0^1(1-\theta)^2\partial_v^3\tilde f(R,\theta v,u')\mathrm{d}\theta$ entire. Conversely (iii) makes $(\partial_1-\partial_2)f$ vanish on $u_1=u_2$. (iii)⇒(iv). Antisymmetry transports the factor to every pair (i,k). By the relations of Lemma 3.4(a), $f(u+\lambda e_i)=M_\lambda(u)f(u)$ with M_λ entire and nowhere zero, so $f(u)=M_\lambda(u-\lambda e_i)f(u-\lambda e_i)$ inherits the factor $(u_i-u_k-\lambda)^3$. (iv)⇒(iii) is the case $(1,2,0)$. □

Lemma 3.6 (division by J^3)**.** *If f is entire on $\mathbb{C}^N$ and satisfies Lemma 3.5(iv), then f/J^3 is entire.*

Proof. Let ℓ,ℓ' be nonconstant affine-linear forms on $\mathbb{C}^N$. (D1) If q is entire and $q=(\ell-\lambda)^3\tilde q_\lambda$ with $\tilde q_\lambda$ entire for every $\lambda\in\Lambda_u$, then $q/\vartheta_1(\ell)^3$ is entire: near a point with $\ell=\lambda$, $\vartheta_1(\ell)=(\ell-\lambda)g(\ell)$ with g holomorphic and nonzero (Lemma 3.3(c)), so the quotient is $\tilde q_\lambda/g(\ell)^3$. (D2) If $q=hg$ with h,g entire, $q=(\ell'-\mu)^3\tilde q$ with $\tilde q$ entire, and $h\not\equiv 0$ on $\mathsf{H}:=\{\ell'=\mu\}$, then $g=(\ell'-\mu)^3\tilde g$ with $\tilde g$ entire: choose ξ on which the linear part of ℓ' equals 1; an entire function has the factor $(\ell'-\mu)^3$ iff it and its first two derivatives along ξ vanish on H (Taylor's formula with integral remainder along ξ); by Leibniz, if $\partial_\xi^r g|_{\mathsf{H}}=0$ for $r<m\leq 2$ then $0=\partial_\xi^m(hg)|_{\mathsf{H}}=h|_{\mathsf{H}}\,\partial_\xi^m g|_{\mathsf{H}}$, so $\partial_\xi^m g$ vanishes on the dense open subset $\{h\neq 0\}$ of the connected hyperplane H, hence on H. Enumerate the pairs $p_1,\dots,p_M$, $M=N(N-1)/2$, with forms $\ell_p:=u_i-u_k$ for $p=(i,k)$, and put $q_0:=f$, $q_r:=q_{r-1}/\vartheta_1(\ell_{p_r})^3$. By induction, q_r is entire and has the factor $(\ell_{p_t}-\mu)^3$ for every $t>r$ and $\mu\in\Lambda_u$: q_r is entire by (D1); and (D2) applies to $q_{r-1}=\vartheta_1(\ell_{p_r})^3q_r$ on $\{\ell_{p_t}=\mu\}$, on which ℓ_{p_r} is nonconstant (distinct pairs give independent forms), so $\vartheta_1(\ell_{p_r})^3\not\equiv 0$ there. Then $f/J^3=q_M$ is entire. □

3.5 The quotient and the count

Lemma 3.7. *Let $N\geq 2$, $N_\phi=3N$, $\Psi\in\mathcal{K}_N$, $f=\widehat{\Psi}$. Then $f/J^3=A(U)$ for some $A\in\Theta_3^{(N)}$.*

Proof. $\mathcal{A} := f/J^3$ is entire by Lemmas 3.5 and 3.6, and by Lemma 3.3(e) (off the coincidence set, then everywhere by continuity)

$$\mathcal{A}(u+e_k) = (-1)^{N-1}\mathcal{A}(u), \qquad \mathcal{A}(u+\tau e_k) = (-1)^{N-1}e^{-3\pi\mathrm{i}(2U+\tau)}\mathcal{A}(u).$$

Fix u and $i \neq k$ and put $g(w) := \mathcal{A}(u + w(e_i - e_k))$; U is constant along this line. Then $g(w+1) = g(w)$, and, applying the k-relation backwards at $v - \tau e_k$ and then the i-relation, $\mathcal{A}(v + \tau e_i - \tau e_k) = \mathcal{A}(v)$ for every v, so $g(w+\tau) = g(w)$. Thus g is entire and doubly periodic, hence constant (Liouville). Moving along one direction $e_k - e_1$ at a time, $\mathcal{A}(u+c) = \mathcal{A}(u)$ whenever $\sum_k c_k = 0$, so $\mathcal{A}(u) = \mathcal{A}(Ue_1) =: A(U)$ with A entire; its relations are those of $\mathcal{A}$ at $u = se_1$. □

Lemma 3.8. *Let $N \geq 1$, $N_\phi = 3N$ and $A \in \Theta_3^{(N)}$. Then $f_A(u) := A(U)J(u)^3$ is the holomorphic part of a unique $\Psi_A \in \bigwedge^N \mathbb{C}^{N_\phi}$, $\Psi_A \in \mathcal{K}_N$, and $A \mapsto \Psi_A$ is linear and injective.*

Proof. f_A is entire and antisymmetric. By Lemma 3.3(d),(f), $f_A(u+e_k) = f_A(u)$ and $f_A(u + \tau e_k) = (-1)^{N-1}e^{-3\pi\mathrm{i}(2U+\tau)}(-1)^{N-1}e^{-3\mathrm{i}\pi(N-1)\tau - 6\pi\mathrm{i}(Nu_k - U)}f_A(u) = e^{-3\pi\mathrm{i}N(2u_k+\tau)}f_A(u)$, so $f_A = \widehat{\Psi_A}$ for a unique Ψ_A by Lemma 3.4(a). For $N \geq 2$, $f_A = (u_1-u_2)^3 A(U)[\vartheta_1(u_1-u_2)/(u_1-u_2)]^3 \prod_{(i,k)\neq(1,2)} \vartheta_1(u_i-u_k)^3$ with an entire cofactor, so $\Psi_A \in \mathcal{K}_N$ by Lemma 3.5; for $N = 1$ there is nothing to prove. If $A \not\equiv 0$ then $f_A \not\equiv 0$. □

Proof of Theorem 3.1. For $N = 1$ every one-particle state is in $\mathcal{K}_1$, and $\mathbb{C}^3$ is three-dimensional; the holomorphic parts are the space $\Theta_3[0; e^{-3\mathrm{i}\pi\tau}] = \Theta_3^{(1)}$, and $J = 1$. For $N \geq 2$, Lemmas 3.7 and 3.8 give mutually inverse linear maps between $\mathcal{K}_N$ and $\Theta_3^{(N)}$ (injectivity of $\Psi \mapsto \widehat{\Psi}$), so $\dim \mathcal{K}_N = 3$ by Lemma 3.3(f), with basis $\Psi_j := \Psi_{A_j}$. By Lemma 3.4(c), $\mathcal{F}(T_y)$ shifts U by $N/N_\phi = 1/3$ and leaves J invariant, so $\mathcal{F}(T_y)\Psi_j = e^{2\pi\mathrm{i}a_j}\Psi_j$, and $N_\phi a_j = jN + N(N-1)/2$. $\mathcal{F}(T_x)$ multiplies by $e^{2\pi\mathrm{i}U + \mathrm{i}\pi\tau/3}$ and shifts U by $\tau/3$; completing the square in the series, $\vartheta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z+\tau|3\tau) = e^{-\mathrm{i}\pi\tau/3 - 2\pi\mathrm{i}(z+b)/3}\vartheta\left[\begin{smallmatrix} a+1/3 \\ b \end{smallmatrix}\right](z|3\tau)$, and $\vartheta\left[\begin{smallmatrix} a+1 \\ b \end{smallmatrix}\right] = \vartheta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right]$; with $z = 3U$ this gives $\mathcal{F}(T_x)\Psi_j = e^{-2\pi\mathrm{i}b/3}\Psi_{j+1}$ with $b = (N-1)/2$. The last statement is the relation to the torus explained above. □

Proof of Proposition 3.2. Let Ψ be in the joint kernel. Lemmas 3.5 and 3.6 do not use $N_\phi = 3N'$, so $\mathcal{A} := \widehat{\Psi}/J^3$ is entire with the relations of Lemma 3.3(e) (with N' in place of N). Put $m := N_\phi - 3N' < 0$ and $g(w) := \mathcal{A}(u + w(e_1 - e_2))$ for fixed u. As in Lemma 3.7, $g(w+1) = g(w)$, and the same two-step computation with the general relation gives $g(w+\tau) = e^{-2\pi\mathrm{i}m(u_1-u_2+2w+\tau)}g(w)$. So $g \in \Theta_{2m}[0; e^{-2\pi\mathrm{i}m(u_1-u_2+\tau)}]$ with $2m \leq -2$, and $g = 0$ by Lemma 3.3(a). Hence $\mathcal{A} \equiv 0$, $\widehat{\Psi} \equiv 0$ and $\Psi = 0$. At $N_\phi = 3N$ a sector with $N' > N \geq 1$ particles has $N' \geq 2$ and $N_\phi < 3N'$. □

4 The pair Gram matrix and the projector form

On two particles $H_{\mathbb{T}} = h = \sum_n |v_n\rangle\langle v_n|$ is not a projection: the pair functions v_n of Proposition 2.11 are not orthonormal. Their Gram matrix is computed exactly in this section; its smallest eigenvalue enters the main theorem through the two-body term. We also record the normal-ordered square of a lifted pair operator, which is the starting point of Section 7.

4.1 The Gram matrix

Let $G_{nn'} := \langle v_n, v_{n'}\rangle$ for $n, n' \in \mathbb{Z}_{2N_\phi}$, and put

$$\mathfrak{f}(s) := (1-s)e^{-s/2}, \qquad s_{jk} := \tfrac{1}{4}(j^2L_x^2 + k^2L_y^2) \quad ((j,k) \in \mathbb{Z}^2).$$

Lemma 4.1 (the Gram matrix in closed form)**.** *$G_{nn'} = 0$ unless $n' \equiv n \pmod{N_\phi}$, and for integers n and m*

$$G_{n,n+mN_\phi} = \sum_{j \equiv m \ (2)} \sum_{k \in \mathbb{Z}} e^{\mathrm{i}\pi k(n+jN_\phi/2)}\, \mathfrak{f}(s_{jk}). \tag{4.1}$$

Proof. By the proof of Proposition 2.11, v_n is $\sqrt{2c_0}$ times the tensor $\sum_{A+B=n} f(A-B)\psi_{[A]} \otimes \psi_{[B]}$. Hence $G_{nn'} = 2c_0 \sum f(A-B)f(A'-B')$ over $A+B=n$, $A'+B'=n'$, $A' \equiv A$, $B' \equiv B \pmod{N_\phi}$. Writing $A' = A + pN_\phi$, $B' = B + qN_\phi$ forces $(p+q)N_\phi = n' - n$, which proves the first claim. For $n' = n + mN_\phi$ we have $p+q = m$ and $A' - B' = (A-B) + jN_\phi$ with $j = p - q$, which runs once over the integers $j \equiv m \pmod 2$. With $d = A - B$, which runs over $d \equiv n \pmod 2$,

$$G_{n,n+mN_\phi} = \sum_{j \equiv m \ (2)} S_j, \qquad S_j := 2c_0 \sum_{d \equiv n \ (2)} f(d) f(d + jN_\phi).$$

Put $M := jN_\phi$ and $u := d + M/2$. Then $f(d)f(d+M) = (u^2 - M^2/4)e^{-\kappa^2 u^2/2} e^{-\kappa^2 M^2/8}$, and u runs over the coset $n + M/2 + 2\mathbb{Z}$. Poisson summation on a coset, $\sum_{u \in c+2\mathbb{Z}} g(u) = \frac{1}{2}\sum_{k\in\mathbb{Z}} \hat{g}(k/2) e^{\mathrm{i}\pi k c}$ with $\hat{g}(\xi) = \int g(u) e^{-2\pi \mathrm{i} u\xi} \mathrm{d}u$, and

$$\int_{\mathbb{R}} (u^2 - M^2/4) e^{-\kappa^2 u^2/2} e^{-\mathrm{i}\pi k u} \mathrm{d}u = \frac{\sqrt{2\pi}}{\kappa^3}\Big(1 - \frac{\pi^2 k^2}{\kappa^2} - \frac{\kappa^2 M^2}{4}\Big) e^{-\pi^2 k^2/(2\kappa^2)}$$

give, since $\pi k/\kappa = kL_y/2$ and $\kappa M = jL_x$, $S_j = \sum_k e^{\mathrm{i}\pi k(n+jN_\phi/2)}(1 - s_{jk}) e^{-s_{jk}/2}$. □

Let $\Delta(\ell) := 2(\ell^2/4 - 1)e^{-\ell^2/8}$ for $\ell > 0$, let $\mathcal{X} := \{(0,0), (\pm 1, 0), (0, \pm 1)\}$ and

$$\Upsilon(L_x, L_y) := \sum_{(j,k) \in \mathbb{Z}^2 \setminus \mathcal{X}} |\mathfrak{f}(s_{jk})|.$$

Lemma 4.2 (the levels)**.** *The matrix G is block diagonal with 2×2 blocks on $\{n, n + N_\phi\}$, and its entries depend on n only through $n \bmod 2$.*

(a) If N_ϕ is even, G has the four eigenvalues $1 - (-1)^s \Delta(L_y) \mp \Delta(L_x) + r_{s,\pm}$, $s \in \{0,1\}$, each of multiplicity $N_\phi/2$, with eigenvectors $e_n \pm e_{n+N_\phi}$, $n \equiv s$.

(b) If N_ϕ is odd, G has the two eigenvalues $1 \pm \sqrt{\Delta(L_x)^2 + \Delta(L_y)^2} + r_\pm$, each of multiplicity N_ϕ.

(c) In both cases $|r| \le \Upsilon(L_x, L_y)$, and for $L_x, L_y \ge 3$

$$\Upsilon(L_x, L_y) \le 1.20\,(L_x^2 + L_y^2)\, e^{-(L_x^2+L_y^2)/8} + 2.02\,(L_x^2 e^{-L_x^2/2} + L_y^2 e^{-L_y^2/2}).$$

Consequently the extreme eigenvalues $\lambda_{\min}, \lambda_{\max}$ of G satisfy $\delta_G := \max\{\lambda_{\max} - 1, 1 - \lambda_{\min}\} \le \Delta(L_x) + \Delta(L_y) + \Upsilon(L_x, L_y)$.

Proof. The block structure and the dependence on $n \bmod 2$ are read off (4.1). Put $d_s := G_{nn}$ and $o_s := G_{n,n+N_\phi}$ for $n \equiv s \pmod 2$. Both are real: d_s is a diagonal entry of the Hermitian matrix G; for even N_ϕ every phase in o_s is ± 1; and for odd N_ϕ the terms of o_s with odd k cancel in the pairs $(j,k), (-j,k)$, as shown in (b), while those with even k have phases ± 1. In d_s the term $(0,0)$ is 1 and the terms $(0, \pm 1)$ give $2(-1)^s \mathfrak{f}(L_y^2/4) = -(-1)^s \Delta(L_y)$; in o_s the terms $(\pm 1, 0)$ give $2\mathfrak{f}(L_x^2/4) = -\Delta(L_x)$. The remaining terms of d_s (even j) and of o_s (odd j) run over disjoint subsets of $\mathbb{Z}^2 \setminus \mathcal{X}$, so $|d_s - 1 + (-1)^s \Delta(L_y)| + |o_s + \Delta(L_x)| \le \Upsilon$.

(a) For even N_ϕ, n and $n+N_\phi$ have the same parity s; the block is $\left[\begin{smallmatrix} d_s & o_s \\ o_s & d_s \end{smallmatrix}\right]$ with eigenvalues $d_s \pm o_s$ on $e_n \pm e_{n+N_\phi}$, and there are $N_\phi/2$ blocks of each parity.

(b) For odd N_ϕ, n and $n+N_\phi$ have opposite parities. Moreover $o_s =: o$ does not depend on s: for odd j and odd N_ϕ, the terms with odd k cancel in the pairs $(j,k),(-j,k)$ because $\cos(\pi k j N_\phi/2)=0$, and for even k the phase $e^{\mathrm{i}\pi k n}$ equals 1. The block is $\left[\begin{smallmatrix} d_s & o \\ o & d_{1-s} \end{smallmatrix}\right]$, with eigenvalues $\frac{1}{2}(d_0+d_1) \pm \sqrt{\frac{1}{4}(d_0-d_1)^2+o^2}$, the same for all N_ϕ blocks. With the leading terms above and $|\sqrt{a^2+b^2}-\sqrt{a'^2+b'^2}| \le |a-a'|+|b-b'|$ the error is at most $\max_s |d_s - 1 + (-1)^s \Delta(L_y)| + |o + \Delta(L_x)| \le \Upsilon$.

(c) Put $a := L_x^2/4$ and $b := L_y^2/4$, both at least $9/4$. Off $\mathcal{X}$, $s_{jk} \ge 9/2$, so $|\mathfrak{f}(s)| \le s e^{-s/2}$. The terms with $j=0$, $|k|\ge 2$ sum to at most $2\cdot 4be^{-2b}/(1-q)$, where $q \le \frac{9}{4}e^{-5b/2} \le 0.0081$ bounds the ratio of consecutive terms; this is at most $2.02\, L_y^2 e^{-L_y^2/2}$, and likewise for $k=0$, $|j|\ge 2$. The terms with $j,k \ne 0$ sum to $\Sigma_1(a)\Sigma_0(b)+\Sigma_0(a)\Sigma_1(b)$ with $\Sigma_0(a)=\sum_{j\ne 0} e^{-j^2 a/2} \le 2e^{-a/2}/(1-e^{-3a/2})$ and $\Sigma_1(a)=\sum_{j\ne 0} j^2 a e^{-j^2 a/2} \le 2ae^{-a/2}/(1-4e^{-3a/2})$ (use $j^2-1 \ge 3(j-1)$ and $j^2 \le 4^{j-1}$); with $e^{-3a/2} \le e^{-27/8}$ this is at most $4(a+b)e^{-(a+b)/2}/((1-e^{-27/8})(1-4e^{-27/8})) \le 1.20(L_x^2+L_y^2)e^{-(L_x^2+L_y^2)/8}$. The bound on δ_G follows from (a)–(c) and $\sqrt{a^2+b^2} \le a+b$. □

4.2 The projector form

Proposition 4.3 (the projector form)**.** $\mathrm{spec}(h) \subset \{0\} \cup [\lambda_{\min}, \lambda_{\max}]$*, so* $h^2 \ge \lambda_{\min} h$*. If* $\delta_G < 1$*, then* $\dim V_{\mathbb{T}} = 2N_\phi$*, and with* P_V *the orthogonal projection onto* $V_{\mathbb{T}}$ *and* $H' := \mathcal{L}_2(P_V)$*,*

$$\lambda_{\min} P_V \le h \le \lambda_{\max} P_V, \qquad \lambda_{\min} H' \le H_{\mathbb{T}} \le \lambda_{\max} H', \qquad \ker H' = \ker H_{\mathbb{T}}.$$

Proof. Let $\mathsf{V} : \mathbb{C}^{2N_\phi} \to \bigwedge^2 \mathfrak{h}$ have the columns v_n. Then $h = \mathsf{V}\mathsf{V}^*$, whose nonzero spectrum is that of $G = \mathsf{V}^*\mathsf{V}$; this gives the first claim, and $h^2 - \lambda_{\min} h \ge 0$ by the functional calculus. If $\delta_G < 1$ then $G > 0$, V is injective, $\dim V_{\mathbb{T}} = 2N_\phi$, and with the isometry $W := \mathsf{V} G^{-1/2}$ one has $h = WGW^*$ and $P_V = WW^*$, whence the first sandwich. The lift preserves order (Lemma 2.6(iii)), which gives the second, and the kernels coincide because $\lambda_{\min} > 0$. □

4.3 The square of a lifted pair operator

For an operator h on $\bigwedge^2 \mathfrak{h}$, extended by hA_2 to $\mathfrak{h}^{\otimes 2}$, and $\{i,j\} \subset \{1,\dots,n\}$, h_{ij} denotes h acting on the tensor positions i,j; $h_{ij} = h_{ji}$.

Lemma 4.4 (the normal-ordered square)**.** *For every self-adjoint* h *on* $\bigwedge^2 \mathfrak{h}$*, on* $\mathcal{F}$*,*

$$\mathcal{L}_2(h)^2 = \mathcal{L}_2(h^2) + \mathcal{L}_3(X_3(h)) + \mathcal{L}_4(W_4(h\otimes h)W_4^*), \qquad X_3(h) := \sum_{(ij)\ne(kl)} h_{ij}h_{kl} \ \text{ on } \bigwedge^3 \mathfrak{h}, \tag{4.2}$$

the sum running over the six ordered pairs of distinct pairs in $\{1,2,3\}$*. Moreover* $X_3(h) = W_3(h\otimes \mathbf{1})(W_3^*W_3 - \mathbf{1})(h\otimes \mathbf{1})W_3^*$*. For* $h = P_V$ *a projection this is the identity* $H'^2 = H' + \mathcal{L}_3\big(W_3(W_3^*W_3-\mathbf{1})W_3^*\big) + \mathcal{L}_4(W_4W_4^*)$*, with* W_3*,* W_4 *restricted to* $V_{\mathbb{T}}\otimes\mathfrak{h}$ *and* $V_{\mathbb{T}}\otimes V_{\mathbb{T}}$*, of [1, Lemma 2.1 (Normal-ordered square)].*

Proof. On $\bigwedge^n \mathfrak{h}$, $\mathcal{L}_2(h) = \sum_{|S|=2} h_S$ by Lemma 2.6(ii), so $\mathcal{L}_2(h)^2 = \sum_{S,S'} h_S h_{S'}$. The terms with $S = S'$ give $\sum_S (h^2)_S = \mathcal{L}_2(h^2)$. The terms with $|S\cap S'| = 1$ are grouped by the triple $Q = S\cup S'$ and give $\sum_{|Q|=3}(X_3)_Q$, where X_3 commutes with permutations of its three positions; this is $\mathcal{L}_3(X_3)$. The terms with $S\cap S' = \emptyset$ are grouped by $Q = S\cup S'$, $|Q|=4$, and give $\mathcal{L}_4(X_4)$ with $X_4 := \sum h_S h_{S'}$

over the six ordered pairs of disjoint pairs in $\{1,2,3,4\}$. On antisymmetric tensors $W_4^*\Psi = \sqrt{6}\Psi$ and $W_4 = \sqrt{6}A_4$ (Lemma 2.6(iv)), and $A_4 h_{12} h_{34}\Psi = \frac{1}{6}X_4\Psi$, because each ordered pair of disjoint pairs is the image of $(\{1,2\},\{3,4\})$ under four permutations; hence $X_4 = W_4(h\otimes h)W_4^*$ on $\bigwedge^4\mathfrak{h}$. Similarly $W_3(h\otimes\mathbf{1})W_3^* = 3A_3h_{12} = h_{12}+h_{13}+h_{23}$ and $W_3(h^2\otimes\mathbf{1})W_3^* = (h^2)_{12}+(h^2)_{13}+(h^2)_{23}$ on $\bigwedge^3\mathfrak{h}$, so $X_3 = (W_3(h\otimes\mathbf{1})W_3^*)^2 - W_3(h^2\otimes\mathbf{1})W_3^*$, which is the second formula. For $h = P_V$, $h^2 = h$ and the restrictions are as stated. □

5 Local modes on the torus

The comparison operator of Section 6 is built, at every base point R, from the periodizations of the planar orbitals and pair states about R. This section defines them and bounds their Gram matrix.

5.1 Periodized orbitals and pair states

For $p \geq 0$ and $R \in \mathbb{R}^2$ let v_p^{*R} be the planar two-particle state with relative orbital η_1 and centre-of-mass orbital η_p about $\sqrt{2}R$, that is $v_p^{*R}(r_1,r_2) := \eta_1(\rho)\,\eta_p^{\sqrt{2}R}(\zeta)$ with $\rho = (r_1-r_2)/\sqrt{2}$, $\zeta = (r_1+r_2)/\sqrt{2}$; it is antisymmetric. Let

$$v_p^*(i,j) := (i-j)\sqrt{\frac{p!}{2^p\,i!\,j!}}\;\mathbf{1}_{i+j=p+1} \qquad (0\leq i<j).$$

Lemma 5.1 (periodized local modes)**.** *Let $\chi_m^R := \Pi\eta_m^R$ and $\tilde{u}_p(R) := \Pi^{(2)}v_p^{*R}$.*

(a) $\chi_m^R \in \mathfrak{h}$, $\tilde{u}_p(R) \in \bigwedge^2\mathfrak{h}$, and $\langle\psi_j,\chi_m^R\rangle = \sqrt{\kappa}\sum_{n\equiv j} e^{-\mathrm{i}\kappa n R_y}\mathcal{H}_m(\kappa n - R_x)$.

(b) $v_p^{*R} = \sum_{i<j} v_p^*(i,j)\,\eta_i^R\wedge\eta_j^R$, and hence $\tilde{u}_p(R) = \sum_{i<j} v_p^*(i,j)\,\chi_i^R\wedge\chi_j^R$.

(c) For $\gamma\in\Lambda$ there is a real $\theta = \theta(R,\gamma)$, independent of m, with $\chi_m^{R+\gamma} = e^{\mathrm{i}\theta}\chi_m^R$; then $\tilde{u}_p(R+\gamma) = e^{2\mathrm{i}\theta}\tilde{u}_p(R)$.

Proof. (a) η_m^R is an LLL function with Gaussian decay about R, so Lemma 2.5 applies, and $\langle\psi_j,\chi_m^R\rangle = \overline{\langle\chi_m^R,\psi_j\rangle} = \int_{\mathbb{R}^2}\overline{\psi_j}\,\eta_m^R = \sum_{n\equiv j}\langle\phi_n,\eta_m^R\rangle_{\mathbb{R}^2}$, which is the stated sum by (2.5). $\Pi^{(2)}$ preserves antisymmetry. (b) At $R=0$, in the Bargmann form (Lemmas 2.4(i) and 2.7(i)), v_p^{*0} corresponds to $wZ^p/\sqrt{p!}$ with $w = (z_1-z_2)/\sqrt{2}$, $Z = (z_1+z_2)/\sqrt{2}$. In $(z_1-z_2)(z_1+z_2)^p$ the coefficient of $z_1^i z_2^j$, $i+j=p+1$, is $\binom{p}{i-1} - \binom{p}{i} = p!(i-j)/(i!j!)$; passing to the normalized tensors $z_1^i z_2^j/\sqrt{i!j!}$ and then to the normalized wedges ($\eta_i\wedge\eta_j$ corresponds to $(\eta_i\otimes\eta_j - \eta_j\otimes\eta_i)/\sqrt{2}$) gives the coefficient $v_p^*(i,j)$. For general R apply $t_R\otimes t_R$, which acts as $t_{\sqrt{2}R}$ on ζ and trivially on ρ (Lemma 2.7(ii)) and multiplies both sides by $e^{\mathrm{i}R_xR_y}$ (by (2.4)). The second identity follows because $\Pi^{(2)}$ is linear and $\Pi^{(2)}(\eta_i\wedge\eta_j) = \Pi\eta_i\wedge\Pi\eta_j$. (c) By (2.4) and Lemma 2.3(ii), $\eta_m^{R+\gamma}$ is an m-independent phase times $t_\gamma\eta_m^R$, and $\Pi t_\gamma = \pm\Pi\tilde{t}_\gamma = \pm\Pi$ by reindexing. The statement for $\tilde{u}_p$ follows from (b). □

5.2 The local Gram matrix

For $R\in\mathbb{R}^2$ let $\Gamma(R)_{mm'} := \langle\chi_m^R,\chi_{m'}^R\rangle$, $m,m'\geq 0$, and let $\Gamma^{(m)}(R)$ be its truncation to $0\leq m,m'\leq m$. For $r\geq 0$ let

$$\mu_m(r) := \|\mathcal{D}^{(m)}(r/\sqrt{2})\|, \qquad \mathcal{E}_m(\mathbb{T}) := \sum_{\gamma\in\Lambda\setminus\{0\}}\mu_m(|\gamma|);$$

by Lemma 2.4(iii), $\|\mathcal{D}^{(m)}(\beta)\| = \mu_m(\sqrt{2}|\beta|)$ for every $\beta\in\mathbb{C}$.

Lemma 5.2 (the local Gram matrix)**.** *For every $R \in \mathbb{R}^2$ and $m, m' \geq 0$,*

$$\Gamma(R)_{mm'} = \sum_{\gamma \in \Lambda} \varsigma_\gamma \, e^{\mathrm{i}\,\gamma\wedge R} \, \langle m|\mathcal{D}(\beta(\gamma))|m'\rangle, \qquad \varsigma_{(jL_x,kL_y)} := (-1)^{jkN_\phi}, \tag{5.1}$$

with $\beta(\gamma) = (\gamma_x - \mathrm{i}\gamma_y)/\sqrt{2}$. Consequently

$$\|\Gamma^{(m)}(R) - \mathbf{1}\| \leq \mathcal{E}_m(\mathbb{T}), \qquad \Gamma(R)_{mm} \leq 1 + \sum_{\gamma \in \Lambda\setminus 0} |\langle m|\mathcal{D}(\beta(\gamma))|m\rangle|.$$

Proof. By Lemma 2.5, $\Gamma(R)_{mm'} = \int_{\mathbb{R}^2} \overline{\eta_m^R}\, \Pi \eta_{m'}^R = \sum_\gamma \langle \eta_m^R, \tilde{t}_\gamma \eta_{m'}^R\rangle_{\mathbb{R}^2}$, absolutely convergent. By (2.4) the phases $e^{\mp \mathrm{i} R_x R_y/2}$ cancel, and $t_R^* t_\gamma t_R = t_{-R} t_\gamma t_R = e^{\mathrm{i}\,\gamma\wedge R} t_\gamma$ by Lemma 2.3(ii). Lemma 2.4(ii) gives $\langle \eta_m, t_\gamma \eta_{m'}\rangle = \langle m|\mathcal{D}(\beta(\gamma))|m'\rangle$. The term $\gamma = 0$ is $\delta_{mm'}$, and the truncation of each other term has norm $\mu_m(|\gamma|)$. □

5.3 Bounds on the lattice sums

Lemma 5.3. *For $m \geq 0$ and $r \geq 0$ put $\mathfrak{g}_m(r) := (r^2/2 + m)^m e^{-r^2/4}/m!$.*

(a) For $m \leq 23$, μ_m is nonincreasing on $[16, \infty)$. Consequently, for every rectangular torus whose shorter period is at least L, with $L \geq 16$, and every $m \leq 23$,

$$\mathcal{E}_m(\mathbb{T}) \leq \mathcal{E}_m(L) := \sum_{(j,k)\in\mathbb{Z}^2\setminus 0} \mu_m(L\sqrt{j^2+k^2}),$$

and $\mathcal{E}_m(L)$ is nonincreasing in $L \geq 16$.

(b) For $m \geq 1$ and $r \geq 0$, $\mu_m(r) \leq (m+1)\,\mathfrak{g}_m(r)$, and $\mathfrak{g}_m$ is decreasing for $r^2 > 2m$. Hence, for every rectangular torus whose shorter period is at least L with $L^2 \geq 2m$,

$$\mathcal{E}_m(\mathbb{T}) \leq \widehat{\mathcal{E}}_m(L) := 4(m+1)\,\mathfrak{g}_m(L)\Big(1 + \sum_{n\geq 2}(n+1)\,n^m\, e^{-(n-1)L^2/4}\Big) \xrightarrow[L\to\infty]{} 0,$$

and the same bound holds for $\mathcal{E}_m(L)$.

(c) For every $m \geq 0$ and $|\beta| = r/\sqrt{2}$, $|\langle m|\mathcal{D}(\beta)|m\rangle| \leq \min\{1, \mathfrak{g}_m(r)\}$; $\mathfrak{g}_m(r)$ is nondecreasing in m, and $\min\{1, \mathfrak{g}_z(r)\}$ is nonincreasing in r. Hence, for every rectangular torus whose shorter period is at least $L > 0$,

$$\sigma_z(\mathbb{T}) := \sup_R \max_{m\leq z}(\Gamma(R)_{mm} - 1) \leq \Xi_z(L) := \sum_{(j,k)\in\mathbb{Z}^2\setminus 0} \min\{1, \mathfrak{g}_z(L\sqrt{j^2+k^2})\}.$$

Proof. (a) Take $\beta = r/\sqrt{2} > 0$ and put $x := \beta^2 = r^2/2$, so that $r \geq 16$ means $x \geq 128$. By Lemma 2.4(iii), for $a, b \leq m$ the entry $\langle a|\mathcal{D}(\beta)|b\rangle$ equals $\pm\sqrt{n!/(n+\alpha)!}\, x^{\alpha/2} e^{-x/2} \mathrm{L}_n^{(\alpha)}(x)$ with $n = \min(a,b)$, $\alpha = |a-b|$, and the sign $(-1)^{b-a}$ if $a < b$. The monic polynomials $p_i := (-1)^i i!\, \mathrm{L}_i^{(\alpha)}$ satisfy $p_{i+1} = (x - (2i+\alpha+1))p_i - i(i+\alpha)p_{i-1}$, so, expanding a determinant along its last row, $p_n(x) = \det(x - M_n)$ for the symmetric tridiagonal matrix M_n with diagonal entries $2i+\alpha+1$ $(0 \leq i < n)$ and off-diagonal entries $\sqrt{i(i+\alpha)}$ $(1 \leq i < n)$. Hence the zeros $x_1, \dots, x_n$ of $\mathrm{L}_n^{(\alpha)}$ are real; they are positive (for $x \leq 0$ all terms of $\mathrm{L}_n^{(\alpha)}(x)$ have the same sign), and by Gershgorin's theorem and $\sqrt{i(i+\alpha)} \leq i + \alpha/2$ they are at most $\max_i(4i+2\alpha+2) \leq 4n+2\alpha-2 = 2a+2b-2 \leq 90$.

So for $x \geq 128$ the polynomial $\mathrm{L}_n^{(\alpha)}(x)$ has the sign $(-1)^n$ of its leading coefficient, and every entry of the column b of $\mathcal{D}^{(m)}(\beta)$ has the sign $(-1)^b$; thus $\mathcal{D}^{(m)}(\beta) = |\mathcal{D}^{(m)}(\beta)|\,\mathrm{diag}((-1)^b)$ with the entrywise modulus $|\mathcal{D}^{(m)}(\beta)|$, and $\mu_m(r) = \||\mathcal{D}^{(m)}(\beta)|\|$. The logarithmic derivative of the modulus of an entry is $\frac{\alpha}{2x} - \frac{1}{2} + \sum_i \frac{1}{x-x_i}$. By convexity of $t \mapsto 1/(x-t)$ on $[0, 90]$, $\frac{1}{x-t} \leq \frac{1}{x} + \frac{t}{x(x-90)}$, and $\sum_i x_i = \mathrm{tr}\, M_n = n(n+\alpha)$; hence for $x \geq 128$

$$\frac{\alpha}{2x} + \sum_i \frac{1}{x - x_i} \leq \frac{\alpha + 2n}{2x} + \frac{n(n+\alpha)}{x(x-90)} \leq \frac{46}{256} + \frac{529}{128 \cdot 38} < 0.29 < \frac{1}{2}.$$

So every entry of $|\mathcal{D}^{(m)}(\beta)|$ is nonincreasing in $r \geq 16$. For entrywise nonnegative matrices the spectral norm is $\max\{v^{\mathsf{T}} A u : u, v \geq 0,\ \|u\| = \|v\| = 1\}$, which is monotone in the entries. For a torus with periods $L_x, L_y \geq L \geq 16$, $|(jL_x, kL_y)| \geq L\sqrt{j^2+k^2} \geq 16$ for $(j,k) \neq 0$, which gives the bound and the monotonicity in L.

(b) For $a \geq b$ write $n = b$, $\alpha = a - b$. The coefficients of $\mathrm{L}_n^{(\alpha)}(-x) = \sum_i \binom{n+\alpha}{n-i} x^i/i!$ are dominated by those of $(x+n+\alpha)^n/n!$, since $(n+\alpha)!/(\alpha+i)! \leq (n+\alpha)^{n-i}$; and $|\mathrm{L}_n^{(\alpha)}(x)| \leq \mathrm{L}_n^{(\alpha)}(-x)$ for $x \geq 0$ termwise. Hence the entry is at most $\sqrt{b!/a!}\, x^{(a-b)/2} e^{-x/2} (x+a)^b/b! \leq (x+m)^{(a+b)/2} e^{-x/2}/\sqrt{a!b!}$, which is nondecreasing in a and in b as long as $x + m \geq a + 1$, so at most $(x+m)^m e^{-x/2}/m! = \mathfrak{g}_m(r)$. The spectral norm of an $(m+1) \times (m+1)$ matrix is at most $m+1$ times its largest entry. $\frac{\mathrm{d}}{\mathrm{d}r} \log \mathfrak{g}_m(r) = r(\frac{m}{r^2/2+m} - \frac{1}{2}) < 0$ for $r^2 > 2m$; so $\mu_m(|(jL_x, kL_y)|) \leq (m+1)\, \mathfrak{g}_m(L\sqrt{j^2+k^2})$ when $L^2 \geq 2m$. Group the lattice points by $n = j^2 + k^2$: there are 4 with $n = 1$ and at most $2(2\sqrt{n}+1) \leq 4(n+1)$ for each $n \geq 2$, and $\mathfrak{g}_m(L\sqrt{n}) \leq n^m e^{-(n-1)L^2/4} \mathfrak{g}_m(L)$.

(c) $|\langle m|\mathcal{D}(\beta)|m\rangle| \leq 1$ by unitarity, and $= e^{-x/2}|\mathrm{L}_m^{(0)}(x)| \leq e^{-x/2}\mathrm{L}_m^{(0)}(-x) \leq \mathfrak{g}_m(r)$ by (b) with $\alpha = 0$. Next $\mathfrak{g}_{m+1}/\mathfrak{g}_m \geq (x+m+1)/(m+1) \geq 1$. Finally $\frac{\mathrm{d}}{\mathrm{d}r} \log \mathfrak{g}_z > 0$ only for $r^2 < 2z$, and there $\mathfrak{g}_z(r) \geq \mathfrak{g}_z(0) = z^z/z! \geq 1$; so $\min\{1, \mathfrak{g}_z\}$ equals 1 on $[0, \sqrt{2z}]$ and is decreasing beyond. With Lemma 5.2 and $|(jL_x, kL_y)| \geq L\sqrt{j^2+k^2}$ this gives the bound on σ_z. □

Remark 5.4. The bound (b) is far from sharp near the thresholds of the main theorem. For $m = 22$ and $m = 23$, the indices of the binding four-body term, it gives $\widehat{\mathcal{E}}_{22}(L) \approx 9.82,\ 0.0255$ and $3.6 \cdot 10^{-5}$ and $\widehat{\mathcal{E}}_{23}(L) \approx 77.9,\ 0.221$ and $3.4 \cdot 10^{-4}$ at $L = 16$, 17 and 18, against the values $\mathcal{E}_{22}(17) = 1.13 \cdot 10^{-6}$ and $\mathcal{E}_{23}(17) = 4.73 \cdot 10^{-6}$ computed from the exact norms (Table 1). It is used only for the limit $L \to \infty$ and its rate; the numerical values in Section 9 use the exact norms μ_m and part (a).

6 The torus Moyal identities and the comparison operator

On the sphere the comparison operator is averaged over rotations, and Schur's lemma for $SU(2)$ reduces it to fixed spin blocks. On the torus we average the base point R of the local modes over the torus. The magnetic translations of the centre-of-mass torus then play the role of the rotations, and the resulting identities are exact.

6.1 Magnetic translations of the centre-of-mass torus

Lemma 6.1. *Let $\mathbb{T}'$ be a rectangular torus with periods L'_x, L'_y, lattice Λ', $M = L'_x L'_y/2\pi \in \mathbb{N}$ flux quanta and lowest Landau level $\mathfrak{u}$, and let Π' be its periodization. For $\delta \in \Lambda'/M$, t_δ is unitary on $\mathfrak{u}$ and $t_\delta \Pi' f = \Pi' t_\delta f$ for every Gaussian-decaying LLL function f. An operator on $\mathfrak{u}$ that commutes with all t_δ, $\delta \in \Lambda'/M$, is a multiple of the identity.*

Proof. The first statement is Lemma 2.3(iv) for $\mathbb{T}'$, and $t_\delta \tilde{t}'_{\gamma'} = e^{\mathrm{i}\,\delta\wedge\gamma'}\tilde{t}'_{\gamma'}t_\delta$ with $\delta\wedge\gamma' \in 2\pi\mathbb{Z}$. Let C commute with $\theta'_y := t_{(0,-L'_y/M)}$ and $\theta'_x := t_{(L'_x/M,0)}$. By Lemma 2.3(iv), θ'_y has the M distinct eigenvalues $e^{2\pi\mathrm{i}j/M}$ on the orbitals ψ'_j, so $C\psi'_j = c_j\psi'_j$, and $\theta'_x\psi'_j = \psi'_{j+1}$ forces $c_{j+1} = c_j$. □

6.2 The Moyal identity

For $k \geq 1$ a *relative state* is a function a of the relative Jacobi coordinates $\rho \in (\mathbb{R}^2)^{k-1}$ which is an LLL function of each of them and decays like a Gaussian. For $R \in \mathbb{R}^2$ and $p \geq 0$ we write $a \otimes \varphi_p^R$ for the planar k-particle function $(\zeta,\rho) \mapsto a(\rho)\,\eta_p^{\sqrt{k}R}(\zeta)$: relative state a, centre-of-mass orbital of degree p about the point R (the centre of mass $k^{-1}\sum r_i$ equals $\zeta/\sqrt{k}$). By Lemma 2.7(i) it is an LLL function of each particle and decays like a Gaussian about R in all variables, so $\Pi^{(k)}(a\otimes\varphi_p^R) \in \mathfrak{h}^{\otimes k}$.

Theorem 6.2 (torus Moyal identity)**.** *Let $k \geq 1$, let a, b be relative states and $p, q \geq 0$. The operator $|\Pi^{(k)}(a\otimes\varphi_p^R)\rangle\langle\Pi^{(k)}(b\otimes\varphi_q^R)|$ on $\mathfrak{h}^{\otimes k}$ is a Λ-periodic continuous function of R, and*

$$\frac{k}{2\pi}\int_{\mathsf{C}} |\Pi^{(k)}(a\otimes\varphi_p^R)\rangle\langle\Pi^{(k)}(b\otimes\varphi_q^R)|\,\mathrm{d}R = \delta_{pq}\,\mathfrak{P}_k\big(|a\rangle\langle b|\otimes\mathbf{1}\big) = \delta_{pq}\,\mathcal{I}_a^*\mathcal{I}_b. \tag{6.1}$$

Proof. Let Π' be the periodization of the centre-of-mass torus $\mathbb{T}_k = \mathbb{R}^2/\sqrt{k}\Lambda$. For $\Psi \in \mathfrak{h}^{\otimes k}$, by Lemma 2.5 on $\mathbb{T}$, Fubini in Jacobi coordinates, and Lemma 2.5 on $\mathbb{T}_k$ (recall $\mathcal{I}_b\Psi \in \mathfrak{u}_k$),

$$\langle\Pi^{(k)}(b\otimes\varphi_q^R),\Psi\rangle = \int_{(\mathbb{R}^2)^k} \overline{b(\rho)\,\eta_q^{\sqrt{k}R}(\zeta)}\,\Psi = \int_{\mathbb{R}^2}\overline{\eta_q^{\sqrt{k}R}}\,\mathcal{I}_b\Psi = \langle\Pi'\eta_q^{\sqrt{k}R},\mathcal{I}_b\Psi\rangle_{\mathfrak{u}_k}.$$

Hence, as a form in (Φ,Ψ), the left side of (6.1) is $\langle\mathcal{I}_a\Phi,\mathcal{Q}_{pq}\mathcal{I}_b\Psi\rangle$ with

$$\mathcal{Q}_{pq} := \frac{1}{2\pi}\int_{\sqrt{k}\mathsf{C}} |\Pi'\eta_p^{\zeta_0}\rangle\langle\Pi'\eta_q^{\zeta_0}|\,\mathrm{d}\zeta_0$$

(substitute $\zeta_0 = \sqrt{k}R$). *Periodicity.* For $\gamma' \in \sqrt{k}\Lambda$, $\eta_p^{\zeta_0+\gamma'}$ is a p-independent phase times $t_{\gamma'}\eta_p^{\zeta_0}$ (by (2.4) and Lemma 2.3(ii)), and $\Pi' t_{\gamma'} = \pm\Pi'$; so the integrand is periodic, which also proves the first assertion. *Schur.* For $\delta \in \sqrt{k}\Lambda/(kN_\phi)$, $t_\delta\eta_p^{\zeta_0}$ is a p-independent phase times $\eta_p^{\zeta_0+\delta}$ and t_δ commutes with Π' (Lemma 6.1); so $t_\delta\mathcal{Q}_{pq}t_\delta^*$ is the integral of the translated periodic integrand over a period cell, which is $\mathcal{Q}_{pq}$. By Lemma 6.1, $\mathcal{Q}_{pq} = c_{pq}\mathbf{1}$ on $\mathfrak{u}_k$. *The scalar.* $c_{pq} = \operatorname{tr}\mathcal{Q}_{pq}/(kN_\phi)$ and $\operatorname{tr}|\Pi'\eta_p^{\zeta_0}\rangle\langle\Pi'\eta_q^{\zeta_0}| = \langle\Pi'\eta_q^{\zeta_0},\Pi'\eta_p^{\zeta_0}\rangle = \sum_{\gamma'\in\sqrt{k}\Lambda}\varsigma'_{\gamma'}e^{\mathrm{i}\,\gamma'\wedge\zeta_0}\langle q|\mathcal{D}(\beta(\gamma'))|p\rangle$, exactly as in Lemma 5.2, the series converging absolutely. For $\gamma' = (j\sqrt{k}L_x, l\sqrt{k}L_y) \neq 0$ the integral of $e^{\mathrm{i}\,\gamma'\wedge\zeta_0} = e^{\mathrm{i}jk L_x\zeta_{0,y}}e^{-\mathrm{i}lkL_y\zeta_{0,x}}$ over $\sqrt{k}\mathsf{C}$ vanishes, because $\sqrt{k}L_x\cdot\sqrt{k}L_y = 2\pi kN_\phi$; the term $\gamma' = 0$ gives $|\sqrt{k}\mathsf{C}|\,\delta_{pq} = 2\pi kN_\phi\,\delta_{pq}$. Hence $c_{pq} = \delta_{pq}$, and Lemma 2.9(c) gives the second equality. □

The identity holds for every k with the normalization built into the definition of $\mathfrak{P}_k$. For $k = 2$ the normalization of h itself was identified in Proposition 2.11, so no further normalization step is needed.

Corollary 6.3 (pair level)**.** *For $p, q \geq 0$, $\frac{1}{\pi}\int_{\mathsf{C}}|\tilde{u}_p(R)\rangle\langle\tilde{u}_q(R)|\mathrm{d}R = \delta_{pq}\,h$. Consequently $\tilde{u}_p(R) \in V_{\mathbb{T}}$ for every p and R, and*

$$\frac{1}{\pi}\int_{\mathsf{C}} B(\tilde{u}_p(R))^*B(\tilde{u}_p(R))\,\mathrm{d}R = H_{\mathbb{T}} \qquad \text{for every } p \geq 0.$$

Proof. Take $k = 2$ and $a = b = \eta_1$; then $\Pi^{(2)}(\eta_1\otimes\varphi_p^R) = \tilde{u}_p(R)$ and $\mathfrak{P}_2(|\eta_1\rangle\langle\eta_1|\otimes\mathbf{1}) = h$. If $w \perp V_{\mathbb{T}}$, then $\int_{\mathsf{C}}|\langle w,\tilde{u}_p(R)\rangle|^2\mathrm{d}R = \pi\langle w,hw\rangle = 0$, and $R \mapsto \langle w,\tilde{u}_p(R)\rangle$ is continuous, so it vanishes. The last identity follows by applying the linear map $\mathcal{L}_2$ (Lemma 2.6) to the first, the integrand being continuous with values in a finite-dimensional space. □

6.3 Planar coupled bases

The certificate of [1] is written in terms of coupling coefficients. For $0 < u < 1$, $v := \sqrt{1-u^2}$ and integers $0 \le z \le T$, $0 \le p \le T$, the coefficients $s^{(u)}_{z,T,p}$ are defined by the polynomial identity

$$\frac{(uX - vY)^z (vX + uY)^{T-z}}{\sqrt{z!(T-z)!}} = \sum_{p=0}^{T} s^{(u)}_{z,T,p} \, \frac{X^p Y^{T-p}}{\sqrt{p!(T-p)!}} \tag{6.2}$$

[1, Lemma 3.3 (Fixed-deficit coupling limit)], and are zero for indices outside these ranges. The substitution $(X, Y) \mapsto (uX - vY, vX + uY)$ is orthogonal, hence unitary for the Bargmann inner product, so for fixed T the real matrix $(s^{(u)}_{z,T,p})_{z,p}$ is orthogonal.

Three particles. Use the Jacobi coordinates $\rho_1 = (r_1 - r_2)/\sqrt{2}$, $\rho_2 = (r_1 + r_2 - 2r_3)/\sqrt{6}$ and $\zeta = (r_1 + r_2 + r_3)/\sqrt{3}$, and put $s_{z,T,p} := s^{(1/\sqrt{3})}_{z,T,p}$,

$$\xi_z(\rho) := \eta_1(\rho_1)\, \eta_z(\rho_2), \qquad |z, T\rangle^R := \xi_z \otimes \varphi^R_{T-z} \quad (0 \le z \le T).$$

Permutations of the particles act on ρ only (Lemma 2.7(iii)); we write $W_3 \xi_z := \sqrt{3} A_3 \xi_z$, so that $W_3 |z, T\rangle^R = (W_3 \xi_z) \otimes \varphi^R_{T-z}$.

Lemma 6.4. *For $p, j \ge 0$ and $R \in \mathbb{R}^2$, $v_p^{*R} \otimes \eta_j^R = \sum_{z=0}^{p+j} s_{z,p+j,p} |z, p+j\rangle^R$, where v_p^{*R} acts on the particles $1, 2$ and η_j^R on particle 3.*

Proof. At $R = 0$, in the Bargmann form, $v_p^{*0} \otimes \eta_j$ is $w\, X^p Y^j / \sqrt{p!j!}$ with $w = (z_1 - z_2)/\sqrt{2}$, $X = (z_1 + z_2)/\sqrt{2}$, $Y = z_3$. For $u = 1/\sqrt{3}$, $uX - vY = (z_1 + z_2 - 2z_3)/\sqrt{6}$ and $vX + uY = (z_1 + z_2 + z_3)/\sqrt{3}$ are the Bargmann forms of ρ_2 and ζ, so $|z, T\rangle^0$ is w times the left side of (6.2); invert the orthogonal matrix. For general R apply $t_R^{\otimes 3}$, which acts as $t_{\sqrt{3}R}$ on ζ and trivially on ρ and multiplies both sides by $e^{3\mathrm{i} R_x R_y/2}$. □

Four particles. Use $\rho_1 = (r_1 - r_2)/\sqrt{2}$, $\rho_2 = (r_3 - r_4)/\sqrt{2}$, $\rho_3 = (r_1 + r_2 - r_3 - r_4)/2$ and $\zeta = (r_1 + r_2 + r_3 + r_4)/2$. For odd r and $D \ge r$ let

$$\omega_{D,r}(\rho) := \eta_1(\rho_1)\, \eta_r(\rho_2)\, \eta_{D-r}(\rho_3), \qquad |D, T; r\rangle^R := \omega_{D,r} \otimes \varphi^R_{T-D} \quad (T \ge D),$$

and, following [1, Section 4.2, (23)], $S_r(D, T; p, j, k) := \sqrt{2}\, s^{(1/\sqrt{2})}_{r,j+k,j}\, s^{(1/\sqrt{2})}_{D-r,T-r,p}$, interpreted as zero if $r > j + k$, for $p + j + k = T$. Let $\mathcal{R}_D$ be the set of odd integers in $[1, D]$.

Lemma 6.5. *For $p \ge 0$ and $j \ne k$, with $T = p + j + k$, $S_r(D, T; p, j, k) = -S_r(D, T; p, k, j)$ and $v_p^{*R} \otimes (\eta_j^R \wedge \eta_k^R) = \sum_{D \le T} \sum_{r \in \mathcal{R}_D} S_r(D, T; p, j, k) |D, T; r\rangle^R$, with v_p^{*R} on the particles $1, 2$ and $\eta_j^R \wedge \eta_k^R$ on $3, 4$. The vectors $|D, T; r\rangle^R$, $r \in \mathcal{R}_D$, $D \le T$, are orthonormal; in particular, for $b = (p, j, k)$ with $p + j + k = D$ and $j < k$, the matrix $S_{rb} := S_r(D, D; b)$ satisfies $SS^{\mathsf{T}} = \mathbf{1}$.*

Proof. At $R = 0$, in the Bargmann form, $\eta_j \wedge \eta_k$ on the particles $3, 4$ is $(z_3^j z_4^k - z_3^k z_4^j)/\sqrt{2j!k!}$. By (6.2) with $u = v = 1/\sqrt{2}$, $X = z_3$, $Y = z_4$ and $m = j + k$, the normalized monomial $z_3^j z_4^{m-j}/\sqrt{j!(m-j)!}$ is $\sum_r s^{(1/\sqrt{2})}_{r,m,j}\, \omega_{34}^r \Omega_{34}^{m-r} / \sqrt{r!(m-r)!}$ with $\omega_{34} = (z_3 - z_4)/\sqrt{2}$, $\Omega_{34} = (z_3 + z_4)/\sqrt{2}$; exchanging X and Y multiplies the left side of (6.2) by $(-1)^r$, so $s^{(1/\sqrt{2})}_{r,m,m-j} = (-1)^r s^{(1/\sqrt{2})}_{r,m,j}$, and the antisymmetric combination keeps the odd r with the coefficient $\sqrt{2}\, s^{(1/\sqrt{2})}_{r,m,j}$. Next couple the pair centres $X' =$

$(z_1+z_2)/\sqrt{2}$ (degree p) and $Y'=\Omega_{34}$ (degree $m-r$) by (6.2) with $u=1/\sqrt{2}$: the relative variable $(X'-Y')/\sqrt{2}$ is the Bargmann form of ρ_3 and $(X'+Y')/\sqrt{2}$ that of ζ, and the relative degree $d=D-r$ comes with the coefficient $s^{(1/\sqrt{2})}_{D-r,T-r,p}$. With the factor $w=(z_1-z_2)/\sqrt{2}$ of v_p^{*0} this is the expansion. The vectors $|D,T;r\rangle^0$ are products of normalized monomials in the orthonormal Jacobi coordinates, hence orthonormal; translate by $t_R^{\otimes 4}$ for general R. The antisymmetry in (j,k) was shown above. Finally, at fixed T both $\{v_p^{*R}\otimes(\eta_j^R\wedge\eta_k^R): p+j+k=T,\ j<k\}$ and $\{|D,T;r\rangle^R : r\in\mathcal{R}_D,\ D\le T\}$ are orthonormal bases of the same space (in the Bargmann form: w times the polynomials of degree T in the remaining variables that are odd under $z_3\leftrightarrow z_4$), so the change of basis is orthogonal; its rows with $D=T$ are the rows of S, whence $SS^{\mathsf{T}}=\mathbf{1}$. □

We write $W_4\omega_{D,r}:=\sqrt{6}A_4\,\omega_{D,r}$; then $W_4|D,T;r\rangle^R=(W_4\omega_{D,r})\otimes\varphi^R_{T-D}$, and $W_4|D,D;r\rangle^0$ is the vector w^*_{Dr} of [1, Section 4.2, (27)].

6.4 The certificate

The following data and inequalities are taken from [1, Section 4 and Appendix A]; the exact rows are reproduced in Appendix A. There are seven *rows* $\nu=1,\dots,7$, each with an integer $t_\nu\in\{0,\dots,7\}$, a real λ_ν and real coefficients α^ν_{pj} supported in

$$\mathcal{S}_\nu:=\{(p,j)\in\mathbb{Z}^2: 0\le p\le 7,\ 0\le j\le 8,\ 0\le p+j-t_\nu\le 8\}.$$

Put $\eta_*:=\sum_\nu\lambda_\nu^2=93527408868499/10^{14}$ and $\varepsilon:=3\cdot10^{-6}$. For $z\ge0$ define

$$e_z:=\sum_\nu\sum_{T=\max(z,t_\nu)}^{15}\Big[2\lambda_\nu\, s_{z,T,t_\nu}\sum_{p+j=T}\alpha^\nu_{pj}s_{z,T,p}+\Big(\sum_{p+j=T}\alpha^\nu_{pj}s_{z,T,p}\Big)^2\Big], \tag{6.3}$$

so that $e_z=0$ for $z\ge16$, and for $1\le D\le23$ and $r,s\in\mathcal{R}_D$

$$(E_D)_{rs}:=-\sum_\nu\sum_{\substack{(p,j),(q,l)\in\mathcal{S}_\nu\\ T\ge D}}\alpha^\nu_{pj}\alpha^\nu_{ql}\,S_r(D,T;p,j,k)\,S_s(D,T;q,l,i), \tag{6.4}$$

where $i=p+j-t_\nu$, $k=q+l-t_\nu$ and $T=p+j+k$. Let $(e_r)_{r\in\mathcal{R}_D}$ be the standard basis of $\mathbb{C}^{\mathcal{R}_D}$, so that $|e_1\rangle\langle e_1|$ is the matrix with $(1,1)$ entry 1 and all other entries zero, and let G_D be the Gram matrix of the vectors $W_4\omega_{D,r}$, $r\in\mathcal{R}_D$. The certificate consists of the inequalities

$$e_z<\tfrac{3}{2}(3z-1)(-\tfrac12)^z\qquad\text{for } z\in\{2,4,5,\dots,15\}, \tag{6.5}$$

$$G_D\,(2|e_1\rangle\langle e_1|-E_D+\varepsilon\mathbf{1})\,G_D\ge0\qquad\text{for } 1\le D\le23, \tag{6.6}$$

which are proved in [1, Appendix A] by exact rational arithmetic. We use them as data.

Lemma 6.6 (compression; [1, Lemma 5.1 (Compression and annihilator bounds)])**.** *Let W be a linear map between finite-dimensional Hilbert spaces, $G=W^*W$, and C self-adjoint on the domain of W. Then $GCG\ge0$ if and only if $WCW^*\ge0$. In particular (6.6) is equivalent to $\sum_{r,s\in\mathcal{R}_D}(2|e_1\rangle\langle e_1|-E_D+\varepsilon\mathbf{1})_{rs}|W_4\omega_{D,r}\rangle\langle W_4\omega_{D,s}|\ge0$.*

Proof. Both conditions say that the quadratic form of C is nonnegative on $\operatorname{ran}G=\operatorname{ran}W^*$. Apply this to the map $e_r\mapsto W_4\omega_{D,r}$. □

6.5 The comparison operator and its parts

Definition 6.7 (the comparison operator)**.** For $R \in \mathbb{R}^2$ and each row ν let

$$F_\nu(R) := \lambda_\nu\, B(\tilde u_{t_\nu}(R)) + \sum_{(p,j)\in\mathcal{S}_\nu} \alpha^\nu_{pj}\, c(\chi^R_{p+j-t_\nu})^*\, c(\chi^R_j)\, B(\tilde u_p(R)),$$

$$K_{\mathbb{T}} := \frac{1}{\pi}\int_{\mathsf{C}} \sum_\nu F_\nu(R)^* F_\nu(R)\,\mathrm{d}R.$$

By Lemma 5.1(c), $F_\nu(R+\gamma) = e^{-2\mathrm{i}\theta} F_\nu(R)$ for $\gamma \in \Lambda$, so the integrand is a function on $\mathbb{T}$. Clearly $K_{\mathbb{T}} \geq 0$; by Corollary 6.3, every summand of $F_\nu(R)$ annihilates $\ker H_{\mathbb{T}}$.

These are the rows of [1, Section 4, (14)] with every planar mode replaced by its periodization; each summand removes two particles and lowers the total orbital degree by $t_\nu + 1$.

Proposition 6.8 (the parts of $K_{\mathbb{T}}$)**.** *Let* $Y_z := |W_3\xi_z\rangle\langle W_3\xi_z| \otimes \mathbf{1}$ *and* $C_3 := 72\sum_\nu \|\alpha^\nu\|^2$*, where* $\|\alpha^\nu\|^2 = \sum_{p,j}(\alpha^\nu_{pj})^2$*. Then* $K_{\mathbb{T}} = K^{(2)}_{\mathbb{T}} + K^{(3)}_{\mathbb{T}} + K^{(4)}_{\mathbb{T}}$ *with*

(a) $K^{(2)}_{\mathbb{T}} = \eta_*\, H_{\mathbb{T}}$*;*

(b) $K^{(3)}_{\mathbb{T}} = \frac{2}{3}\mathcal{L}_3\mathfrak{P}_3\big(\sum_{z\leq 15} e_z Y_z\big) + \mathcal{C}_3$ *with* $\pm\mathcal{C}_3 \leq C_3\,\mathcal{E}_8(\mathbb{T})(1+\mathcal{E}_8(\mathbb{T}))H_{\mathbb{T}}$*;*

(c) $K^{(4)}_{\mathbb{T}} = \frac{1}{2}\mathcal{L}_4\mathfrak{P}_4\big(\sum_{D=1}^{23}\sum_{r,s\in\mathcal{R}_D}(E_D)_{rs}|W_4\omega_{D,r}\rangle\langle W_4\omega_{D,s}| \otimes \mathbf{1}\big)$*;*

(d) with $\mathcal{T}_4 := \mathcal{L}_4\mathfrak{P}_4\big(\sum_{D=1}^{23}|W_4\omega_{D,1}\rangle\langle W_4\omega_{D,1}| \otimes \mathbf{1}\big)$*,*

$$\tfrac{1}{2}\mathcal{L}_4\mathfrak{P}_4\Big(\sum_{D=1}^{23}\sum_{r\in\mathcal{R}_D}|W_4\omega_{D,r}\rangle\langle W_4\omega_{D,r}| \otimes \mathbf{1}\Big) \leq 1222\,(1+\mathcal{E}_{23})^2 H_{\mathbb{T}}, \qquad \mathcal{T}_4 \leq 2444\,(1+\mathcal{E}_{23})^2 H_{\mathbb{T}},$$

with $\mathcal{E}_{23} = \mathcal{E}_{23}(\mathbb{T})$*, and* $K^{(4)}_{\mathbb{T}} \leq \mathcal{T}_4 + 1222\,\varepsilon\,(1+\mathcal{E}_{23})^2 H_{\mathbb{T}}$*.*

Proof. Fix R and a row; write $t = t_\nu$, $\alpha_b = \alpha^\nu_{pj}$ for $b = (p,j)$, $i = i(b) := p + j - t$, $c_m := c(\chi^R_m)$, $B_p := B(\tilde u_p(R))$ and $\Gamma = \Gamma(R)$. Since $\{c(f), c(g)^*\} = \langle f, g\rangle$, $c_i c_k^* = \Gamma_{ik} - c_k^* c_i$, and normal ordering gives

$$F^*F = \lambda^2 B_t^* B_t + \sum_b \lambda\alpha_b\big[(c_i B_t)^* c_j B_p + \text{h.c.}\big] + \sum_{b,b'}\alpha_b\alpha_{b'}\Gamma_{ik}(c_j B_p)^* c_l B_q - \sum_{b,b'}\alpha_b\alpha_{b'} O^*_{p,j,k} O_{q,l,i},$$

with $b' = (q,l)$, $k = i(b')$ and $O_{p,j,k} := c_k c_j B_p$. By (2.7), $c_j B_p = B(\tilde u_p \wedge \chi_j)$ and $O_{p,j,k} = B(\tilde u_p \wedge \chi_j \wedge \chi_k)$, and by Lemma 5.1(b), $\tilde u_p(R) \wedge \chi^R_j = \Pi^{(3)} W_3(v_p^{*R} \otimes \eta^R_j)$ and $\tilde u_p \wedge \chi_j \wedge \chi_k = \Pi^{(4)} W_4(v_p^{*R} \otimes (\eta^R_j \wedge \eta^R_k))$. Let $K^{(2)}_{\mathbb{T}}$, $K^{(3)}_{\mathbb{T}}$, $K^{(4)}_{\mathbb{T}}$ be $\frac{1}{\pi}\int_{\mathsf{C}}$ of the sums over the rows of the two-, three- and four-body terms, and $\mathcal{C}_3$ the part of $K^{(3)}_{\mathbb{T}}$ with $\Gamma_{ik} - \delta_{ik}$ in place of Γ_{ik}.

(a) is Corollary 6.3 with $p = t$.

(b) For a finite-rank planar operator $X = \sum_\mu |x_\mu\rangle\langle y_\mu|$ with Gaussian-decaying x_μ, y_μ write $\Pi^{(3)}[X] := \sum_\mu |\Pi^{(3)} x_\mu\rangle\langle \Pi^{(3)} y_\mu|$. With δ_{ik} in place of Γ_{ik}, the three-body terms summed over the rows are $\mathcal{L}_3(\Pi^{(3)}[W_3 M_3(R) W_3^*])$, where

$$M_3(R) := \sum_\nu\Big[\lambda_\nu \sum_b \alpha_b(|v_t^{*R} \otimes \eta^R_i\rangle\langle v_p^{*R} \otimes \eta^R_j| + \text{h.c.}) + \sum_{b,b':\, i=k}\alpha_b\alpha_{b'}|v_p^{*R} \otimes \eta^R_j\rangle\langle v_q^{*R} \otimes \eta^R_l|\Big];$$

we expand $M_3(R)$ with Lemma 6.4. Every term has equal total degree on both sides ($t+i=p+j$, and $p+j=q+l$ when $i=k$), so $M_3(R)=\sum m_{(z,c),(z',c')}|\xi_z\otimes\varphi_c^R\rangle\langle\xi_{z'}\otimes\varphi_{c'}^R|$ with real coefficients independent of R and supported on $z+c=z'+c'\le 15$. By Theorem 6.2 with $k=3$ and $\frac{1}{\pi}=\frac{2}{3}\cdot\frac{3}{2\pi}$, only $c=c'$, hence $z=z'$, survives, and the result is $\frac{2}{3}\mathcal{L}_3\mathfrak{P}_3(\sum_z[\sum_c m_{(z,c),(z,c)}]Y_z)$. Finally $\sum_c m_{(z,c),(z,c)}=\sum_{T\ge z}\langle|z,T\rangle^0,M_3(0)|z,T\rangle^0\rangle$, and since $\langle|z,T\rangle^0,v_p^{*0}\otimes\eta_j\rangle=s_{z,T,p}$ for $p+j=T$ (Lemma 6.4), this is (6.3). For $\mathcal{C}_3$, let $x_b:=c_jB_p\Psi$ and $M_{bb'}:=\alpha_b\alpha_{b'}(\Gamma-\mathbf{1})_{ik}$. Only the modes $0,\dots,8$ occur in (i,k), so $\|M\|\le\|M\|_{\mathrm{HS}}\le\mathcal{E}_8\|\alpha^\nu\|^2$ by Lemma 5.2, and since $[\langle x_b,x_{b'}\rangle]\ge 0$, $|\sum M_{bb'}\langle x_b,x_{b'}\rangle|\le\|M\|\sum_b\|x_b\|^2$. Next $\|c_jB_p\Psi\|^2\le\Gamma_{jj}\|B_p\Psi\|^2\le(1+\mathcal{E}_8)\|B_p\Psi\|^2$, each $p\le 7$ occurs with at most nine values of j, and $\frac{1}{\pi}\int_{\mathsf{C}}\sum_{p\le 7}\|B(\tilde{u}_p(R))\Psi\|^2\mathrm{d}R=8\langle\Psi,H_{\mathbb{T}}\Psi\rangle$ by Corollary 6.3. Summing over the rows gives $|\langle\Psi,\mathcal{C}_3\Psi\rangle|\le 72\,\mathcal{E}_8(1+\mathcal{E}_8)\sum_\nu\|\alpha^\nu\|^2\langle\Psi,H_{\mathbb{T}}\Psi\rangle$.

(c) The four-body terms summed over the rows are $\mathcal{L}_4(\Pi^{(4)}[W_4M_4(R)W_4^*])$, with the notation of (b) and

$$M_4(R):=-\sum_\nu\sum_{b,b'}\alpha_b\alpha_{b'}|v_p^{*R}\otimes(\eta_j^R\wedge\eta_k^R)\rangle\langle v_q^{*R}\otimes(\eta_l^R\wedge\eta_i^R)|.$$

Expanding with Lemma 6.5, both sides have total degree $T=p+j+k=q+l+i\le 23$; Theorem 6.2 with $k=4$ and $\frac{1}{\pi}=\frac{1}{2}\cdot\frac{4}{2\pi}$ keeps the equal centre-of-mass degrees, hence equal D, with the copy matrix $\sum_{T\ge D}\langle|D,T;r\rangle^0,M_4(0)|D,T;s\rangle^0\rangle$, which is (6.4) because $\langle|D,T;r\rangle^0,v_p^{*0}\otimes(\eta_j\wedge\eta_k)\rangle=S_r(D,T;p,j,k)$.

(d) By Theorem 6.2 with $k=4$ and $p=q=0$,

$$\mathcal{L}_4\mathfrak{P}_4(|W_4\omega_{D,r}\rangle\langle W_4\omega_{D,r}|\otimes\mathbf{1})=\frac{2}{\pi}\int_{\mathsf{C}}B(x_{D,r}(R))^*B(x_{D,r}(R))\,\mathrm{d}R,$$
$$x_{D,r}(R):=\Pi^{(4)}W_4|D,D;r\rangle^R=\sum_b S_{rb}\,\tilde{u}_p(R)\wedge\chi_j^R\wedge\chi_k^R,$$

with $b=(p,j,k)$, $p+j+k=D$, $j<k$ (Lemma 6.5; S is real). Hence $B(x_{D,r})=\sum_b S_{rb}O_b$ with $O_b:=c_kc_jB_p$, and since $SS^{\mathsf{T}}=\mathbf{1}$,

$$\sum_r\|B(x_{D,r})\Psi\|^2\le\sum_b\|O_b\Psi\|^2\le(1+\mathcal{E}_{23})^2\sum_b\|B_p\Psi\|^2,$$

the mode indices $j<k\le D$ being at most 23. For the single vector $r=1$, Cauchy–Schwarz with $\sum_b S_{1b}^2=1$ gives the same bound for $\|B(x_{D,1})\Psi\|^2$. There are $n_D=\sum_{m=1}^D\lceil m/2\rceil=\lfloor(D+1)^2/4\rfloor$ triples b, and $\sum_{D=1}^{23}n_D=1222$. Averaging with Corollary 6.3 gives the two bounds. Finally, by Lemma 6.6 and (6.6), for each D,

$$\sum_{r,s}(E_D)_{rs}|W_4\omega_{D,r}\rangle\langle W_4\omega_{D,s}|\le 2|W_4\omega_{D,1}\rangle\langle W_4\omega_{D,1}|+\varepsilon\sum_r|W_4\omega_{D,r}\rangle\langle W_4\omega_{D,r}|$$

as relative operators, and $\frac{1}{2}\mathcal{L}_4\mathfrak{P}_4(\,\cdot\otimes\mathbf{1})$ preserves order (Lemmas 2.9(c) and 2.6(iii)), which gives the last inequality. □

7 The square of $H_{\mathbb{T}}$

By Lemma 4.4, $H_{\mathbb{T}}^2=\mathcal{L}_2(h^2)+\mathcal{L}_3(X_3(h))+\mathcal{L}_4(W_4(h\otimes h)W_4^*)$. In this section we show that the three-body coefficient $X_3(h)$ is *exactly* the periodized image of the planar three-body coefficient. The reason is structural: $X_3(h)$ is a sum of products of two pair operators sharing exactly one particle, and periodization is multiplicative on such products. Consequently the three-body target transfers to the torus without error, and besides the non-orthonormality term of Proposition 6.8(b) only the planar tail has to be controlled.

7.1 The image-sum calculus

For a set I of particles, $\mathfrak{P}_I(Y)$ denotes the periodized image of an operator Y admissible in the variables of I, and $\mathfrak{P}_I(Y)\otimes\mathbf{1}$ its action on $\mathfrak{h}^{\otimes n}$, $I\subset\{1,\dots,n\}$; on functions, Y acts in the variables of I with the other variables as parameters.

Lemma 7.1 (image-sum representation)**.** *Let Y be admissible on $I\subset\{1,\dots,n\}$, $s\in I$ and $\Psi\in\mathfrak{h}^{\otimes n}$. Then*

$$((\mathfrak{P}_I(Y)\otimes\mathbf{1})\Psi)(r)=\sum_{\gamma\in\Lambda^{I\setminus\{s\}}}\Big(\prod_{i\in I\setminus\{s\}}\tilde t_{\gamma_i}^{(i)}\Big)(Y\Psi)(r),$$

the series converging absolutely and locally uniformly.

Proof. Expanding Ψ in a product basis of $\mathfrak{h}^{\otimes I}\otimes\mathfrak{h}^{\otimes I^c}$ reduces the claim to $n=|I|=k$; relabel so that $s=1$. By Lemma 2.9(a), $Y\Psi$ is an LLL function of each variable decaying like a Gaussian in r_i-r_1, so the series S converges absolutely and locally uniformly, and S is an LLL function of each variable. Reindexing shows $\tilde t_\gamma^{(i)}S=S$ for $i\ge2$. For $i=1$ write $\tilde t_\gamma^{(1)}=\tilde t_\gamma^{\otimes k}\prod_{i\ge2}(\tilde t_\gamma^{(i)})^{-1}$ with $\tilde t_\gamma^{\otimes k}=\prod_i\tilde t_\gamma^{(i)}$; all these operators commute, $\tilde t_\gamma^{\otimes k}$ commutes with Y by (Y2) and fixes Ψ, so $\tilde t_\gamma^{(1)}S=S$ by reindexing. Thus $S\in\mathfrak{h}^{\otimes k}$, and for $\Phi\in\mathfrak{h}^{\otimes k}$, unfolding the variables $r_2,\dots,r_k$ as in Lemma 2.5 gives $\langle\Phi,S\rangle=\int_{\{r_1\in\mathsf{C}\}}\overline{\Phi}\,Y\Psi=\langle\Phi,\mathfrak{P}_k(Y)\Psi\rangle$. □

Lemma 7.2 (products sharing one particle)**.** *Let $I_1=\{s\}\cup I_1'$ and $I_2=\{s\}\cup I_2'$ with $I_1'\cap I_2'=\emptyset$, let Y_1 be admissible on I_1 and Y_2 on I_2, and $I:=I_1\cup I_2$. Then Y_1Y_2 is admissible on I, and on $\mathfrak{h}^{\otimes n}$*

$$\big(\mathfrak{P}_{I_1}(Y_1)\otimes\mathbf{1}\big)\big(\mathfrak{P}_{I_2}(Y_2)\otimes\mathbf{1}\big)=\mathfrak{P}_I(Y_1Y_2)\otimes\mathbf{1}.$$

Proof. On LLL functions the identity in a variable is the integral operator with the kernel P_{pl} (Lemma 2.4(iv)), so the kernel of Y_1Y_2 is $\int Y_1(r_s,r_{I_1'};u,r'_{I_1'})\,Y_2(u,r_{I_2'};r'_s,r'_{I_2'})\,\mathrm{d}u$. It satisfies (Y1) and (Y2), and the Gaussian bounds (Y3) of the two factors, centred at r_s and at u, combine under the integral over u into a Gaussian bound in all r_i-r_s and r'_i-r_s, $i\in I$. By Lemma 7.1 with the anchor s, $(\mathfrak{P}_{I_2}(Y_2)\otimes\mathbf{1})\Psi=\sum_{\gamma\in\Lambda^{I_2'}}\tilde t_\gamma(Y_2\Psi)$, and applying Lemma 7.1 again,

$$(\mathfrak{P}_{I_1}(Y_1)\otimes\mathbf{1})(\mathfrak{P}_{I_2}(Y_2)\otimes\mathbf{1})\Psi=\sum_{\gamma_1\in\Lambda^{I_1'}}\tilde t_{\gamma_1}Y_1\sum_{\gamma_2\in\Lambda^{I_2'}}\tilde t_{\gamma_2}Y_2\Psi=\sum_{\gamma_1,\gamma_2}\tilde t_{\gamma_1}\tilde t_{\gamma_2}\,Y_1Y_2\Psi,$$

because Y_1 acts on the variables of I_1, which are disjoint from I_2', and therefore commutes with $\tilde t_{\gamma_2}$; the interchange of Y_1 with the sum is justified by dominated convergence, the partial sums being bounded uniformly. The right side is $(\mathfrak{P}_I(Y_1Y_2)\otimes\mathbf{1})\Psi$ by Lemma 7.1 with the anchor s and the images of $I_1'\cup I_2'=I\setminus\{s\}$. □

The lemma fails for products sharing two particles: then the image sums of the second factor do not commute with the first, and cross-image terms appear. This is why $h^2\neq h$: by Lemma 4.1, h is not a projection.

7.2 The planar three-body Gram operator

Let $Q_{12}:=(P_1)_{12}$ on planar three-particle functions; its range consists of the functions $\xi_z(\rho)\,g(\zeta)$, $z\ge0$, and their closed span (Section 6). On this range $W_3=\sqrt{3}A_3$.

Lemma 7.3 (the planar coefficients)**.** *On* $\operatorname{ran} Q_{12}$, $W_3^*W_3 - \mathbf{1} = \sum_{z\ge 0} q_z\,|\xi_z\rangle\langle\xi_z| \otimes \mathbf{1}$ *with*

$$q_z = (3z-1)(-\tfrac{1}{2})^z.$$

In particular $\|W_3\xi_z\|^2 = 1 + q_z$*, so* $W_3\xi_z = 0$ *for* $z \in \{0,1,3\}$ *and* $W_3\xi_z \neq 0$ *otherwise.*

Proof. $W_3^*W_3 = 3Q_{12}A_3Q_{12}$. Since $P_{12}Q_{12} = Q_{12}P_{12} = -Q_{12}$, grouping the permutations in pairs σ, σP_{12} gives $3Q_{12}A_3Q_{12} = Q_{12}(\mathbf{1} - P_{13} - P_{23})Q_{12}$, and $P_{12}P_{13}P_{12} = P_{23}$ gives $Q_{12}P_{13}Q_{12} = Q_{12}P_{23}Q_{12}$; so $W_3^*W_3 - \mathbf{1} = -2\,Q_{12}P_{23}Q_{12}$. In the Bargmann form with $w_1 = (z_1 - z_2)/\sqrt{2}$ and $w_2 = (z_1 + z_2 - 2z_3)/\sqrt{6}$ (the forms of ρ_1, ρ_2), the exchange $z_2 \leftrightarrow z_3$ acts by the substitution

$$w_1 \mapsto \tfrac{1}{2}w_1 + \tfrac{\sqrt{3}}{2}w_2, \qquad w_2 \mapsto \tfrac{\sqrt{3}}{2}w_1 - \tfrac{1}{2}w_2,$$

and fixes $(z_1 + z_2 + z_3)/\sqrt{3}$. It preserves the degree, and $\operatorname{ran} Q_{12}$ contains one relative state of each degree $1 + z$, so $Q_{12}P_{23}Q_{12} = \sum_z b_z|\xi_z\rangle\langle\xi_z| \otimes \mathbf{1}$, where b_z is the coefficient of $w_1w_2^z$ in $(\frac{1}{2}w_1 + \frac{\sqrt{3}}{2}w_2)(\frac{\sqrt{3}}{2}w_1 - \frac{1}{2}w_2)^z$, the normalizations $\sqrt{z!}$ being equal on both sides. The two contributions are $\frac{1}{2}(-\frac{1}{2})^z$ and $\frac{\sqrt{3}}{2}\cdot z\frac{\sqrt{3}}{2}(-\frac{1}{2})^{z-1}$, so $b_z = -\frac{1}{2}(3z-1)(-\frac{1}{2})^z$ and $q_z = -2b_z$. □

7.3 The three-body coefficient

Recall $Y_z = |W_3\xi_z\rangle\langle W_3\xi_z| \otimes \mathbf{1}$, $s_{z,T,p} = s_{z,T,p}^{(1/\sqrt{3})}$, and $|q_z| \le 3z2^{-z}$ for $z \ge 1$.

Theorem 7.4 (the square of $H_{\mathbb{T}}$)**.** *On* $\mathcal{F}$*, for every rectangular torus,*

$$H_{\mathbb{T}}^2 = \mathcal{L}_2(h^2) + \sum_{z\ge 0} q_z\,\mathcal{L}_3\mathfrak{P}_3(Y_z) + \mathcal{L}_4(W_4(h\otimes h)W_4^*), \tag{7.1}$$

the series converging absolutely in norm. Moreover, for every $z \ge 0$*,*

$$\mathcal{L}_3\mathfrak{P}_3(Y_z) = \frac{3}{2\pi}\int_{\mathsf{C}} B(y_z(R))^*B(y_z(R))\,\mathrm{d}R, \qquad y_z(R) := \sum_{p=0}^{z} s_{z,z,p}\,\tilde{u}_p(R)\wedge\chi_{z-p}^R. \tag{7.2}$$

Proof. By Lemma 4.4 it suffices to show $X_3(h) = \sum_z q_z\mathfrak{P}_3(Y_z)$ on $\bigwedge^3\mathfrak{h}$. The operator P_1 vanishes on functions that are even in the relative coordinate, so $h = \mathfrak{P}_2(P_1)$ vanishes on symmetric tensors and $h_{ij} = \mathfrak{P}_{\{i,j\}}(P_1)\otimes\mathbf{1}$. Each term $h_{ij}h_{kl}$ of $X_3(h)$ has $|\{i,j\}\cap\{k,l\}| = 1$, so by Lemma 7.2

$$X_3(h) = \mathfrak{P}_3(Y_{\mathrm{pl}}), \qquad Y_{\mathrm{pl}} := \sum_{(ij)\neq(kl)} (P_1)_{ij}(P_1)_{kl},$$

and by Lemma 2.9(b), on $\bigwedge^3\mathfrak{h}$ we may replace Y_{pl} by $A_3Y_{\mathrm{pl}}A_3$. On planar antisymmetric three-particle functions the computation in the proof of Lemma 4.4, with $P_1^2 = P_1$, gives $A_3Y_{\mathrm{pl}}A_3 = W_3(W_3^*W_3 - \mathbf{1})W_3^*$ with W_3 restricted to $\operatorname{ran} Q_{12}$, and Lemma 7.3 turns this into $\sum_z q_zY_z$, because W_3 acts on the relative variables only. The kernels of the Y_z are $(W_3\xi_z)(\rho)\overline{(W_3\xi_z)(\rho')}P_{\mathrm{pl}}(\zeta,\zeta')$. Since $|\eta_z(x)| = |x|^z e^{-|x|^2/4}/\sqrt{2\pi 2^z z!}$ and permutations preserve $|\rho|$, $|(W_3\xi_z)(\rho)| \le C|\rho|^{z+1}e^{-|\rho|^2/4}/\sqrt{2^z z!}$; with $|q_z| \le 3z2^{-z}$ for $z \ge 1$ (and $Y_0 = 0$) and $\sum_z z\,x^z/z! = xe^x$, the series $\sum_z|q_z||Y_z(r;r')|$ is bounded by $C(1+|\rho||\rho'|)^2e^{-(|\rho|^2+|\rho'|^2)/4+|\rho||\rho'|/4}|P_{\mathrm{pl}}(\zeta,\zeta')| \le C'e^{-(|\rho|^2+|\rho'|^2)/9}|P_{\mathrm{pl}}(\zeta,\zeta')|$. So the kernel series converges locally uniformly to the (continuous) kernel of $A_3Y_{\mathrm{pl}}A_3$, under an admissible Gaussian bound, and dominated convergence in (2.9) gives $\mathfrak{P}_3(\sum_z q_zY_z) = \sum_z q_z\mathfrak{P}_3(Y_z)$, absolutely convergent on the finite-dimensional space $\bigwedge^3\mathfrak{h}$. For (7.2), apply Theorem 6.2 with $k = 3$, $a = b = W_3\xi_z$ and $p = q = 0$: $\Pi^{(3)}(W_3\xi_z\otimes\varphi_0^R) = \Pi^{(3)}W_3|z,z\rangle^R = \sum_p s_{z,z,p}\,\tilde{u}_p(R)\wedge\chi_{z-p}^R$ by (6.2) and Lemma 5.1(b), and then lift with $\mathcal{L}_3$. □

In words: the three-body coefficient of $H_{\mathbb{T}}^2$ is the periodized planar three-body target on *every* block z, with the planar coefficients q_z and no correction depending on the torus.

7.4 The three-body comparison and the tail

Proposition 7.5 (three-body comparison)**.** *On $\mathcal{F}$, for every rectangular torus,*

$$\mathcal{L}_3(X_3(h)) - K_{\mathbb{T}}^{(3)} \geq -\big(\delta_{\mathrm{tail}}(\mathbb{T}) + C_3\,\mathcal{E}_8(\mathbb{T})(1+\mathcal{E}_8(\mathbb{T}))\big)H_{\mathbb{T}},$$

where

$$\delta_{\mathrm{tail}}(\mathbb{T}) := \frac{3}{2}\sum_{\substack{z\geq 17\\ z\ \mathrm{odd}}} (3z-1)(z+1)2^{-z}(1+\sigma_z(\mathbb{T})).$$

Moreover $\frac{3}{2}\sum_{z\geq 17,\, z\ \mathrm{odd}}(3z-1)(z+1)2^{-z} = \frac{61}{4096}$, and for every rectangular torus whose shorter period is at least $L>0$,

$$\delta_{\mathrm{tail}}(\mathbb{T}) - \frac{61}{4096} \leq \frac{3}{2}\sum_{\substack{z\geq 17\\ z\ \mathrm{odd}}} (3z-1)(z+1)2^{-z}\,\Xi_z(L) \leq \sum_{(j,k)\in\mathbb{Z}^2\setminus 0} \Omega(L\sqrt{j^2+k^2}),$$

where $\Omega(r) := 36\,e^{-r^2/16} + 9\,(y^2+5y+8)\,2^{-y}$ with $y := r^2/20$. The middle term is nonincreasing in L, and the right side is $O(L^4\,2^{-L^2/20})$ as $L\to\infty$.

Proof. By Theorem 7.4 and Proposition 6.8(b), with $e_z = 0$ for $z \geq 16$,

$$\mathcal{L}_3(X_3(h)) - K_{\mathbb{T}}^{(3)} = \sum_{z\leq 15} (q_z - \tfrac{2}{3}e_z)\mathcal{L}_3\mathfrak{P}_3(Y_z) + \sum_{z\geq 16} q_z\,\mathcal{L}_3\mathfrak{P}_3(Y_z) - \mathcal{C}_3.$$

Near part. $Y_z = 0$ for $z\in\{0,1,3\}$ (Lemma 7.3), and $q_z - \frac{2}{3}e_z > 0$ for $z\in\{2,4,5,\dots,15\}$ by (6.5); so the first sum is $\mathcal{L}_3\mathfrak{P}_3$ of a positive finite-rank relative operator tensored with $\mathbf{1}$, which is positive by Lemmas 2.9(c) and 2.6(iii). *Tail.* For even $z\geq 16$, $q_z > 0$ and the term is positive. For odd $z\geq 17$, $q_z = -(3z-1)2^{-z}$, and by (7.2), $B(y_z(R)) = \sum_p s_{z,z,p}\,c(\chi^R_{z-p})B(\tilde{u}_p(R))$ (by (2.7); s is real); by Cauchy–Schwarz with $\sum_p s_{z,z,p}^2 = 1$ and $\|c(\chi^R_m)\|^2 = \Gamma(R)_{mm} \leq 1+\sigma_z(\mathbb{T})$ for $m\leq z$,

$$0 \leq \langle\Psi, \mathcal{L}_3\mathfrak{P}_3(Y_z)\Psi\rangle \leq \frac{3}{2\pi}\int_{\mathsf{C}}(1+\sigma_z)\sum_{p\leq z}\|B(\tilde{u}_p(R))\Psi\|^2\mathrm{d}R = \frac{3}{2}(1+\sigma_z)(z+1)\langle\Psi, H_{\mathbb{T}}\Psi\rangle,$$

the last step by Corollary 6.3. Summing over odd $z\geq 17$ gives $-\delta_{\mathrm{tail}}H_{\mathbb{T}}$, and $\mathcal{C}_3$ is bounded in Proposition 6.8(b). *The sum.* With $z = 17+2m$, $(3z-1)(z+1) = 900+208m+12m^2$, and $\sum_m 4^{-m} = \frac{4}{3}$, $\sum_m m4^{-m} = \frac{4}{9}$, $\sum_m m^2 4^{-m} = \frac{20}{27}$ give $\frac{3}{2}\cdot 2^{-17}\cdot\frac{3904}{3} = \frac{61}{4096}$. *The excess.* $\sigma_z(\mathbb{T}) \leq \Xi_z(L)$ by Lemma 5.3(c), and $\Xi_z(L)$ is nonincreasing in L. For $z\geq 1$ and $r^2\geq 20z$, $\mathfrak{g}_z(r) \leq e^{-r^2/16}$: with $r^2 = 2zt$, $t\geq 10$, and $z! \geq (z/e)^z$, $\log\mathfrak{g}_z(r) \leq z(1+\log(t+1)-t/2) \leq -zt/8$, since $1+\log(t+1)\leq 3t/8$ for $t\geq 10$. Exchange the sums over z and over the lattice points, bound the odd $z\geq 17$ by all $z\geq 1$ and $\frac{3}{2}(3z-1)(z+1)$ by $\frac{9}{2}z(z+1)$, and fix a lattice point with $r = L\sqrt{j^2+k^2}$. The terms with $z < r^2/20$ have $\min\{1,\mathfrak{g}_z(r)\} \leq e^{-r^2/16}$ and contribute at most $\frac{9}{2}\sum_{z\geq 1} z(z+1)2^{-z}e^{-r^2/16} = 36e^{-r^2/16}$. The terms with $z\geq z_0 := \lceil r^2/20\rceil$ have $\min\{1,\mathfrak{g}_z\}\leq 1$ and contribute at most $\frac{9}{2}\sum_{z\geq z_0} z(z+1)2^{-z} = 9(z_0^2+3z_0+4)2^{-z_0} \leq 9(y^2+5y+8)2^{-y}$, using $y\leq z_0\leq y+1$. This is the bound by $\sum\Omega$. For the asymptotics, Ω is decreasing for $r^2\geq 20$, at most $4(n+1)$ lattice points have $j^2+k^2 = n\geq 2$, and $\Omega(L\sqrt{n}) \leq n^2 e^{-(n-1)(\log 2)L^2/20}\,\Omega(L)$ for $L^2\geq 20$, so the sum is at most a constant times $\Omega(L) = O(L^4 2^{-L^2/20})$. □

The near part is the planar three-body inequality (6.5) transported to the torus by the order-preserving map $\mathcal{L}_3\mathfrak{P}_3$; there is no error term between the torus and the plane on any block, frustrated or not.

8 The four-body Bessel bound

The four-body coefficient of $H_{\mathbb{T}}^2$ is $\mathcal{L}_4(W_4(h\otimes h)W_4^*)$, a product of two pair operators on disjoint pairs, and its comparison with $K_{\mathbb{T}}^{(4)}$ is not exact: the periodized planar target $\mathcal{T}_4$ of Proposition 6.8(d) involves only the channels $D\le 23$ of the relative coordinate of the two pair centres, and these channels are localized. The bound below compares $\mathcal{T}_4$ with $\mathcal{L}_4(W_4(h\otimes h)W_4^*)$ up to a factor $1+\mathcal{E}_{22}(\mathbb{T})$; the proof reduces it to a Gram matrix of the same kind as the local Gram matrix of Section 5, now for the pair centres.

Lemma 8.1 (four-body Bessel bound)**.** *Let* $\mathcal{Z}_4 := \mathfrak{P}_4\big(\sum_{D=1}^{23}|\omega_{D,1}\rangle\langle\omega_{D,1}|\otimes\mathbf{1}\big)$ *on* $\mathfrak{h}^{\otimes 4}$*, and let* $h\otimes h$ *act on the particles* $(1,2)$ *and* $(3,4)$*. Then*

$$\mathcal{Z}_4 \le \big(1+\mathcal{E}_{22}(\mathbb{T})\big)\, h\otimes h.$$

Proof. Let $\mathcal{I}=\mathcal{I}_{\eta_1}:\mathfrak{h}^{\otimes 2}\to\mathfrak{u}_2$ be the fibering map of Proposition 2.11, applied to the particles $(1,2)$ and to the particles $(3,4)$; it maps $\mathfrak{h}^{\otimes 4}$ to $\mathfrak{u}_2\otimes\mathfrak{u}_2$, the two-particle space of the torus $\mathbb{T}_2=\mathbb{R}^2/\sqrt{2}\Lambda$, whose two particles are the pair centres $\zeta_{12}=(r_1+r_2)/\sqrt{2}$ and $\zeta_{34}=(r_3+r_4)/\sqrt{2}$. Their Jacobi coordinates are $\rho_3=(\zeta_{12}-\zeta_{34})/\sqrt{2}$ and $\zeta=(\zeta_{12}+\zeta_{34})/\sqrt{2}$. For $d\ge 0$ let $\mathcal{I}'_d$ be the fibering map of the torus $\mathbb{T}_2$ with the relative state $\eta_d(\rho_3)$; it maps $\mathfrak{u}_2\otimes\mathfrak{u}_2$ to the centre-of-mass space $\mathfrak{u}''$ of $\mathbb{T}_2$, the lowest Landau level of $\mathbb{R}^2/2\Lambda$. Since $\omega_{D,1}=\eta_1(\rho_1)\eta_1(\rho_2)\eta_{D-1}(\rho_3)$, Fubini gives $\mathcal{I}_{\omega_{D,1}}=\mathcal{I}'_{D-1}(\mathcal{I}\otimes\mathcal{I})$, and Lemma 2.9(c) gives

$$\mathcal{Z}_4=\sum_{d=0}^{22}(\mathcal{I}\otimes\mathcal{I})^*(\mathcal{I}'_d)^*\mathcal{I}'_d(\mathcal{I}\otimes\mathcal{I})=(\mathcal{I}\otimes\mathcal{I})^*\mathcal{G}^*\mathcal{G}(\mathcal{I}\otimes\mathcal{I}),\qquad h\otimes h=(\mathcal{I}\otimes\mathcal{I})^*(\mathcal{I}\otimes\mathcal{I}),$$

with $\mathcal{G}:=(\mathcal{I}'_d)_{d\le 22}:\mathfrak{u}_2\otimes\mathfrak{u}_2\to\bigoplus_{d\le 22}\mathfrak{u}''$. Hence $\mathcal{Z}_4\le\|\mathcal{G}\mathcal{G}^*\|\,h\otimes h$.

By (2.10), applied on $\mathbb{T}_2$, $(\mathcal{I}'_{d'})^*g=\sum_{\gamma'\in\sqrt{2}\Lambda}\tilde{t}^{(2)}_{\gamma'}(\eta_{d'}(\rho_3)g(\zeta))$ for $g\in\mathfrak{u}''$, and here $\tilde{t}^{(2)}_{\gamma'}=t^{(2)}_{\gamma'}$: $\mathbb{T}_2$ carries $2N_\phi$ flux quanta, so its lattice translation by $\gamma'=\sqrt{2}(jL_x,kL_y)$ has the sign $(-1)^{jk\cdot 2N_\phi}=1$. A magnetic translation of the second particle by a acts as $t_{-a/\sqrt{2}}$ on ρ_3 and as $t_{a/\sqrt{2}}$ on ζ (as in Lemma 2.7(ii)). Hence, writing $\gamma'=\sqrt{2}\gamma$ with $\gamma\in\Lambda$,

$$\mathcal{I}'_d(\mathcal{I}'_{d'})^*=\sum_{\gamma\in\Lambda}\langle\eta_d,t_{-\gamma}\eta_{d'}\rangle\; t_\gamma\big|_{\mathfrak{u}''},\qquad \mathcal{G}\mathcal{G}^*=\sum_{\gamma\in\Lambda}\mathcal{D}^{(22)}(\beta(-\gamma))\otimes t_\gamma\big|_{\mathfrak{u}''},$$

the series converging absolutely. Each t_γ, $\gamma\in\Lambda$, is unitary on $\mathfrak{u}''$ by Lemma 6.1, since Λ is contained in the lattice $2\Lambda/(4N_\phi)$ of $\mathbb{R}^2/2\Lambda$. The term $\gamma=0$ is the identity, and $\|\mathcal{D}^{(22)}(\beta(-\gamma))\|=\mu_{22}(|\gamma|)$. So $\|\mathcal{G}\mathcal{G}^*\|\le 1+\mathcal{E}_{22}(\mathbb{T})$. □

In these coordinates the relative coordinate ρ_3 of the two pair centres carries unit charge, and it equals the difference of the two pair centres of mass; so the image displacements in the proof are those of a single electron, and $\mathcal{E}_{22}$ is the same lattice sum as for the local Gram matrix of the orbitals $\chi_0^R,\dots,\chi_{22}^R$.

Proposition 8.2 (four-body comparison)**.** *On* $\mathcal{F}$*, for every rectangular torus, with* $\mathcal{E}_m=\mathcal{E}_m(\mathbb{T})$*,*

$$\mathcal{T}_4\le(1+\mathcal{E}_{22})\,\mathcal{L}_4\big(W_4(h\otimes h)W_4^*\big)\quad\text{and}$$
$$\mathcal{L}_4\big(W_4(h\otimes h)W_4^*\big)-K_{\mathbb{T}}^{(4)}\ge-(1+\mathcal{E}_{23})^2\big(1222\,\varepsilon+2444\,\mathcal{E}_{22}\big)H_{\mathbb{T}}.$$

Proof. $W_4\omega_{D,1} = \sqrt{6}A_4\omega_{D,1}$ with A_4 acting on the relative variables, so $|W_4\omega_{D,1}\rangle\langle W_4\omega_{D,1}| \otimes \mathbf{1} = 6A_4(|\omega_{D,1}\rangle\langle\omega_{D,1}| \otimes \mathbf{1})A_4$, and by Lemma 2.9(b) its periodized image is $6A_4\mathfrak{P}_4(|\omega_{D,1}\rangle\langle\omega_{D,1}| \otimes \mathbf{1})A_4$. The operator $\mathfrak{P}_4(|\omega_{D,1}\rangle\langle\omega_{D,1}|\otimes\mathbf{1})$ maps into $\bigwedge^2\mathfrak{h}\otimes\bigwedge^2\mathfrak{h}$, and on that space $6A_4XA_4 = W_4XW_4^*$ on $\bigwedge^4\mathfrak{h}$ (Lemma 2.6(iv)). Hence $\mathcal{T}_4 = \mathcal{L}_4(W_4\mathcal{Z}_4W_4^*)$, and Lemma 8.1, conjugation by W_4 and the lift preserve order. For the second inequality, Proposition 6.8(d) gives $K_{\mathbb{T}}^{(4)} \le \mathcal{T}_4 + 1222\varepsilon(1+\mathcal{E}_{23})^2 H_{\mathbb{T}}$ and $\mathcal{T}_4 \le 2444(1+\mathcal{E}_{23})^2 H_{\mathbb{T}}$; so

$$\mathcal{L}_4(W_4(h\otimes h)W_4^*) - K_{\mathbb{T}}^{(4)} \ge \frac{\mathcal{T}_4}{1+\mathcal{E}_{22}} - \mathcal{T}_4 - 1222\varepsilon(1+\mathcal{E}_{23})^2 H_{\mathbb{T}} \ge -\mathcal{E}_{22}\,\mathcal{T}_4 - 1222\varepsilon(1+\mathcal{E}_{23})^2 H_{\mathbb{T}},$$

using $\mathcal{T}_4 \ge 0$, and the bound on $\mathcal{T}_4$ finishes the proof. □

9 The theorem

This section proves Theorems 1.1 and 1.2. Theorem 9.1 and Corollary 9.2, with the rate of Remark 9.3 and the thresholds of Section 9.2, are the proof of Theorem 1.2; Section 9.4 proves Theorem 1.1 by the same argument with the data of the certificate left general.

9.1 The budget

Recall $\gamma_* = 1 - \eta_* - \frac{61}{4096} - 1222\,\varepsilon = 4616733319001/10^{14}$, where the last equality is exact arithmetic with $\eta_* = 93527408868499/10^{14}$ and $\varepsilon = 3/10^6$, and $C_3 = 72\sum_\nu \|\alpha^\nu\|^2$ (Appendix A).

Theorem 9.1 (uniform gap on rectangular tori)**.** *For every rectangular torus $\mathbb{T}$, on the whole Fock space $\mathcal{F}$,*

$$H_{\mathbb{T}}^2 \ge K_{\mathbb{T}} + \gamma_{\mathbb{T}}\,H_{\mathbb{T}} \ge \gamma_{\mathbb{T}}\,H_{\mathbb{T}},$$

where, with $\mathcal{E}_m = \mathcal{E}_m(\mathbb{T})$, $\lambda_{\min}$ the smallest eigenvalue of the pair Gram matrix and $\delta_{\mathrm{tail}} = \delta_{\mathrm{tail}}(\mathbb{T})$,

$$\begin{aligned}\gamma_{\mathbb{T}} &:= \lambda_{\min} - \eta_* - \delta_{\mathrm{tail}} - C_3\,\mathcal{E}_8(1+\mathcal{E}_8) - (1+\mathcal{E}_{23})^2(1222\,\varepsilon + 2444\,\mathcal{E}_{22}) \\ &= \gamma_* - (1-\lambda_{\min}) - \Big(\delta_{\mathrm{tail}} - \frac{61}{4096}\Big) - C_3\,\mathcal{E}_8(1+\mathcal{E}_8) \\ &\qquad - 1222\,\varepsilon\big((1+\mathcal{E}_{23})^2 - 1\big) - 2444\,\mathcal{E}_{22}(1+\mathcal{E}_{23})^2.\end{aligned} \tag{9.1}$$

Proof. By Theorem 7.4 and Proposition 6.8,

$$H_{\mathbb{T}}^2 - K_{\mathbb{T}} = \big[\mathcal{L}_2(h^2) - \eta_* H_{\mathbb{T}}\big] + \big[\mathcal{L}_3(X_3(h)) - K_{\mathbb{T}}^{(3)}\big] + \big[\mathcal{L}_4(W_4(h\otimes h)W_4^*) - K_{\mathbb{T}}^{(4)}\big].$$

By Proposition 4.3, $h^2 \ge \lambda_{\min} h$, and $\mathcal{L}_2$ preserves order with $\mathcal{L}_2(h) = H_{\mathbb{T}}$; so the first bracket is at least $(\lambda_{\min} - \eta_*)H_{\mathbb{T}}$. The second is bounded by Proposition 7.5 and the third by Proposition 8.2. Finally $K_{\mathbb{T}} \ge 0$ and $H_{\mathbb{T}} \ge 0$. □

All the constants in (9.1) depend on the torus only through its lattice of periods, and none depends on the particle number. They are small when the shorter period is large:

Corollary 9.2 (uniform bound)**.** *For $L \ge 16$ let*

$$\begin{aligned}\underline{\gamma}(L) := \gamma_* - [2\Delta(L) + \Upsilon(L,L)] - \frac{3}{2}\sum_{\substack{z\ge 17\\ z\ \text{odd}}} \frac{(3z-1)(z+1)}{2^z}\,\Xi_z(L) - C_3\,\mathcal{E}_8(L)(1+\mathcal{E}_8(L)) \\ - 1222\,\varepsilon\big((1+\mathcal{E}_{23}(L))^2 - 1\big) - 2444\,\mathcal{E}_{22}(L)(1+\mathcal{E}_{23}(L))^2,\end{aligned}$$

with $\mathcal{E}_m(L)$ and $\Xi_z(L)$ as in Lemma 5.3. Then $\gamma_{\mathbb{T}} \ge \underline{\gamma}(L)$ for every rectangular torus whose shorter period is at least L, the function $\underline{\gamma}$ is nondecreasing on $[16,\infty)$, and $\underline{\gamma}(L) \to \gamma_*$ as $L \to \infty$.

Proof. By Lemma 4.2, $1-\lambda_{\min} \le \delta_G \le \Delta(L_x)+\Delta(L_y)+\Upsilon(L_x,L_y)$. The function Δ is decreasing on $(\sqrt{12},\infty)$, and every term $|\mathfrak{f}(s_{jk})| = (s_{jk}-1)e^{-s_{jk}/2}$ of Υ is decreasing in L_x and L_y, since off $\mathcal{X}$ it has $s_{jk} \ge 9/2 > 3$; so the bound is at most $2\Delta(L)+\Upsilon(L,L)$ and is nonincreasing in L. The tail term is bounded by Proposition 7.5, and $\mathcal{E}_m(\mathbb{T}) \le \mathcal{E}_m(L)$ for $m \in \{8,22,23\}$ by Lemma 5.3(a); all subtracted terms are nonincreasing in L. As $L\to\infty$, $\Delta(L)\to 0$ and $\Upsilon(L,L)\to 0$ (Lemma 4.2(c)), the tail excess tends to zero (Proposition 7.5), and $\mathcal{E}_m(L)\to 0$ (Lemma 5.3(b)). □

Remark 9.3 (the rate of convergence). As $L\to\infty$,

$$0 \le \gamma_* - \underline{\gamma}(L) = O(L^4\, 2^{-L^2/20}).$$

The rate is that of the majorant Ω of Proposition 7.5 for the tail term, and it is not claimed to be sharp. The same rate holds for the transfer principle, Theorem 1.1: for every planar certificate of the same form, the bound $\epsilon_{\mathrm{tr}}(L)$ on $\epsilon_{\mathrm{tr}}(\mathbb{T})$ obtained in the proof of Theorem 9.6 is $O(L^4\, 2^{-L^2/20})$, with a constant that depends on the certificate.

Proof. $\gamma_* - \underline{\gamma}(L)$ is the sum of the five nonnegative terms subtracted in the definition of $\underline{\gamma}$. By the definition of Δ and Lemma 4.2(c), $2\Delta(L)+\Upsilon(L,L) = O(L^2 e^{-L^2/8})$. By Lemma 5.3(b), $\mathcal{E}_m(L) \le \widehat{\mathcal{E}}_m(L) = O(L^{2m}e^{-L^2/4})$, so the three terms containing $\mathcal{E}_8$, $\mathcal{E}_{22}$ and $\mathcal{E}_{23}$ are $O(L^{46}e^{-L^2/4})$. By Proposition 7.5, the tail term $\frac{3}{2}\sum_{z\ge 17,\, z \text{ odd}}(3z-1)(z+1)2^{-z}\,\Xi_z(L)$ is at most $\sum_{(j,k)\ne 0}\Omega(L\sqrt{j^2+k^2})$, which is $O(L^4\,2^{-L^2/20})$. Since $2^{-L^2/20} = e^{-(\log 2)L^2/20}$ and $(\log 2)/20 < 1/8 < 1/4$, the other four terms are $o(L^4\,2^{-L^2/20})$. The proof of Theorem 9.6 bounds the terms of $\epsilon_{\mathrm{tr}}(\mathbb{T})$ by the same three results, with $\mathcal{E}_{M_c}$, $\mathcal{E}_{D_c-1}$ and $\mathcal{E}_{D_c}$ in place of $\mathcal{E}_8$, $\mathcal{E}_{22}$ and $\mathcal{E}_{23}$, and the tail by the same majorant (whose derivation bounds the sum over all $z \ge 1$); this gives the second statement. □

9.2 The constants

Table 1 lists the terms of $\underline{\gamma}(L)$. The first column after L is the bound $2\Delta(L)+\Upsilon(L,L)$ on $1-\lambda_{\min}$; the binding term is the four-body Bessel term, followed by the tail excess, while the pair-level and non-orthonormality terms are negligible.

Thresholds. The computed function $\underline{\gamma}$ crosses 0 at $L = 16.4562$, $1/25$ at $L = 16.8566$, and $\gamma_* - 10^{-5}$ at $L = 18.6843$ (magnetic lengths, to the precision shown). Hence $\underline{\gamma}(L) > 0$ for $L \ge 16.457$, $\underline{\gamma}(L) > 1/25$ for $L \ge 16.857$, and $\underline{\gamma}(L) \ge \gamma_* - 10^{-5}$ for $L \ge 18.685$; by Corollary 9.2 the same holds for $\gamma_{\mathbb{T}}$ on every rectangular torus whose shorter period is at least these values.

Remark 9.4 (on the evaluation of the constants). Every quantity in $\underline{\gamma}(L)$ is given by an explicit expression: Δ is elementary, $\Upsilon(L,L)$ and $\Xi_z(L)$ are lattice sums of elementary functions with Gaussian decay, the sum over odd z converges geometrically (Proposition 7.5), and $\mu_m(r)$ is the spectral norm of an explicit $(m+1)\times(m+1)$ matrix whose entries are Laguerre polynomials times Gaussians (Lemma 2.4(iii)). The reduction from the supremum over all rectangular tori to the evaluation at the radii $L\sqrt{j^2+k^2}$ is rigorous by Lemma 5.3(a), whose proof shows that for $r \ge 16$ every entry of $\mathcal{D}^{(m)}(r/\sqrt{2})$, $m \le 23$, has a fixed sign and a modulus decreasing in r, so that the spectral norm is a monotone function of the entries. What remains is a numerical evaluation. The entries of Table 1 were computed in 60-digit floating-point arithmetic from these expressions, with γ_*, $61/4096$ and η_* as exact rationals, the displacement entries from the Laguerre representation, and the spectral norms in double precision cross-checked against 60-digit eigenvalue computations (relative agreement $3\cdot 10^{-16}$); the lattice sums and the series over z were truncated where the omitted terms are far below the displayed digits. A second, independent implementation

L	$2\Delta(L)+\Upsilon(L,L)$	$\mathcal{E}_8(L)$	$\mathcal{E}_{22}(L)$	$\mathcal{E}_{23}(L)$	tail excess
16.5	$4.5\cdot10^{-13}$	$2.1\cdot10^{-17}$	$1.50\cdot10^{-5}$	$5.73\cdot10^{-5}$	$5.42\cdot10^{-4}$
17	$5.8\cdot10^{-14}$	$5.3\cdot10^{-19}$	$1.13\cdot10^{-6}$	$4.73\cdot10^{-6}$	$1.93\cdot10^{-4}$
17.5	$7.2\cdot10^{-15}$	$1.2\cdot10^{-20}$	$6.95\cdot10^{-8}$	$3.19\cdot10^{-7}$	$8.38\cdot10^{-5}$
18	$8.2\cdot10^{-16}$	$2.2\cdot10^{-22}$	$3.56\cdot10^{-9}$	$1.77\cdot10^{-8}$	$4.22\cdot10^{-5}$
19	$9.0\cdot10^{-18}$	$5.2\cdot10^{-26}$	$5.45\cdot10^{-12}$	$3.16\cdot10^{-11}$	$4.05\cdot10^{-6}$
20	$7.6\cdot10^{-20}$	$7.1\cdot10^{-30}$	$4.18\cdot10^{-15}$	$2.79\cdot10^{-14}$	$4.95\cdot10^{-7}$

L	$C_3\mathcal{E}_8(1+\mathcal{E}_8)$	$1222\,\varepsilon\big((1+\mathcal{E}_{23})^2-1\big)$	$2444\,\mathcal{E}_{22}(1+\mathcal{E}_{23})^2$	$\underline{\gamma}(L)$
16.5	$6.0\cdot10^{-15}$	$4.2\cdot10^{-7}$	$3.67\cdot10^{-2}$	0.0089238941
17	$1.5\cdot10^{-16}$	$3.5\cdot10^{-8}$	$2.75\cdot10^{-3}$	0.0432229058
17.5	$3.3\cdot10^{-18}$	$2.3\cdot10^{-9}$	$1.70\cdot10^{-4}$	0.0459136282
18	$6.2\cdot10^{-20}$	$1.3\cdot10^{-10}$	$8.70\cdot10^{-6}$	0.0461164613
19	$1.5\cdot10^{-23}$	$2.3\cdot10^{-13}$	$1.33\cdot10^{-8}$	0.0461632655
20	$2.0\cdot10^{-27}$	$2.0\cdot10^{-16}$	$1.02\cdot10^{-11}$	0.0461668382

Table 1: The terms of the uniform bound $\underline{\gamma}(L)$ of Corollary 9.2. The tail excess is $\frac{3}{2}\sum_{z\geq17\ \mathrm{odd}}(3z-1)(z+1)2^{-z}\,\Xi_z(L)$, and $\gamma_*=0.04616733319001$.

with different truncations reproduces every entry to the digits it reports (four to seven significant figures for $\underline{\gamma}$). The values are therefore floating-point evaluations of rigorous closed-form bounds, not interval-arithmetic certificates, and the thresholds were located by bisection of the computed function.

9.3 Consequences

Corollary 9.5 (spectrum and kernel)**.** *Let $\mathbb{T}$ be a rectangular torus with $\gamma_{\mathbb{T}}>0$; by Corollary 9.2 this holds whenever the shorter period is at least* 16.457*.*

(a) For every N, $\operatorname{spec}\big(H_{\mathbb{T}}|_{\bigwedge^N\mathfrak{h}}\big)\subset\{0\}\cup[\gamma_{\mathbb{T}},\infty)$.

(b) At $N_\phi=3N$, $H_{\mathbb{T}}|_{\bigwedge^N\mathfrak{h}}\geq\gamma_{\mathbb{T}}(\mathbf{1}-P_N)$, where P_N is the orthogonal projection onto the three-dimensional span of the Laughlin states Ψ_0,Ψ_1,Ψ_2 of Theorem 3.1, and $H_{\mathbb{T}}|_{\bigwedge^{N'}\mathfrak{h}}\geq\gamma_{\mathbb{T}}$ for every $N'>N$.

(c) If $\delta_G<1$, the projector form satisfies $H'^2\geq(\gamma_{\mathbb{T}}/\lambda_{\max})H'$.

In particular, on every rectangular torus whose shorter period is at least 16.857*, the gap above the three-dimensional kernel at $N_\phi=3N$ is larger than $1/25$, uniformly in N and in the aspect ratio.*

Proof. (a) $H_{\mathbb{T}}\geq0$ preserves the particle number, and an eigenvalue λ satisfies $\lambda^2\geq\gamma_{\mathbb{T}}\lambda$ by Theorem 9.1. (b) By (a) and the spectral theorem, $H_{\mathbb{T}}\geq\gamma_{\mathbb{T}}(\mathbf{1}-P_{\ker})$ on each sector; Theorem 3.1 identifies the kernel at N particles and Proposition 3.2 shows that it is zero for $N'>N$. (c) By Proposition 4.3, $H'\geq\lambda_{\max}^{-1}H_{\mathbb{T}}\geq(\gamma_{\mathbb{T}}/\lambda_{\max})(\mathbf{1}-P_{\ker})$ with $\ker H'=\ker H_{\mathbb{T}}$. □

At the square aspect ratio the shorter period is $L=\sqrt{2\pi N_\phi}$, so $\underline{\gamma}>0$ for $N_\phi\geq44$ and $\underline{\gamma}>1/25$ for $N_\phi\geq46$. At filling one third, $N_\phi=3N$: for $N=15$ ($L=16.815$) the bound is $\underline{\gamma}\geq0.0385$, and for $N=16$ ($L=17.366$) it is $\underline{\gamma}\geq0.0456>1/25$. On elongated tori the shorter period is smaller at the same N, and correspondingly larger systems are needed.

9.4 The transfer principle

It remains to prove Theorem 1.1, for a planar certificate of the same form, with the inequalities (1.1) and the constant γ of the definition in Section 1.1. For the certificate of [1], $P_c = 7$, $M_c = 8$, $Z_c = 15$, $D_c = 23$, $\delta_\infty(15) = 61/4096$, $N_4 = 1222$ and $\gamma = \gamma_*$. We restate the theorem in its full form.

Theorem 9.6 (the transfer principle; Theorem 1.1 restated)**.** *Given a planar certificate of the same form with constant γ, for every rectangular torus, on $\mathcal{F}$,*

$$H_{\mathbb{T}}^2 \geq (\gamma - \epsilon_{\mathrm{tr}}(\mathbb{T}))\, H_{\mathbb{T}},$$

where, with $\mathcal{E}_m = \mathcal{E}_m(\mathbb{T})$ and $\sigma_z = \sigma_z(\mathbb{T})$,

$$\epsilon_{\mathrm{tr}}(\mathbb{T}) := (1 - \lambda_{\min}) + \frac{3}{2} \sum_{\substack{z > Z_c \\ z \text{ odd}}} \frac{(3z-1)(z+1)}{2^z} \sigma_z + (M_c+1)(P_c+1) \sum_\nu \|\alpha^\nu\|^2\, \mathcal{E}_{M_c}(1 + \mathcal{E}_{M_c}) + \varepsilon N_4 \big((1+\mathcal{E}_{D_c})^2 - 1\big) + 2N_4\, \mathcal{E}_{D_c-1}(1+\mathcal{E}_{D_c})^2.$$

For every rectangular torus whose shorter period is at least L, $\epsilon_{\mathrm{tr}}(\mathbb{T}) \leq \epsilon_{\mathrm{tr}}(L)$ with $\epsilon_{\mathrm{tr}}(L) \to 0$ as $L \to \infty$. In particular, if $\gamma > 0$, then for every $\gamma' < \gamma$ there is L_0 such that $H_{\mathbb{T}}^2 \geq \gamma' H_{\mathbb{T}}$ on the whole Fock space of every rectangular torus whose shorter period is at least L_0, uniformly in the particle number and the aspect ratio.

Proof. The proofs of Proposition 6.8, Proposition 7.5, Lemma 8.1, Proposition 8.2 and Theorem 9.1 use the specific certificate only through the data of the definition in Section 1.1, and go through verbatim with the following changes. In Proposition 6.8, the modes in (i,k) and j are at most M_c, each pair label $p \leq P_c$ occurs with at most $M_c + 1$ values of j, and $\frac{1}{\pi}\int \sum_{p \leq P_c} \|B(\tilde{u}_p)\Psi\|^2 = (P_c+1)\langle \Psi, H_{\mathbb{T}}\Psi\rangle$, which gives the constant $(M_c+1)(P_c+1)\sum_\nu \|\alpha^\nu\|^2$ with $\mathcal{E}_{M_c}$; the three-body blocks are $z \leq Z_c$, the four-body blocks $D \leq D_c$, with mode indices at most D_c (whence $\mathcal{E}_{D_c}$) and $\sum_{D \leq D_c} n_D = N_4$. In Proposition 7.5 the near part uses the first family in (1.1) (non-strict inequalities suffice, and the blocks $z \in \{0,1,3\}$ vanish), the even blocks $z > Z_c$ are positive and the odd ones are bounded as before. In Lemma 8.1 the pair-centre channels are $d \leq D_c - 1$, whence $\mathcal{E}_{D_c-1}$, and Proposition 8.2 gives $-(1+\mathcal{E}_{D_c})^2(\varepsilon N_4 + 2N_4 \mathcal{E}_{D_c-1}) H_{\mathbb{T}}$. Collecting the terms as in (9.1) gives the first claim. For the second, $1 - \lambda_{\min} \leq 2\Delta(L) + \Upsilon(L,L)$ (proof of Corollary 9.2), the tail is bounded by Proposition 7.5, and $\mathcal{E}_m(\mathbb{T}) \leq \widehat{\mathcal{E}}_m(L)$ for $L^2 \geq 2m$ by Lemma 5.3(b); each bound tends to zero. □

10 What is not proved, and what remains

The transfer principle. Theorem 1.1 separates the two halves of the argument: the planar certificate, a finite computation in the plane, and its passage to rectangular tori, which the theorem carries out for every planar certificate of the form of [1], with errors bounded by explicit functions of the shorter period that decay at a Gaussian rate. It is the part of this paper on which further work can build, in particular on the spinful hollow-core model discussed at the end of this section. For the certificate of [1] the bounds are explicit: evaluated numerically (Remark 9.4), they give the thresholds 16.457, 16.857 and 18.685 magnetic lengths for a positive constant, a constant above $1/25$ and a constant within 10^{-5} of γ_* (Theorem 1.2), and the lower bound $\underline{\gamma}(L)$ tends to γ_* at the Gaussian rate $O(L^4\, 2^{-L^2/20})$ of Remark 9.3. The candidate proof of [10] claims a uniform bound of broader scope, on flat tori of every lattice shape and with every boundary twist, above an existential threshold; the bounds here are restricted to rectangular tori and are explicit.

Oblique tori. Theorem 9.1 is proved for rectangular tori, uniformly in the aspect ratio; it says nothing about tori with an oblique lattice of periods. The kernel theorem (Theorem 3.1 and Proposition 3.2) holds for every modular parameter, but for non-rectangular τ it is a statement about the operators $B_n^{(\tau)}$ of (3.2), and we have not identified $\sum_n (B_n^{(\tau)})^* B_n^{(\tau)}$ with the periodized planar pair projector on an oblique torus. The definition $H_{\mathbb{T}} := \mathcal{L}_2(\mathfrak{P}_2(P_1))$ makes sense for every lattice, and we expect most of the argument to carry over: the proof of the Moyal identity (Theorem 6.2) uses only the irreducibility of the fine-lattice magnetic translations of the centre-of-mass torus and the vanishing of the averaged image terms, and the image-sum calculus of Section 7 never uses the shape of the lattice. What would have to be redone is the part tied to the rectangular Landau gauge: the orbital form (Proposition 2.11), the pair Gram matrix and its levels (Lemmas 4.1 and 4.2), the irreducibility argument of Lemma 6.1 written in an oblique orbital basis, and the lattice sums $\mathcal{E}_m$ and Ξ_z, whose first shell can contain six vectors instead of four. A bound uniform in the modular parameter would allow the modular parameter to be varied within the gapped phase, which is the setting in which the Hall viscosity appears as an adiabatic curvature over the space of flat metrics [12, 13]; this is a separate problem, not addressed here.

Sharper constants. The threshold is set almost entirely by the four-body term $2444\,\mathcal{E}_{22}(1+\mathcal{E}_{23})^2$ and, after it, by the tail excess (Table 1). The first comes from bounding the periodized four-body target $\mathcal{T}_4$ by the pair energy through Cauchy–Schwarz over all n_D triples of each block (Proposition 6.8(d)); the second from the crude bound $|\langle m|\mathcal{D}|m\rangle| \le \min\{1, \mathfrak{g}_m\}$ on the diagonal entries of the local Gram matrix. A sharper estimate of $\mathcal{T}_4$, or the exact Laguerre values in Ξ_z, would lower the threshold; we have not attempted either. The bound on the pair-level term is already far below the others, and so is the non-orthonormality term $C_3\mathcal{E}_8(1+\mathcal{E}_8)$.

Finite tori below the threshold. Below the threshold the method gives nothing. For instance at the square aspect ratio and filling one third the bound is positive only for $N \ge 15$; smaller systems are accessible to exact diagonalization, which is a different kind of statement.

The spinful case. For spin-$\frac{1}{2}$ fermions with the hollow-core (pure V_1) interaction, at $N_\phi = 2N$ flux quanta, the zero modes are the Haldane–Rezayi states. We expect a spinful transfer principle of the form of Theorem 1.1: a planar certificate of the same form for the spinful model, with constant $\gamma > 0$, should give a uniform gap above the zero modes on rectangular tori. What it would need is the spinful version of each step: the spinful pair Gram matrix and its levels; the torus Moyal identities for pair states with spin labels, which should follow from the same Schur argument, since the magnetic translations act on the orbital factor only; and the three-body coefficient of the square of the spinful Hamiltonian, whose exact periodization should follow from Lemma 7.2, which does not depend on internal labels, but whose planar coefficients replace the q_z by spin-dependent blocks that have to be computed. None of this is done here. In the contrapositive, such a principle would show that no planar certificate of this form can exist for the spinful hollow-core model if a uniform gap above its zero modes on rectangular tori is excluded.

A The certificate data

This appendix records the data of the planar certificate of [1] that the proof consumes, in the form given there (its Appendix A and the file `verification/rows.txt` of its source directory). Nothing

in this appendix is re-derived; the inequalities (6.5) and (6.6) are proved in [1, Appendix A] by exact rational arithmetic.

A.1 The rows

Let $P := 10^7$. Each row below starts with the integers t_ν and $\ell_\nu = P\lambda_\nu$; an entry `pj:a` specifies the integer a^ν_{pj}, with p and j read as single digits; omitted entries are zero, and indented lines continue the preceding row. The coefficients are

$$\alpha^\nu_{pj} = \frac{a^\nu_{pj}}{P}\sqrt{\frac{2^{p-t_\nu}\, t_\nu!\,(p+j-t_\nu)!}{p!\,j!}}, \tag{A.1}$$

[1, Appendix A.1, (39)]; every nonzero entry has $0 \le p + j - t_\nu \le 8$, so the support lies in $\mathcal{S}_\nu$.

```
0 7181518 05:-849651 07:-2958744 15:751659 16:-43084 17:-146216 32:720949 35:-127514
  42:222437 43:-262822 53:-117700 61:-55024 71:-166049
1 -112991 05:142772 06:-142342 07:527055 08:50742 16:197917 18:-18343 24:5764 27:-21183
  32:75832 35:-6115 45:2660 53:-9455 61:-27091 63:-3494 71:479 72:-15331
1 -4247311 05:-12207250 06:-14064191 07:-14063663 08:-15662679 16:483831 18:3987214
  24:-814558 27:888887 32:966462 35:145978 45:-171722 53:34350 61:389319 63:97603
  71:412447 72:-53786
2 3438746 14:6689266 15:6620205 16:6625065 17:4969272 18:4878238 27:-1387380 36:-586188
  37:344087 42:-1120634 45:-318530 46:559169 55:157455 61:1207103 63:211617 64:33240
  71:577001 73:-142578
5 -98645 05:-2719776 08:-8519913 14:-949523 17:-1952105 18:-1562606 28:-8919966 32:248826
  35:-1017623 37:3656005 38:-2081871 42:-33163 43:-305905 46:35677 47:-1320749
  48:-1040378 53:502094 55:1060633 58:-998003 61:-113617 63:184919 64:209239 65:-68198
  67:-266297 72:-606696 73:-563620 74:437345 76:-432189
5 -3160276 05:-1698129 08:10575875 14:-252576 17:-8002616 18:22592844 28:7370640
  32:289294 35:1400647 37:1987740 38:4030699 42:-1731401 43:-1034062 46:-1106371
  47:-1257051 48:1461735 53:1408149 55:1136847 58:1878501 61:-2838516 63:-734099
  64:1062416 65:11745 67:611312 72:-480776 73:20459 74:316925 76:37092
7 1441784 07:3441286 08:-6097155 37:-1002399 47:-251321 58:1143225 66:-17397 71:-1007464
  73:-662286 76:-344621 78:-1115284
```

A.2 The constants

- $P^2\eta_* = \sum_\nu \ell_\nu^2 = 93527408868499$, so $\eta_* = 93527408868499/10^{14} = 0.93527408868499$.
- $\varepsilon = 3/10^6$, and $\sum_{D=1}^{23} n_D = \sum_{m=1}^{12} m^2 + \sum_{m=1}^{11} m(m+1) = 1222$ with $n_D = \lfloor (D+1)^2/4 \rfloor$.
- $\frac{3}{2}\sum_{z\ge 17,\, z \text{ odd}} (3z-1)(z+1)2^{-z} = 61/4096$ (Proposition 7.5).
- $\gamma_* = 1 - \eta_* - 61/4096 - 1222\varepsilon = 4616733319001/10^{14} = 0.04616733319001 > 1/25$.
- $\sum_\nu \|\alpha^\nu\|^2$ is a rational number determined by (A.1); its value is 3.9764514174 to the digits shown, so $C_3 = 72\sum_\nu \|\alpha^\nu\|^2 = 286.3045\ldots$; only the bound $C_3 \le 286.31$ is used.
- Only the orbital modes $0, \ldots, 8$ and the pair labels $0, \ldots, 7$ occur in the rows; the three-body blocks are $z \le 15$ and the four-body blocks $D \le 23$.

A.3 The three-body inequalities

For $z \in \{2, 4, 5, \dots, 15\}$ the margins $m_z := \frac{3}{2}(3z-1)(-\frac{1}{2})^z - e_z$ are rational, with e_z given by (6.3). In [1, Appendix A.2] they are computed exactly, by a formula in which all radicals cancel; the integer parts $\lfloor 10^6 m_z \rfloor$ recorded there are

z	2	4	5	6	7	8	9	10	11	12	13	14	15
$\lfloor 10^6 m_z \rfloor$	1298313	211317	249	250	249	250	250	250	249	250	250	6470	250

All are positive, which is (6.5).

A.4 The four-body inequalities

For $1 \le D \le 23$ and $r, s \in \mathcal{R}_D$ put $u_r := 1/(r!(D-r)!)$. Then $(E_D)_{rs} = \sqrt{u_r u_s}\, Y_{rs}$ and $(G_D)_{rs} = \sqrt{u_r u_s}\, Z_{rs}$ with rational matrices Y, Z given by explicit finite sums over the rows, and the matrix in (6.6) equals $D_{\sqrt{u}} M D_{\sqrt{u}}$ with $M = ZBZ$, $B = \mathrm{diag}((2\mathbf{1}_{r=1} + \varepsilon)u_r) - D_u Y D_u$ [1, Appendix A.3, (42)–(46)]. Positivity of the rational symmetric matrix M is verified there through the signs of the coefficients of $\det(x\mathbf{1} + M)$, computed exactly by the Leverrier–Faddeev recurrence: all coefficients after the leading one are nonnegative, so M has no negative eigenvalue. The recorded sign patterns (`+` positive, `0` zero, in descending degree) are

D	signs	D	signs	D	signs
1	0	9	++000	17	++++++000
2	0	10	+0000	18	+++++0000
3	00	11	+++000	19	+++++++000
4	00	12	++0000	20	++++++0000
5	+00	13	++++000	21	++++++++000
6	000	14	+++0000	22	+++++++0000
7	+000	15	+++++000	23	+++++++++000
8	+000	16	++++0000		

which is (6.6).

B Notation

The table lists the symbols that recur across sections, with the place where each is defined. Symbols used in one section only, and symbols local to a proof, are defined where they are used.

symbol	meaning	defined in
$\mathbb{T}$, Λ, C; L_x, L_y, L	the torus $\mathbb{R}^2/\Lambda$, its lattice of periods, a period cell; the periods; the shorter period	Def. 2.1
N_ϕ, κ	number of flux quanta, $L_x L_y = 2\pi N_\phi$; $\kappa = 2\pi/L_y$	Def. 2.1
$\mathfrak{h}$, $\mathfrak{h}_{\mathrm{pl}}$	lowest Landau level of the torus; planar lowest Landau level	Def. 2.1, (2.2)
w, $a \wedge b$	complex coordinate $x + \mathrm{i}y$; $a_x b_y - a_y b_x$	Section 2
ϕ_n, ψ_j, $[n]$	Landau-gauge orbitals ($n \in \mathbb{Z}$); torus orbitals ($j \in \mathbb{Z}_{N_\phi}$); class of n in $\mathbb{Z}_{N_\phi}$	Lem. 2.2
$\mathcal{H}_m$	normalized Hermite functions	(2.5)

symbol	meaning	defined in
t_a, $\tilde{t}_\gamma$, t_δ; θ_x, θ_y	magnetic translations; lattice translations, $\gamma \in \Lambda$; fine-lattice translations, $\delta \in \Lambda/N_\phi$; the translations $\psi_j \mapsto \psi_{j+1}$, $\psi_j \mapsto e^{2\pi \mathrm{i} j/N_\phi}\psi_j$	(2.2), Lem. 2.3
$\mathcal{B}$, $\lvert m\rangle$, $\mathcal{U}$	Bargmann space and its monomial basis; the unitary $\mathcal{B} \to \mathfrak{h}_{\mathrm{pl}}$	Lem. 2.4
$\mathcal{D}(\beta)$, $\mathcal{D}^{(m)}(\beta)$, $\beta(a)$	displacement operator; its truncation to the levels $0, \dots, m$; $(a_x - \mathrm{i}a_y)/\sqrt{2}$	(2.3), Lem. 2.4
η_m, η_m^R	symmetric-gauge orbitals about the origin and about R	Lem. 2.4, (2.4)
$P_{\mathrm{pl}}(r, r')$	kernel of the orthogonal projection onto $\mathfrak{h}_{\mathrm{pl}}$	Lem. 2.4(iv)
$\Pi^{(k)}$, Π	periodization of planar k-particle functions; $\Pi = \Pi^{(1)}$	Lem. 2.5
$\mathcal{F}$, $\mathcal{F}(u)$	fermionic Fock space over $\mathfrak{h}$; second quantization of a unitary u	§2.4
$B(v)$, $c(f)$, c_j	adjoint of wedge multiplication by v; annihilation operators, $c_j = c(\psi_j)$	(2.7)
A_n; P_σ	antisymmetrizer on $\mathfrak{h}^{\otimes n}$; permutation of the variables	§2.4; before Lem. 2.5
$\mathcal{L}_k$, X_S	lift of an operator on $\bigwedge^k \mathfrak{h}$ to $\mathcal{F}$; X acting on the tensor positions in S	(2.8)
W_3, W_4	the wedge maps $\bigwedge^2 \mathfrak{h} \otimes \mathfrak{h} \to \bigwedge^3 \mathfrak{h}$ and $\bigwedge^2 \mathfrak{h} \otimes \bigwedge^2 \mathfrak{h} \to \bigwedge^4 \mathfrak{h}$	Lem. 2.6(iv)
ζ, ρ; $\mathbb{T}_k$, $\mathfrak{u}_k$; $\tilde{t}_\gamma^{\otimes k}$	centre-of-mass and relative Jacobi coordinates; the centre-of-mass torus $\mathbb{R}^2/\sqrt{k}\Lambda$ and its lowest Landau level; the simultaneous lattice translation	Lem. 2.7
$\mathfrak{P}_k(Y)$, D	periodized image of an admissible operator Y; a fundamental domain of the diagonal action	Def. 2.8
$\mathcal{I}_a$	fibering map, the partial inner product with the relative state a	Lem. 2.9(c)
P_1, h, $H_{\mathbb{T}}$, $V_{\mathbb{T}}$	planar projector onto relative angular momentum one; $h = \mathfrak{P}_2(P_1)$; the torus V_1 Hamiltonian $H_{\mathbb{T}} = \mathcal{L}_2(h)$; $V_{\mathbb{T}} = \operatorname{ran} h$	Def. 2.10
c_0, $f(d)$, v_n, b_n	$\kappa^3/\sqrt{2\pi}$; $d\, e^{-\kappa^2 d^2/4}$; pair functions; pair operators	Prop. 2.11
τ, $B_n^{(\tau)}$	modular parameter; theta-function pair operators	(3.2)
Ψ_j	the three Laughlin states at $N_\phi = 3N$	Thm. 3.1
G, $\mathfrak{f}$, s_{jk}	pair Gram matrix; $\mathfrak{f}(s) = (1-s)e^{-s/2}$; $\frac{1}{4}(j^2L_x^2 + k^2L_y^2)$	Lem. 4.1
Δ, Υ; $\lambda_{\min}$, $\lambda_{\max}$, δ_G	$\Delta(\ell) = 2(\ell^2/4 - 1)e^{-\ell^2/8}$; remainder of the pair Gram levels; extreme eigenvalues of G and their deviation from 1	Lem. 4.2
P_V, H'	projection onto $V_{\mathbb{T}}$; $\mathcal{L}_2(P_V)$	Prop. 4.3
$X_3(h)$	three-body coefficient of $\mathcal{L}_2(h)^2$	Lem. 4.4
χ_m^R, v_p^{*R}, $\tilde{u}_p(R)$	periodized orbitals about R; planar pair states; periodized pair states	Lem. 5.1
$\Gamma(R)$, $\Gamma^{(m)}(R)$; μ_m, $\mathcal{E}_m(\mathbb{T})$	local Gram matrix and its truncation; norms of truncated displacement matrices and their lattice sum	Lem. 5.2
$\mathcal{E}_m(L)$, $\widehat{\mathcal{E}}_m(L)$, $\mathfrak{g}_m$; σ_z, Ξ_z	bounds on $\mathcal{E}_m(\mathbb{T})$; the majorant $(r^2/2 + m)^m e^{-r^2/4}/m!$; excess of the diagonal of $\Gamma(R)$ and its bound	Lem. 5.3
φ_p^R	centre-of-mass orbital of degree p about R	before Thm. 6.2
$s_{z,T,p}^{(u)}$, $s_{z,T,p}$; ξ_z, $\lvert z, T\rangle^R$	coupling coefficients, $s_{z,T,p} = s_{z,T,p}^{(1/\sqrt{3})}$; three-body relative states and coupled basis	(6.2), Lem. 6.4
$\omega_{D,r}$, $\lvert D, T; r\rangle^R$, S_r, $\mathcal{R}_D$	four-body relative states, coupled basis and coefficients; the odd integers in $[1, D]$	Lem. 6.5

symbol	meaning	defined in
ν, t_ν, λ_ν, α^ν_{pj}, $\mathcal{S}_\nu$	the rows of the certificate and their data	§1.1, before (6.3), App. A
η_*, ε; e_z, E_D, G_D	$\sum_\nu \lambda_\nu^2$; the allowance $3/10^6$; the three-body coefficients, four-body matrices and four-body Gram matrices of the certificate	(6.3), (6.4)
$F_\nu(R)$, $K_{\mathbb{T}}$	the rows at the base point R; the comparison operator	Def. 6.7
$K_{\mathbb{T}}^{(k)}$, Y_z, C_3, $\mathcal{C}_3$, $\mathcal{T}_4$	k-body part of $K_{\mathbb{T}}$; $\lvert W_3\xi_z\rangle\langle W_3\xi_z\rvert \otimes \mathbf{1}$; $72\sum_\nu\lVert\alpha^\nu\rVert^2$; non-orthonormality term; periodized four-body target	Prop. 6.8
q_z	planar three-body coefficients $(3z-1)(-\frac{1}{2})^z$	Lem. 7.3
δ_{tail}, Ω	tail constant of the three-body comparison; majorant of its excess	Prop. 7.5
$\gamma_{\mathbb{T}}$, γ_*, $\underline{\gamma}$	the torus constant; the planar constant; the uniform lower bound	Thm. 9.1, Cor. 9.2
$\epsilon_{\mathrm{tr}}(\mathbb{T})$, $\epsilon_{\mathrm{tr}}(L)$; P_c, M_c, Z_c, D_c, N_4	the error of the transfer principle and its bound; the cutoffs and the four-body count of a certificate of the same form	§1.1, Thm. 1.1

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